

THE SPECTRUM OF β PEGASI*

DOROTHY N. DAVIS†

Mount Wilson Observatory

Received February 10, 1947

ABSTRACT

Wave lengths, intensities, and identifications of 10,000 lines in the spectrum of β Pegasi, $\lambda\lambda$ 3400–8839 (Table 5), have been determined from spectrograms with dispersion of 2.9 Å/mm, or 5.8 Å/mm longward from λ 4900. The precision of wave lengths varies from ± 0.005 Å in the ultraviolet to ± 0.026 Å in the infrared. The visually estimated intensities are almost linearly related to equivalent widths. The technique of identifying atomic and molecular contributors is described, with emphasis upon the special problems presented by a high-dispersion M-type spectrum.

Forty-five elements are certainly present in the atmosphere of β Pegasi, while *eleven more* are doubtfully present. The unobserved elements are classified according to whether laboratory or stellar data are inadequate or whether ultimate lines are accessible. Brief notes on individual spectra of special interest, including the rare earths, are given in Part II.

As a guide to physical conditions in the atmosphere of β Pegasi the highest observed excitation potentials for several elements are listed in Table 3.

TiO is the most abundant molecule; *MgH*, *SiH*, and *AlH* are next in importance. Considerably less abundant are the following molecules: *ZrO*, *ScO*, *YO*, *CrO*, *AlO*, *BO*, *C₂*, *CH*, *CN*, *SiF*, *SiN*, and, possibly, *MgO*. It is doubtful that *BH*, *PH*, *FeO*, *MgF*, *SrH*, and *SrO* are present.

Typical of normal M-type giants is the bright star, β Pegasi,¹ which is essentially nonvariable both in light and in radial velocity. The rich spectrum is characterized by sharp lines. The spectral resolution attainable with the coudé spectrograph at the Mount Wilson Observatory is sufficiently high for certain identification of about 95 per cent of all features for which laboratory data are available. Lines as close as 0.10 Å can be resolved and measured micrometrically. On twenty fold, high-contrast enlargements even more detail can be discerned, so that some line pairs as close as 0.08 Å are seen resolved. Since it does not appear likely that better resolution will be achieved within a decade, it seems worth while to publish in as concise a form as possible the results of a detailed investigation. Such thorough and time-consuming studies may not be justified for many other stars, but it *does* seem worth while to attempt completeness for one example of each spectral class.

Another reason for studying β Pegasi is that it has nearly the same surface temperature as the supergiant α Scorpii, which was investigated earlier by the writer.² A comparison of the two spectra is interesting from many points of view.

The most compelling reason for selecting β Pegasi lies in the fact that this star appears to be completely normal. While it is exceedingly interesting to study peculiar stars and their anomalous behavior, one must first have knowledge of normal stars in order to evaluate and interpret unusual stars. It is hoped that the star β Pegasi provides a reliable basis of comparison.

I. THE OBSERVATIONAL DATA

All the spectrograms for this study of β Pegasi were obtained by Dr. W. S. Adams at the coudé focus of the 100-inch telescope. A complete list of the plates used in this investigation appears in Table 1. The radial velocities and probable errors are based upon unblended lines only. The range is perhaps greater than would be expected if the velocity

* Contributions from the Mount Wilson Observatory, Carnegie Institution of Washington, No. 733.

† Now at National Technical Laboratories.

¹ HD 217906; $\alpha = 22^{\text{h}}58^{\text{m}}9^{\text{s}}$; $\delta = +27^{\circ}32'$ (1900); mag. 2.6; sp. M2; abs. mag. -0.6 .

² *Ap. J.*, 87, 335, 1938, and 89, 41, 1939.

were constant. The present data should not be regarded as conclusive evidence for the variability of the velocity of β Pegasi, because no effort was made to eliminate personal or instrumental errors in order to obtain absolute values. Furthermore, the velocities given in Table 1 were obtained from different groups of lines throughout the spectral range from λ 3400 to λ 8800.

TABLE 1
SPECTROGRAMS OF β PEGASI

Plate No. (Ce)	Date	Average Dispersion (A/Mm)	No. of Lines	Radial Velocity (Km/Sec)	Probable Error of Mean (Km/Sec)
671.....	1933 July 10	5.7	211	+ 7.2	± 0.13
2083.....	1939 Aug. 23	2.9	117	9.2	.06
2139.....	1939 Oct. 21	5.8	37	9.4	.18
2376.....	1940 July 13	2.9	26	9.6	.08
2396.....	1940 Aug. 20	2.9	71	10.3	.06
2621.....	1941 Oct. 8	5.8	50	8.1	.16
2666.....	1941 Nov. 6	5.8	7	7.4	.34
2899.....	1942 Oct. 31	5.8	175	+ 6.9	± 0.11

TABLE 2
ACCURACY OF WAVE LENGTHS

Region Measured	Plate No. (Ce)	No. of Lines	Probable Error of Single Measure- ment of Wave Length
$\lambda\lambda$ 3392-3876.....	2396	71	± 0.0060 A
3786-4307.....	2083	55	.0053
4291-4325.....	2376	26	.0056
4312-4950.....	2083	62	.0071
4823-5285.....	2621	50	.021
4974-5367.....	671	81	.020
5288-6146.....	2621		
5373-5775.....	671	88	.022
5780-6188.....	671		
6064-6266.....	2666		
6192-6595.....	671	42	.0165
6301-6781.....	2139	37	.0236
6531-7608.....	2899	150	.0262
7016-8046.....	2666	7	.025
7619-8839.....	2899	25	± 0.020

ACCURACY OF WAVE LENGTHS

Although the primary object of this investigation was not to obtain highly accurate wave lengths, sufficient plates were measured to yield an accuracy satisfactory for identification. In most regions, duplicate plates were measured in order to establish the reality of the faintest lines, as well as to improve wave lengths. In all, about eighteen thousand lines and features were measured.

In Table 2 are listed the regions for which each plate was used; the last two columns contain the probable errors and the numbers of unblended lines used in the analysis. No

probable errors are given for the plates on which comparison lines were so faint or infrequent that the stellar wave lengths had to be computed from constants derived from measurements of unblended stellar lines. The corresponding correction-curves show that accidental errors have about the same magnitude as the errors in near-by regions, which were measured on plates having good comparison lines.

Because the probable errors in Tables 1 and 2 are "internal," they serve only as a basis for comparing the accuracy of measurements on different plates. They indicate that over the region $\lambda\lambda$ 3700–5000 the wave lengths of the best lines are good to $\pm 0.01\text{\AA}$ or better. In the visual and infrared regions, where the dispersion is only half of that in the violet, errors in the wave lengths of good atomic lines may be as large as $\pm 0.03\text{\AA}$. Wave lengths of hazy features are somewhat less trustworthy.

Systematic displacements of certain resonance lines were noted, but could not be studied thoroughly enough to warrant full publication at this time. The largest residuals from the mean velocity are shown by the absorption centers of the doubly reversed H and K lines of *Ca* II; they average -12.5 km/sec. The *Na* I lines are displaced by -3.6 km/sec, while the *K* I lines show an average displacement of only -2.4 km/sec. The resonance lines of *Al* I are apparently in their normal positions. The very broad line of *Ca* I, λ 4227, may be slightly displaced to the red.

INTENSITIES

Originally it was planned to obtain equivalent widths for all the lines in this spectrum. Precise photometric measurement of eleven thousand lines presents formidable problems, not the least of which is human endurance. A few microphotometer tracings were measured, but accuracy was everywhere seriously limited by the fact that the plates had been very well exposed in order to facilitate accurate wave-length measurement and the detection of weak lines. As a result, the continuous background is so dense that the energy absorbed by a given line cannot be measured with much accuracy. It is the purpose of this section to show that the writer's visual intensities are reasonably consistent and should serve fairly well in lieu of equivalent widths.

In estimating intensity the aim was to select a number which would represent the *total* energy subtracted from the continuous background by a single line. After some experience, the writer has developed a well-expanded visual-intensity scale which is almost linearly related to equivalent widths. For example, a line of intensity 10 is taken as twice as strong as a line of intensity 5; that is, the one line absorbs twice as much energy from the continuous background as does the other. One step in the scale corresponds to an intensity change such that two lines with intensities differing by one unit are so similar that a difference can barely be distinguished. A difference of two intensity units indicates that there is no question about the relative intensities of the corresponding lines. After some practice, such a scale seems simple, natural, and reproducible. By starting with the faintest detectable line and building up the scale step by step, it is possible to assign reasonably accurate intensities to all the lines which can be seen on one enlargement of a portion of the plate. Twenty fold enlargements were found to be best for this type of work. It was also found that by starting with the faintest lines and working up to the strongest it was possible to set up, independently on each enlargement, intensity scales which led to intensity estimates with accidental errors of less than two units. This figure was established by making several sets of estimates of the same region on enlargements of two different plates.

A somewhat independent check on the writer's intensity scale is afforded by comparing visual estimates from an enlargement of a sky plate with C. W. Allen's equivalent widths of solar lines³ and the revised Rowland disk intensities. The correlation-curves in

³ *Mem. Mt. Stromlo Commonwealth Solar Obs.*, Vol. 1, No. 5, Part II, 1934.

Figure 1,*A*, show that a strictly linear relation holds in the range from 1 to 12 on the visual scale. The average deviation of individual points from the curve is less than one unit in the writer's scale. Figure 1,*B*, shows a nearly linear relation between the two visual scales. Deviations are, however, somewhat large. In view of the fact that deviations for the same group of lines are small in Figure 1,*A*, it may be concluded that the residuals indicated in Figure 1,*B*, are largely due to the contracted nature of the Rowland scale.

The intensities in Table 5 are of varying degrees of accuracy. The most reliable intensities are those in the region $\lambda\lambda$ 4000–4800. The estimates made by inspection with the microscope while the plates were on the measuring engine could not be compared over wide regions of the spectrum because the field of view was limited. For the region $\lambda\lambda$ 3400–4800 the micrometer estimates were supplemented by inspection of twenty fold

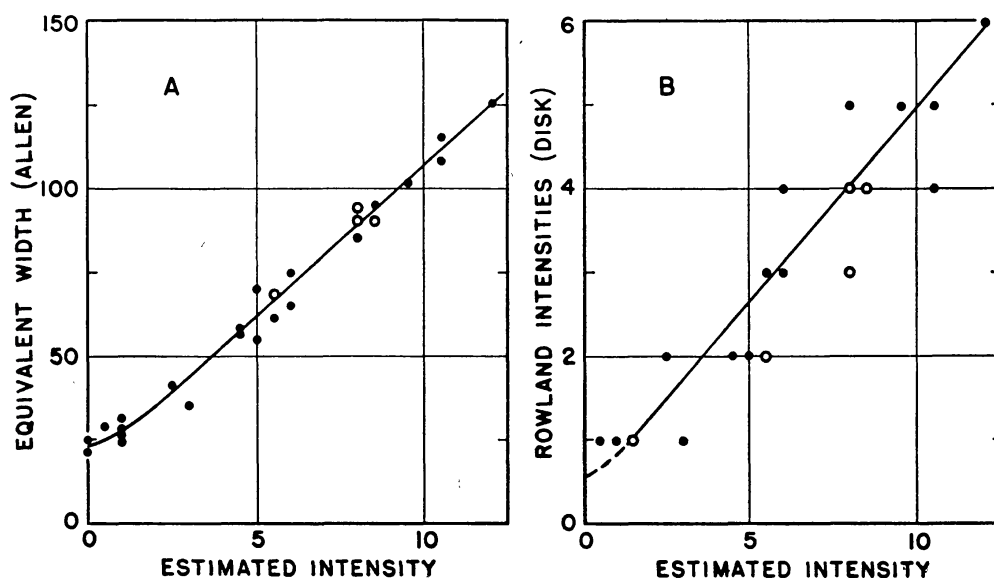


FIG. 1.—Intensities of lines in the solar spectrum estimated from a sky plate, Ce 2261, compared with: *A*, Allen's widths; *B*, intensities from the Revised Rowland Table. Open circles have greater weight than dots.

enlargements. Two sets of estimates were made at different times from enlargements of plate Ce 2376 for the region $\lambda\lambda$ 4000–4800, which was also covered once with enlargements of the contrasty plate Ce 2083. The region $\lambda\lambda$ 3400–4000 was covered twice with enlargements of plate Ce 2396. Obvious internal adjustments made it possible to place the intensities of all lines from λ 3600 to λ 4800 on a reasonably homogeneous scale.

Intensities in the region $\lambda\lambda$ 3400–3600 are not so reliable as the others, mainly because of the rapid change in the continuous background. For the region $\lambda\lambda$ 4800–8836 no attempt was made to adjust the intensities. The quality of the plates does not warrant such a procedure for the region beyond λ 6300.

The approximate relation between visually estimated intensities and equivalent widths in the blue and ultraviolet regions is shown in Figure 2, where typical correlation-curves are shown for three different plates. The deviations are larger than for Figure 1,*A*, because equivalent widths are difficult to determine accurately in late-type spectra. Furthermore, the intensity range is larger in Figure 2 than in Figure 1. Still, the correlation can be seen to approach linearity for the visual-intensity range from 5 to 20, where most of the stellar lines fall. Accidental errors are relatively small in this range. The equivalent width of a line (in $m\text{\AA}$) is approximately ten times the visual intensity.

In estimating intensities the writer attempted always to take into consideration the appearance of lines. Descriptive terms, such as "hazy," "sharp," etc., were freely recorded in the original ledgers, but the value of publishing such terms is doubtful, if due regard has been paid to all subjective descriptive data when identifications were made. The numerical values of the intensities (for approximate verbal meanings see Part IV) include an adjustment for line appearance. For example, a wide, shallow line which appears to absorb the same amount of energy as a sharp, intense line was assigned the same numerical intensity as the latter. As a result, two lines of equal numerical intensity but of different character may have very different central intensities, while their total intensities are the same.

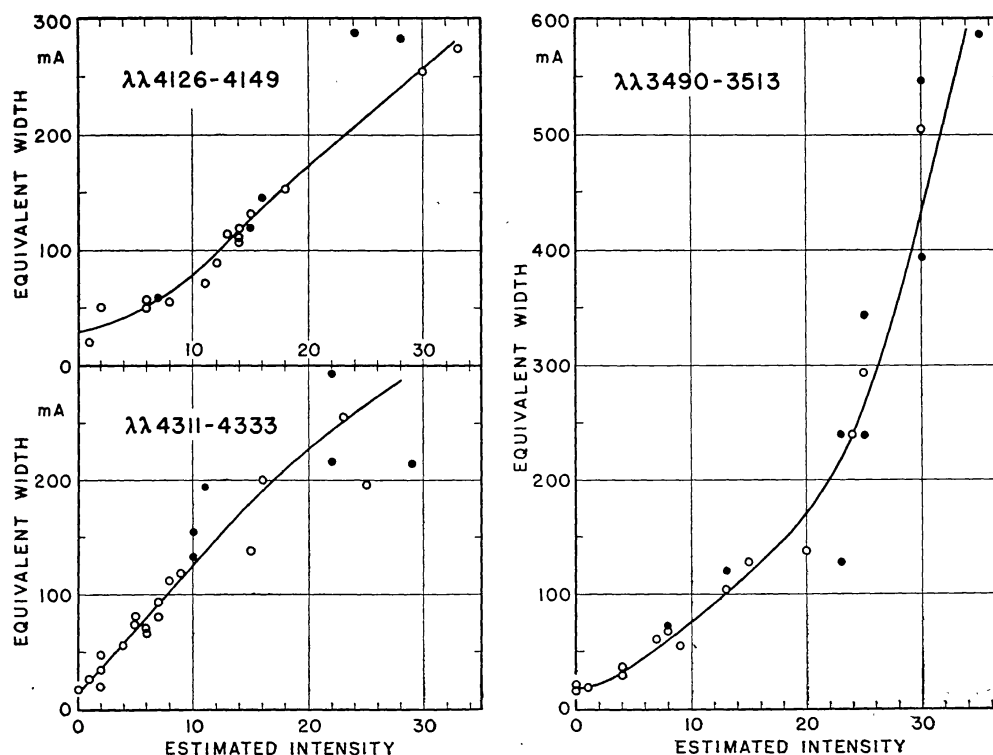


FIG. 2.—Correlation of visually estimated intensities and equivalent widths for β Pegasi. Open circles have greater weight than dots. The region $\lambda\lambda$ 3490–3513 from plate Ce 2396; $\lambda\lambda$ 4126–4149 and $\lambda\lambda$ 4311–4333 from plate Ce 2376.

II. THE ATOMIC LINES

TECHNIQUE OF IDENTIFICATION

The most complete source of information on atomic lines is the *Revised Multiplet Table (RMT)*,⁴ which has been adopted as the basic source in this study. The usual technique of identification involves consideration of agreement in wave length, excitation potential, transition probability, and laboratory and stellar intensities. This simple procedure is not, however, sufficient for the rich, high-dispersion spectrum of β Pegasi. Certain refinements which were especially useful in this investigation will be described. Extensive as it is, the *RMT* does not contain all the atomic lines which appear in the spectrum of β Pegasi. This is largely due to the fact that, in order to avoid prohibitive

⁴C. E. Moore, "A Multiplet Table of Astrophysical Interest, Revised Edition," *Princeton Contr.*, No. 20, 1945.

size and expense, many lines had to be omitted from the published *RMT*. Atomic spectra for which no spectroscopic analysis exists were omitted in entirety. The writer was fortunate in obtaining from Miss Moore large portions of the original manuscript on which the *RMT* is based. Several *Ti*, *V*, and *Zr* multiplets which were either wholly or partly omitted from the *RMT* are present in β Pegasi. Many of these lines appear in the *MIT* table,⁵ which accounts for the fact that the latter is frequently cited as a source in the third column of Table 5.

A few stellar lines coincide with atomic lines which are in the solar spectrum but do not appear in either the *RMT* or the *MIT* table. Such lines can often be found in the original laboratory lists, references to which are given in the *RMT*.

In spite of the high dispersion employed in this study, many lines are not completely resolved. Usually the number of significant contributors to a single line can be narrowed down to two. When more than three contributors appear to be involved, they are generally not all acknowledged in Table 5; in fact, a third contributor is sometimes omitted if it adds less than 10 per cent to the intensity of the line in question. This policy is based partly on typographical considerations and partly on the fact that one aim of this study is to show what lines are sufficiently unblended to be used for a determination of the curve of growth. Hence, since a line with two contributors is already unsuitable for photometric work, there is no need to specify more, unless a given line is critical in determining the presence of a spectroscopic term or element.

For difficult cases of blending it was helpful to use multiplet intensity tables.⁶ If a multiplet has only one unblended line, the intensity of the leading line can be predicted, whence theoretical stellar intensities of all other lines in the multiplet may be computed. (Better results are obtained, of course, if the multiplet in question has more than one unblended line, thus making it possible to obtain a better average for the leading line.) The line of intensity 10 at λ 4776.33 will serve as an example. As for all multiply blended lines, a card was prepared giving the information shown in the accompanying tabulation.

λ 4776.311.....	<i>Co</i>	158	x_6	2
.34.....	<i>Fe</i>	1206	x_1	?
.364.....	<i>V</i>	113	x_4	4

The first three columns give the laboratory wave length, element, and multiplet number for the only three lines in the *RMT* which could be responsible for the stellar line in question. Suppose we wish to know which contributor is most important and how much of the observed intensity is due to the high-level *Fe* line. The *Co* and *V* lines, both in the main diagonals of normal multiplets in which the leading lines are of intensity 10 or less, are x_6 and x_4 lines, respectively; whence their theoretical intensities are 2 and 4. This leaves about four intensity units for the *Fe* line, the leading line of a multiplet in which the other lines are blended or absent. Since this is a rather high intensity for such a high-level line, in view of the fact that its laboratory intensity is only "(1n)," there must be another contributor, and so the final identification would read "*V*(113)*Co*(158)*Fe*(1206)—." (The final dash was omitted from the published identification for reasons described in the preceding paragraph.)

PREDICTED LINES

It is well known that the *Fe* spectrum is rich in predicted lines observable in the solar spectrum. Many of the same predicted lines are present in the spectrum of β Pegasi.

⁵ G. R. Harrison and others, *M.I.T. Wavelength Tables* (New York: Wiley, 1939). Hereafter cited as "*MIT*" table.

⁶ The notation used here is that of H. N. Russell, *Mt. W. Contr.*, No. 537; *Ap. J.*, 83, 129, 1936.

Predicted low-level lines of $Ti\ I$ and $Ti\ II$ are also abundantly present. Multiplet (40) of $Ti\ I$ is made up entirely of predicted lines, which are all present and, with one exception, unblended in β Pegasi. An amazingly large percentage of high-level predicted $Fe\ I$ lines also are present, a good example being multiplet (1272).

A few stellar lines were found to be previously unpredicted lines which were computed by the writer from known spectroscopic levels. The possibilities have by no means been exhausted.

UNIDENTIFIED LINES

After the possibilities offered by the above methods were exhausted, a few lines remained unidentified. The majority of these lines coincide closely with unidentified solar lines. There are, however, so many solar lines of intensity -3 that many coincidences must be due to pure chance. The temptation to equate every unidentifiable stellar line with any unidentified solar line of the same wave length has been resisted. If, however, an unidentified solar line is certainly real or if it is strengthened in the sun-spot spectrum, it may reasonably be expected to appear in an M-type spectrum.

Other unidentified stellar lines are undoubtedly due to unobserved, but predictable, lines of well-known atoms or molecules. None of them are in the *MIT* table, for that source has already been exhausted by the writer, but some may be found in original laboratory lists. It would be a tremendous task to comb through all the lines in Kayser's *Handbuch* and in recent research papers. This was done for a few exceptionally abundant elements like Ti , V , and Zr ; but for most elements further searching would yield diminishing returns.

DOUBLING OF RESONANCE LINES AND OTHER LOW-LEVEL LINES

The most remarkable example of doubling is provided by the enormously strong, wide resonance lines of $Ca\ II$. Strong emission at their centers is divided by relatively narrow absorption (see Fig. 3, *a*). Contours for these lines have been determined from microphotometer tracings. The nature and extent of double reversal of the $Ca\ II$ lines in spectra of K- and M-type stars have been studied and will be described in a future communication.

Low-level lines of $Cr\ I$, $Fe\ I$, $Ti\ I$, $Ti\ II$, $Sc\ II$, and $Ni\ II$ show definite or merely incipient doubling. No doubling was observed, however, for the resonance lines of $Ca\ I$ and $Al\ I$. This very interesting problem deserves further study. It should be noted that the symmetrical doubling observed in the spectrum of β Pegasi differs from the asymmetrical broadening and doubling which have been observed in the spectra of M-type supergiants.⁷ Undoubtedly, some connection exists between the doubling and the effective atmospheric level at which the central part of the line originates.

UNUSUAL STRENGTH OF SATELLITE LINES IN MULTIPLETS

By means of Russell's tables⁶ the theoretical intensities of the first and second satellite lines were computed for several well-developed multiplets. It was often found that, while intensities in the main diagonal followed closely the theoretical intensities, the lines in the satellite branches, particularly the z -branch, were somewhat stronger than the theoretical values. Until more is known about the curve of growth, it would be dangerous to draw any conclusions.

EXCITATION CONDITIONS IN THE ATMOSPHERE OF β PEGASI

In a project of this type it is difficult to avoid becoming submerged in a sea of details. Many minor and apparently unrelated facts have been brought out by this study. The

⁷ W. S. Adams, *Mt. W. Contr.*, No. 638; *Ap. J.*, **93**, 11, 1941; L. Spitzer, Jr., *Mt. W. Contr.*, No. 619; *Ap. J.*, **90**, 494, 1939.

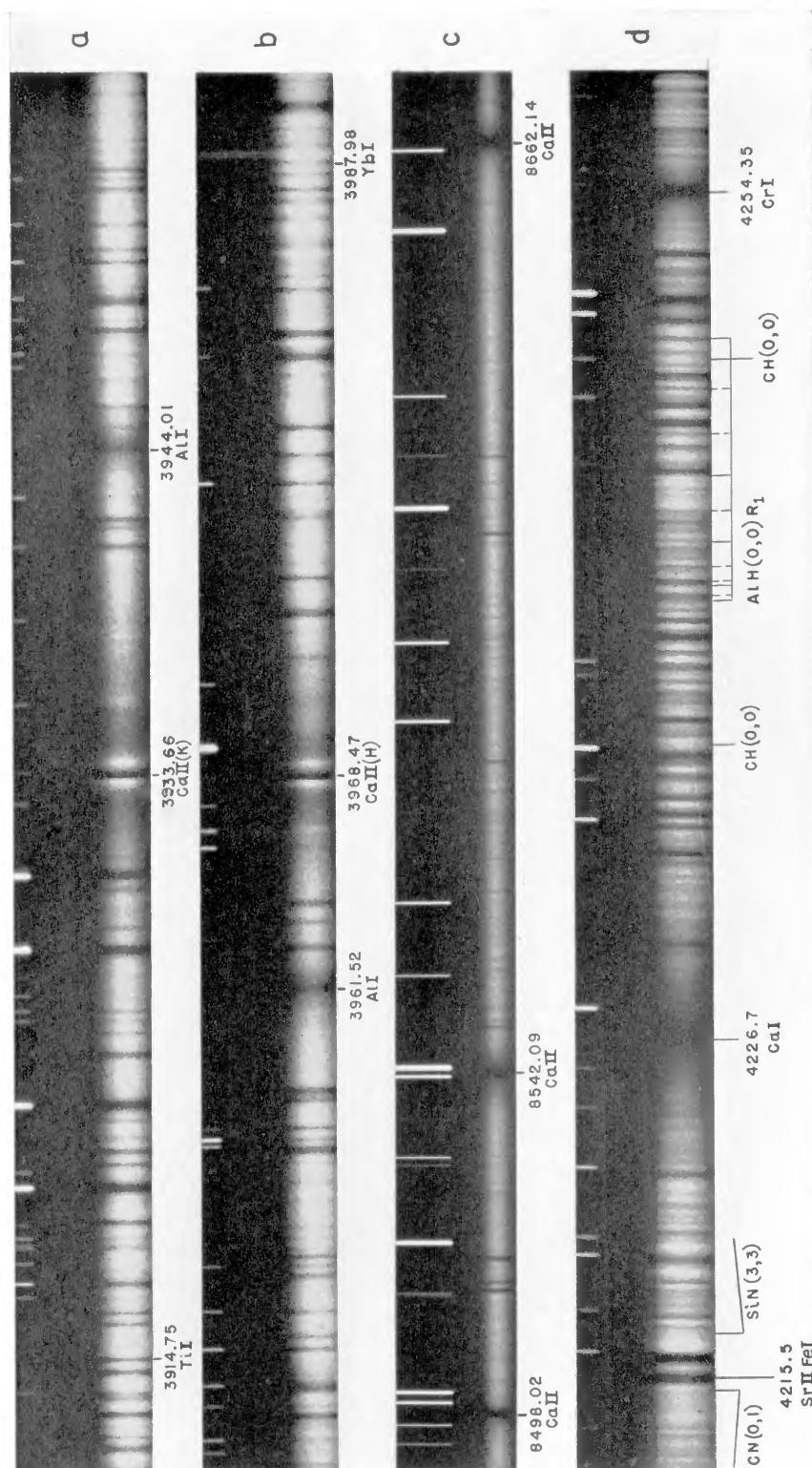


FIG. 3.—Calcium lines in the spectrum of β Pegasi

TABLE 3
ATOMIC LINES OF HIGHEST EXCITATION POTENTIAL OBSERVED IN β PEGASI

SINGLY IONIZED ELEMENTS																
ELE- MENT	NEUTRAL ELEMENTS							SINGLY IONIZED ELEMENTS								
	I.A.	Mult. No.	E.P. (Volts)	E.P. I.P.	β Pegasi		Sun		I.A.	Mult. No.	E.P. (Volts)	E.P. I.P.	β Pegasi		Sun	
					Int.	Blend?	D	S					Int.	Blend?	D	S
H^*	3889.051	2	10.15	0.749	5											
Na	4668.560	12	2.10	.410	(6)	TiO										
Mg	8736.0	39P	5.92	.779	12											
Al	8772.88	9	4.00	.671	20											
Si	8752.02	43	5.85	.720	3											
Ca	4685.265	51	2.92	.480	11											
Sc	4709.336	22	2.29	.34	0											
Ti	5766.330	309	3.28	.482	(3)	TiO										
V	5817.532	142	3.08	.459	0											
Cr	4170.202	278	4.08	.605	3											
Mn	8740.93	49	4.42	.597	1											
Fe	7351.56	1275	4.93	.628	1											
Co	5342.703	190	4.00	.510	0											
Ni	5595.735	206	3.88	.510	(3)	?										
Cu	8092.634	6	3.80	.495	(5)	TiO										
Zn	4810.534	2	4.06	.435	(1)	TiO										
Sr	5540.051	9	2.25	.397	0											
Y	6845.24	16	2.36	.363	3											
Zr	4719.12	66	1.85	.268	3											
Rh	3958.865	7	0.96	0.12	0											
Ba																
La																
Ce																
Pr																
Nd																
Sm																
Eu																
Gd																

* λ 3889.05 is not a distinct line in the sun. It probably accounts for a haze from λ 3888.5 to λ 3889.5.

† There is a strong possibility that some $Fe I$ lines of higher E.P. are present in β Pegasi; for example, λ 7547.89 and λ 7588.30, both of which are predicted, multiplet (1306).

‡ A line of higher E.P. almost certainly present in β Pegasi is λ 5381.776, 4.22v, multiplet (196).

§ The line, λ 5340.051, is either absent from the solar spectrum or masked by C_2 lines.

|| A $Sm II$ line of higher E.P. very probably present in β Pegasi is λ 7042.24, but the wave lengths do not agree well enough. Furthermore, this line is either absent from the solar spectrum or masked by atmospheric lines.

next phase is to correlate observations and to strive for an accurate over-all description of M-type spectra. A curve of growth would partly fill this need. An analysis based upon calibrated intensities, absolute f -values, and transition probabilities cannot, however, be undertaken by the writer at present.

Some indication of physical conditions in the atmosphere of β Pegasi is afforded by a tabulation of the highest excitation potential observed in the spectra of certain elements. Table 3 gives also the wave lengths and multiplet numbers of the observed lines which correspond to these levels. The table is divided into two parts, one for neutral and the other for singly ionized elements.

Intensities are given in Table 3 for the solar counterparts of the lines observed in β Pegasi. Immediately following the intensities are indicated the blending elements or molecules, if any. In order to obtain a representative value of the highest observed excitation potential, it is necessary to include a few blended lines in this table. If they had been omitted because of blending, the excitation potential of a line of lower excitation potential, which would have been chosen for its purity, would have been quite without meaning as far as excitation conditions are concerned. A blended line was, however, included only when its identification was certain. The amount which the element in question contributes to the observed absorption, given in parentheses in the intensity column, was estimated by studying other lines in the same multiplet and/or studying the intensity distribution in multiplets (or molecular spectra) responsible for blending.

It is interesting to note that the elements of high atomic weight (that is, from Ba on) are represented in this table only by lines arising from the ionized state. A few lines of the neutral rare earths may be present in β Pegasi, but they are weak and doubtful.

The column headed "E.P./I.P.," which gives the ratio of the excitation potential to the first ionization potential above the given state, is included for comparison with a similar tabulation in the forthcoming paper on the infrared solar spectrum which Mr. Babcock kindly showed to the writer in advance of publication. For the first long period of the periodic table the ratio E.P./I.P. progresses slowly to a maximum at Fe and declines thereafter, averaging about 0.6 for this group. For elements of higher atomic weight the ratio is smaller.

While the ratio E.P./I.P. for the neutral elements varies through a range of approximately only three times, the Boltzmann factor, $e^{-E/kT}$, shows an enormous range. Before they are multiplied by the ratio of the statistical weights, the Boltzmann factors (with $T = 3100^\circ K$) vary from 2.49×10^{-10} for Mg to 9.9×10^{-4} for Zr , a range⁸ of a million times! This enormous range illustrates the well-known fact that temperature is not the only factor which determines whether a level is sufficiently well populated to give rise to observable lines.

OCCURRENCE OF ELEMENTS IN β PEGASI

The certain identification of forty-five elements in the atmosphere of β Pegasi resulted from this study. The presence of eleven other elements is doubtful, their critical lines being a little too heavily masked for certainty. It would seem, however, that they must be present because, without them, some lines would be difficult to explain.

Though the remaining thirty-six elements have not been observed in β Pegasi, they should not be regarded as definitely absent. Some cannot be observed for reasons beyond human control, while others are unobservable because of insufficient data. The unobserved elements form the last two groups of Table 4. The distinction between "Insufficient stellar data" and "Ultimate lines inaccessible" is that the former might be observed if stellar observations were extended beyond the wave-length limits of this study, while the latter groups could never be observed on the earth's surface because of atmospheric absorption.

⁸ The true extreme values 3.29×10^{-17} for H and 2.76×10^{-2} for Rh have been purposely ignored because H is so abundant and Rh so rare.

NOTES ON INDIVIDUAL SPECTRA

Certain interesting details about atomic spectra observed in β Pegasi are arranged below in order of atomic number. Several elements have been omitted from the discussion, not because they were unobserved, but because there was nothing unusual about their occurrence in β Pegasi.

H.—*H α* and *H β* have intense, “square” cores 1.2A and 0.5A wide, respectively. *H γ* and *H δ* have fairly wide, symmetrical wings and high central intensities.

Li I.—The resonance doublet at λ 6707, although searched for on several plates, was not found. No normal lines appear in the expected position in the stellar spectrum. If

TABLE 4
OCCURRENCE OF ELEMENTS IN β PEGASI

	Group I. Elements Present—56									
	<i>H</i> <i>K</i> <i>Cu</i> <i>Pd</i> <i>Dy</i>	<i>B*</i> <i>Ca</i> <i>Zn</i> <i>Ba</i> <i>Tm</i>	<i>C*</i> <i>Sc</i> <i>Ga</i> <i>La</i> <i>Yb</i>	<i>N*</i> <i>Ti</i> <i>Sr</i> <i>Ce</i> <i>Os</i>	<i>O*</i> <i>V</i> <i>Y</i> <i>Pr</i> <i>Pb</i>	<i>F*</i> <i>Cr</i> <i>Zr</i> <i>Nd</i>	<i>Na</i> <i>Mn</i> <i>Cb</i> <i>Sm</i>	<i>Mg</i> <i>Fe</i> <i>Mo</i> <i>Eu</i>	<i>Al</i> <i>Co</i> <i>Ru</i> <i>Gd</i>	<i>Si</i> <i>Ni</i> <i>Rh</i> <i>Tb</i>
Certain.....										
Doubtful.....	<i>Li</i> <i>Tl</i>	<i>P*</i>	<i>Rb</i>	<i>In</i>	<i>Ho</i>	<i>Er</i>	<i>Hf</i>	<i>W</i>	<i>Re</i>	<i>Ir</i>
	Group II. Insufficient Data—17									
	<i>Be</i> <i>Pt</i>	<i>S</i> <i>Bi</i>	<i>Ge</i>	<i>Ag</i>	<i>Cd</i>	<i>Sn</i>	<i>Sb</i>	<i>Cs</i>	<i>Lu</i>	<i>Ta</i>
Insufficient stellar data.....										
Insufficient laboratory data.....	<i>Ma</i>	85	87	<i>Il</i>	<i>Th</i>					
	Group III. Unobserved—19									
	<i>U</i>									
Ultimate lines accessible.....										
Ultimate lines inaccessible..	<i>Ho</i> <i>Xe</i>	<i>Ne</i> <i>Au</i>	<i>Cl</i> <i>Hg</i>	<i>A</i>	<i>As</i>	<i>Se</i>	<i>Br</i>	<i>Kr</i>	<i>Te</i>	<i>I</i>
Not expected.....	<i>Po</i>	<i>Rn</i>	<i>Ra</i>	<i>Ac</i>	<i>Pa</i>					

* Elements observed only in molecular combinations.

the *Li* lines are shifted as much to the violet as are the *Na* lines, then the line at λ 6707.53, which is only partly due to *TiO*, may be partly due to *Li*.

Na I.—The D lines are a little unsymmetrical (see Fig. 4,*c*), the shortward side being sharper. They are about 1 A wide.

Mg I.—The great abundance of *Mg* in β Pegasi is indicated not only by very strong low-level lines but also by several predicted lines from the high 3¹D level.

Al I.—The symmetry of the very strong, broad resonance lines (see Fig. 3,*a,b*) is not determinate because of blending in the wings of the H and K lines.

Si I.—In addition to the familiar lines in the violet, several high-level, infrared lines are present. Lines arising in the 3d¹D° state all have wave lengths which coincide

with solar values rather than with laboratory values. Mr. Babcock has pointed out⁹ that this is an outstanding example of the effect of laboratory excitation conditions upon wave lengths.

K I.—The strong resonance lines are easily distinguished from the intense lines of the atmospheric A band. Only one other line in the principal series, λ 4044.14, is unblended.

Ca I.—The symmetrical wings of the intense resonance line (see Fig. 3,*d*) may easily be seen over a region 10 Å wide.

Ca II.—The strongest and most unusual lines in the spectrum of β Pegasi are due to *Ca II*. The total equivalent width of the two resonance lines is 44.3 ± 1.5 Å. The wings overlap (see Fig. 3,*a,b*) and cover a region more than 100 Å in width. This would be an excellent region for further investigation of the theories of line formation by the method of A. D. Thackeray.¹⁰ The infrared lines of *Ca II* are the strongest observed in the infrared region (see Fig. 3,*c*).

Ti II.—The spectrum of *Ti II*, like that of *Ti I*, is exceptionally well developed in β Pegasi. Levels of excitation potential higher than 1 volt are represented by strong lines. Lines barely visible in the laboratory, indicated in the *RMT* by “tr,” have intensities as high as 12 in β Pegasi! Predicted lines are also well represented. Above 2 volts there is a distinct drop in occurrence and intensity.

Fe I.—Approximately two-fifths of the lines in the spectrum of β Pegasi belong to the well-developed *Fe I* spectrum. As expected, low-level lines are intense, but certain high-level lines are unexpectedly strong. Levels having excitation potentials greater than 4 volts have been detected with certainty. Particularly interesting are the strong infrared multiplets (60) and (1154). Hundreds of predicted lines present in the solar spectrum are also present in β Pegasi.

Fe II.—The strongest *Fe II* line in β Pegasi, λ 4233.16, has an intensity of only 3. If correctly identified, the *Fe II* line with the greatest laboratory intensity is only 2 in β Pegasi. Its weakness is probably due to its relatively high excitation potential.

Co I.—Intensities in the first multiplet follow closely the laboratory intensities, which is interesting because this is an intercombination multiplet. On the whole, the *Co I* spectrum is well developed in β Pegasi.

Ni I.—Some of the low-level lines are very strong and wide. They are predominant features of the ultraviolet spectrum of β Pegasi.

Ga I.—Both of the lines listed in the *RMT* are blended in β Pegasi. The line at λ 4172.05 (see Fig. 4,*a*) coincides with a very strong stellar line, which could not possibly be due to *Fe* alone. Since the same line is strengthened in α Herculis, it seems highly probable that *Ga* is present in both stars.

Rb I.—Both ultimate lines fall in regions where *TiO* lines are numerous. The *Rb I* line at λ 7800.23 is 0.06 Å from a stellar line which coincides with a feature in the laboratory spectrogram of *TiO*, which may very well contain *Rb* as an impurity. The same laboratory plate shows strong lines due to two other alkalis, *Na* and *K*. The *Rb I* lines near λ 4200 are hopelessly masked. We may conclude only that *Rb* is doubtfully present in β Pegasi. At least, it is not clearly absent.

Sr I.—The line of highest excitation potential in β Pegasi is not observed in the sun. This might indicate greater abundance of *Sr* in β Pegasi, but it is more probable that the moderate ionization potential of *Sr I* causes the neutral element to be seriously depleted in the sun.

Zr I.—Nearly every *Zr I* line listed in the *RMT* may be found in β Pegasi. More *Zr I* lines are included in the original list,¹¹ but none which originate from levels higher

⁹ In conversation.

¹⁰ *Mt. W. Contr.*, No. 555; *Ap. J.*, **84**, 433, 1936.

¹¹ C. C. Kiess and Harriet K. Kiess, *Bur. Standards J. Res.*, **6**, 621, 1931.

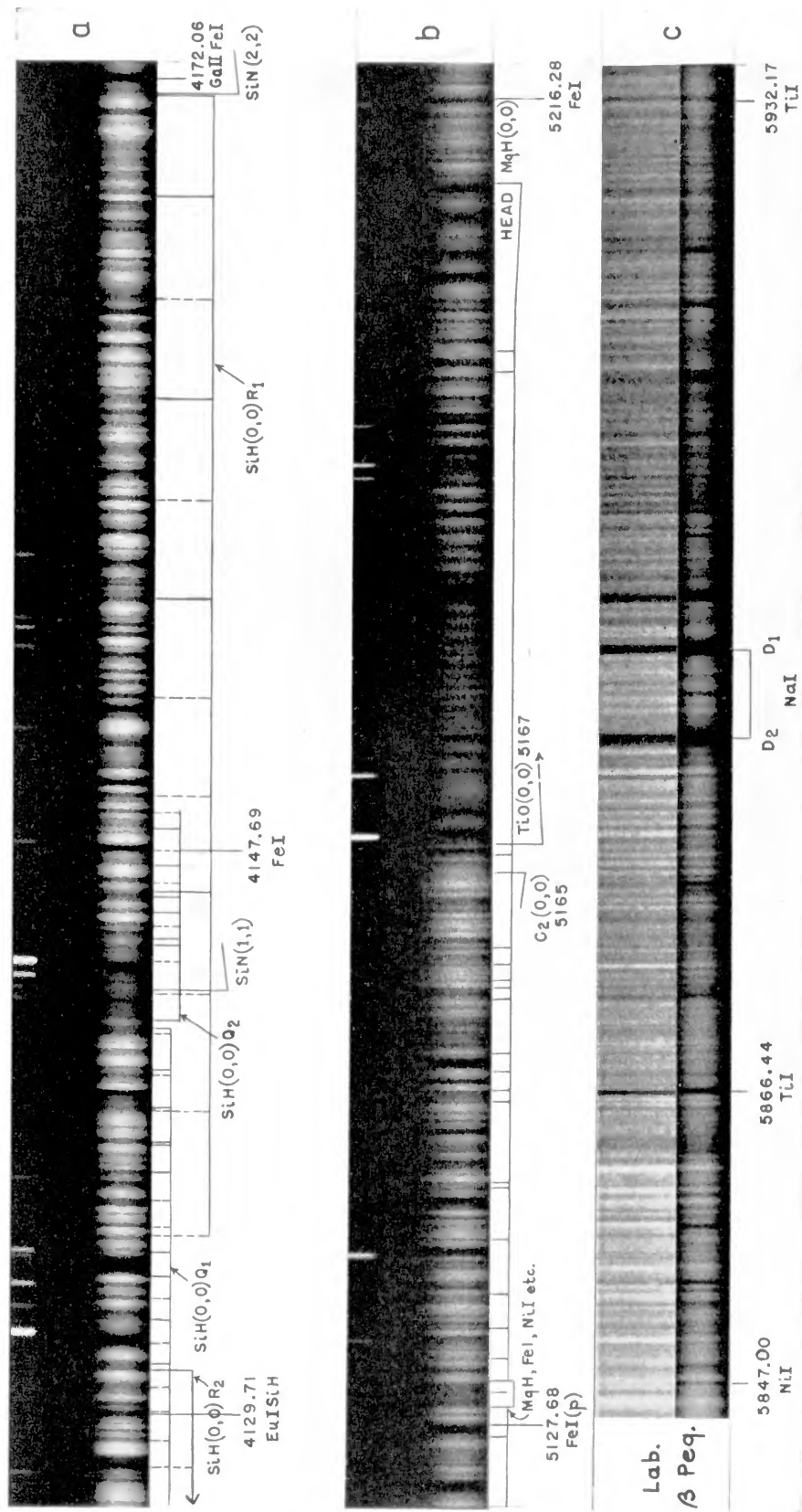


FIG. 4.—Selected molecular spectra in β Pegasi. In *a*, broken lines indicate molecular lines which are blended with atomic lines. In *c*, the laboratory spectrum of TiO shows also some atomic lines of Ti I and Na I .

than 1.87 volts were found in β Pegasi. Lines of Zr II also are well represented in β Pegasi.

Cb I.—All Cb I lines appear fuzzy in β Pegasi.

Ru I.—This is another element whose lines are far more prevalent in β Pegasi than in the sun. A few lines not listed in the *RMT* are definitely present in β Pegasi. No observable Ru II lines are present.

Rh I.—The low-level lines are considerably stronger in β Pegasi than in the sun.

Pd I.—The first multiplet is strengthened in β Pegasi, but in the second multiplet the leading line, supposedly the ultimate line of the Pd I spectrum, is weak.

In I.—The ultimate line has an intensity of only 6, yet it is distinctly stronger than in the sun.

Cs I.—The wave length of the ultimate line is 0.16 Å shorter than that of the nearest stellar line, λ 8521.26, the intensity of which is only -2 . If Cs shares in the velocity shift observed for Na and K , a stellar line should be on the other side, where no line is observed. There would be no hope of finding any weaker lines in the infrared unless a fine-grained emulsion were used.

Ba II.—The great strength of the Ba II and Zr I lines suggests a possible connection between β Pegasi and stars with S-type spectra.

La I.—The ultimate line, as well as other strong lines, is absent.

La II.—The lowest state is not well represented in β Pegasi. The next level, a^1D , provides the strongest La II lines in β Pegasi. Higher levels contribute very little, if at all.

Ce II.—The lines of Group I are stronger in β Pegasi than are the lines of Group II. There are more lines in Group II, but fewer show in β Pegasi. The first multiplet of Group I is well developed in β Pegasi, although none of the lines is above average intensity.

Pr II.—Because the majority of the strongest Pr II lines fall in the region dominated by the H and K lines or in regions masked by TiO lines, very few unblended lines are observable. There can be no doubt that Pr II contributes substantially to many blends. Very puzzling is the broad, hazy line at λ 4408.84, centered on a strong Pr II line but much wider than expected of a single line. The rest of the absorption may be due to an unidentified molecule. Many of the strongest Pr II lines are unclassified, but only two coincide with unblended stellar lines.

Nd II.—The majority of the unblended lines arise from the lowest state. Above an excitation potential of 0.7 volt few Nd II lines show in β Pegasi. About one-fourth of the unblended lines are unclassified spectroscopically. Lines of temperature class II show well. Class IV lines are less prevalent.

Sm I.—The line of greatest laboratory intensity is absent. The ultimate line and other strong lines are masked. The strong line at λ 3756.411 coincides with an unblended stellar line of intensity 0, but this Sm I line coincides with a Sm II line, multiplet (44). Of the two other lines in the Sm I multiplet, one is absent and the other is 0.07 Å from a stellar line of intensity 0. There is also a Dy II line at this wave length. It does not seem very likely that neutral Sm is in the atmosphere of β Pegasi.

Sm II.—The lowest-lying levels are well represented. Some lines are unexpectedly strong in β Pegasi. Multiplet (37) is particularly strong. On the other hand, several lines which have high laboratory intensities are weak or absent from the spectrum of β Pegasi. It is evident that laboratory intensities are not very well correlated with stellar intensities. Lines of excitation potential greater than 0.7 volt do not appear. Several unclassified lines are present.

Eu I.—The ultimate line and another strong line are masked. The zero-level line at λ 4627.22 coincides with a stellar line of intensity 3, which is probably due entirely to TiO .

Eu II.—The ultimate line has an intensity of 6, but a line at λ 3907.10 which is

weaker in the laboratory is twice as strong as the ultimate line in β Pegasi. Another unusually strong line is λ 4205.05. Lines from levels above 1.37 volts are probably absent. None of the unclassified lines appear to be present.

Gd I.—The ultimate line is hopelessly masked by the *Ca* line at λ 4227. Because the ionization potential is higher than for other rare earths, there is some hope of finding *Gd I* in β Pegasi. Five lines which may be present are: $\lambda\lambda$ 4345.46, 5617.91, 5696.22, 5856.22, and 6991.92. No very definite conclusion may be drawn at present.

Gd II.—The four lowest levels appear definitely to be contributing in β Pegasi. There are a few puzzling anomalies in intensity. Above an excitation potential of 1.06 volts, all lines having strong laboratory intensities are absent or very weak. Of the spectroscopically unclassified lines, only two of temperature class IV appear, and none of class V.

Tb II.—The strongest class V line is present and evidently unblended. Of the four strongest class IV lines two are apparently unblended. All the strongest lines, except λ 4005.55, are present in R Andromedae.¹² We may conclude that *Tb II* is definitely present in β Pegasi.

Dy I.—The three strongest lines all coincide closely with lines in the spectrum of β Pegasi: $\lambda\lambda$ 4186.80, 4194.83, and 4211.70. This is the most definite evidence for the presence of a neutral rare earth in β Pegasi.

Dy II.—Of all the rare earths, *Dy II* contributes the strongest lines to the spectrum of β Pegasi. The two strongest lines are at λ 4077.97 and λ 4103.31. A large proportion of the lines in Table 2 of the paper by King and Moore¹³ is present. The stellar intensities are in good accord with laboratory intensities, except for a few lines that are stronger than expected.

Ho I.—The majority of the strongest arc and furnace lines are masked by strong stellar lines of known origin. Two stellar lines which are difficult to identify with any other element coincide with *Ho I* lines at λ 4103.84 and λ 4163.03. The former line is strengthened in the sun-spot spectrum and shows in R Andromedae.¹² The presence of *Ho I* is doubtful.

Ho II.—Of the three strongest lines, one coincides with a *Ho I* line, which is masked in β Pegasi by *Co I*, *Nd II*, and *Ce II*. The second falls near a strong *Co I* line in the wing of the very strong *Fe I* line at λ 4045. The third falls near a stellar line at λ 3891.07, which is due partly to *V I*. This coincidence looks promising but is probably insufficient evidence for the presence of *Ho II*.

Er I or Er II.—There are several good coincidences with stellar lines. One of the strongest lines, λ 3766.26, is absent, however. Nothing more can be said about *Er* at present.

Tm I.—Five stellar lines coincide with the strongest *Tm I* lines. Three of these lines are very weak in the solar spectrum and appear in R Andromedae.¹²

Tm II.—The strongest line of the spectrum falls near a predicted *Ti I* line which is not given in the *RMT*. All except four of the other strong lines are blended. Higher levels are apparently inactive. Three unclassified lines are evidently unblended in β Pegasi.

Yb I.—The two strongest lines are present, one being the ultimate line. All lines having arc intensity greater than 20 were searched for. Only one unblended line was found.

Yb II.—The ultimate line is moderately strong in β Pegasi, the intensity being 14. Several close coincidences were observed with lines having arc intensities greater than 20. Only the lowest levels appear to be contributing.

Lu I.—All except one of the strongest low-level lines are either masked or absent. The unblended line is at λ 4124.73 and has an intensity of 1.

Lu II.—The single zero-level line at λ 3507.39 is masked by two lines which are close together in β Pegasi. There is absorption between them, but this is probably due to *Fe*.

¹² P. W. Merrill, *Mt. W. Contr.*, No. 730; *Ap. J.*, **105**, 360, 1947.

¹³ *Mt. W. Contr.*, No. 681; *Ap. J.*, **98**, 33, 1943.

None of the higher-level lines appear to be present. The presence of Lu II or Lu I in β Pegasi is very doubtful.

Hf I.—All the low-level lines are masked in the observed region.

Hf II.—The strongest low-level line, λ 3399.80, coincides with half of a double stellar line, probable intensity 2. There are no other convincing coincidences.

W I.—The strongest lines coincide with stellar lines, one being unblended, intensity only 1. None of the lines in the first five multiplets in the *RMT* appear to be present.

Re I.—The ultimate line is apparently unblended, intensity 3. Two other lines in the same multiplet contribute to blends. One of the two strong lines in the first multiplet is absent; hence the evidence for *Re* I must be considered doubtful.

Os I.—Three zero-level lines are in the observed range. One line, λ 3528.60, is apparently unblended in β Pegasi, intensity 5. The only line from the next level apparently contributes to the double line at λ 3752.44. One of the three observable lines from the next level is unblended in β Pegasi.

Ir I.—The first line of the first multiplet is unblended, intensity 3. Other lines are masked or absent, which makes the identification of *Ir* in β Pegasi doubtful.

Tl I.—The ultimate line is masked by a powerful *TiO* band. The strong line at λ 3519.24 appears to be unblended, intensity 1.

U I.—The strongest low-level lines are all absent, except for λ 6395.42, which coincides with an identified stellar line, intensity 1. This is probably a chance coincidence. Since *U* is a rare-earth type of element with a low ionization potential, one would expect the spectrum of *U* II to have a better chance of appearing. The great atomic weight and radioactivity of *U* appear, however, to be prohibitive.

III. THE MOLECULAR SPECTRUM

One of the reasons for selecting β Pegasi for study was that the atmosphere of this star has a temperature which is theoretically an optimum for the existence of several diatomic molecules, as may be seen from the molecular abundance-curves computed by M. Nicolet.¹⁴ Except for tightly bound molecules like *CH*, *CN*, and *C₂*, the maxima for the most likely molecules lie between 3000° and 3100° K. Although several molecules remain to be studied, enough has been done to show that the theoretical predictions are moderately well followed out in β Pegasi.

As expected, *TiO* is the dominating molecule in this star. Of all other molecules, *MgH*, *SiH*, and *AlH* are by far the most abundant. This is also to be expected because the constituents are the most abundant atoms in stellar atmospheres.

According to Figure 12 of Nicolet's paper,¹⁴ the order of strength of the metallic oxides should be *ScO*, *VO*, *MnO*, *ZrO*, *CrO*. The observed order of intensity in β Pegasi is more nearly *ZrO*, *ScO*, *YO*, *CrO*. The apparent absence of *VO* and *MnO* may be due to the very open nature of the spectral structure, as compared with the other spectra, which are characterized by closely packed lines near some of the heads.

TECHNIQUE OF IDENTIFYING MOLECULAR ABSORPTION

With high-dispersion stellar spectra it is not sufficient simply to look for band heads. If a rotational analysis exists, its presence is established by noting wave-length coincidences. Due regard must be paid to theoretical intensity distribution, a function mainly of temperature and transition probabilities. Intensity formulae, if not too complicated, may be differentiated to obtain a relation between temperature and the quantum number for which intensity should be a maximum. With this relation as a guide it may not be difficult to decide about the presence of a molecule. If the molecule is not very abundant and has a complex spectrum with many rotational lines, it is worth while to compute the number of chance coincidences to be expected, according to the method outlined

¹⁴ *Mem. Soc. R. d. sci. Liège*, Ser. 4, 2, 89, 1937.

previously by the writer.¹⁵ If the number of coincidences is greater than the number expected by chance and if the coincidences and absences are uncorrelated with the theoretical intensity distribution, it is not possible to conclude that the molecule is present.

If only a vibrational analysis is available, the task of identification is much greater and the results are relatively uncertain. It has been found that the best procedure under such circumstances is to obtain a laboratory spectrogram of the emitting molecule and from it prepare a glass positive with approximately the same dispersion as the stellar spectrogram. With a Hartmann spectrocomparator, the glass positive may be compared directly with the stellar plate. A. S. King and R. B. King took the desired laboratory spectrograms. Glass positives were prepared by the writer for the following molecular spectra: *AlO*, *BaO*, *CN*, *CaO*, *CbO*, *CrO*, *FeO*, *MgO*, *MnO*, *ScO*, *TiO*, *YO*, and *ZrO*. Identifications may be further checked by comparison with other stellar spectra.

The procedure described above provided the only successful method for recognizing the very numerous *TiO* lines, which constitute practically the entire stellar spectrum in several wide regions. Only three *TiO* bands have been submitted to rotational analysis. Countless other *TiO* bands are present in the spectrum of β Pegasi and of all other M-type stars, and could be recognized only by direct comparison with positives throughout the spectrum from λ 4300 to λ 8300 Å (see Fig. 4,c).

Much inconvenience has been caused by the lack of rotational analyses, which are usually conducted by physicists only sufficiently far to identify the transition type, verify the combination principle, and determine molecular constants. It is true that, after these data are substantiated, the physicist has little to gain by further laborious measurements and assignment of quantum numbers. Furthermore, some molecular spectra of high astrophysical importance are quite unattractive to physicists because they are normal or unnecessary for checking theory. The astrophysicist, on the other hand, needs a *complete* picture of a given molecular spectrum in order to identify stellar features. Extensive analyses are needed, particularly for *TiO*, *ZrO*, *ScO*, *YO*, and *CrO*.

Laboratory data are often incomplete because of the mode of excitation. If laboratory sources are at low temperatures, higher molecular levels which are effective in stellar atmospheres are simply not populated and fail to give lines in the laboratory spectra. This is known to be true in the spectra of *SiH* and *AlH*, for which the writer had to resort to the risky procedure of extrapolating combination differences in order to predict the positions of lines which were observed in the spectrum of β Pegasi, although not observable in the laboratory spectrum.

NOTES ON INDIVIDUAL MOLECULAR SPECTRA

A thorough investigation of all the molecules to be expected in β Pegasi was planned but has not yet been completed, except in a qualitative manner. It is to be hoped that quantitative results will soon be forthcoming. The following paragraphs give briefly the most interesting facts about the molecules which have been investigated. The information is arranged alphabetically, according to chemical symbol.

AlH.—The (0, 0) transition, beginning at λ 4241, is one of the strongest molecular contributors to the spectrum of β Pegasi (see Fig. 3,d). The (1, 1), (1, 0), and (0, 1) transitions are far less conspicuous. Some photometric work on this spectrum was mentioned earlier.¹⁶ Lines in the R branch of the (0, 0) band are marked in Figure 3,d.

AlO.—The (0, 0) head begins at λ 4842.6, where there is some very weak absorption that cannot be attributed to any known atom or to *TiO*. The entire absorption due to lines from $K = 1$ to $K = 23$ appears as only a hazy line of intensity 0 or 1 in β Pegasi. The P branch starts around λ 4846, where some weak, unresolved structures, probably due to *TiO* or the R branch of *AlO*, appear in β Pegasi. This evidence and other convinc-

¹⁵ *Ap. J.*, **86**, 109, 1937; and **94**, 276, 1941.

¹⁶ D. N. Davis, *Pub. A.A.S.*, **10**, 48, 1940.

ing evidence of AlO absorption in the sun¹⁷ and in M-type stars make it probable that the $^2\Sigma$ state of AlO is the ground state, although B. Rosen¹⁸ thought differently.

BH.—This elusive molecule is probably not in β Pegasi. Coincidences are about as frequent as might be expected by chance. Only a few very weak stellar lines of intensities from -3 to 0 could possibly be assigned to BH .

BO.—Many BO bands are well resolved and have been measured and thoroughly analyzed,¹⁹ but BO is apparently only a weak contributor to the spectrum of β Pegasi. A few lines of the $(0, 0)$ band at λ 4227 attain, at most, an intensity of about 1 (see Fig. 3,d). A few of the other bands appear to coincide with weak, hazy features in the spectrum of β Pegasi.

BaO.—The $(0, 0)$ transition is evidently favored by the relative positions of the potential curves. Unfortunately, it falls in a region dominated by TiO . The $(1, 1)$ band has high intensity on laboratory plates and coincides with a group of very weak features near λ 6041, which cannot be attributed to TiO .

C₂.—The $(0, 0)$ head falls in the green, just 1 Å from the place where the $(0, 0)$ band of TiO begins (see Fig. 4, b). Fortunately, the two heads are degraded in opposite directions. A broad, hazy feature at λ 5164.6 is probably due entirely to a piling-up of the P branch in the C_2 spectrum. The R branch has a head near λ 5148, but no stellar absorption which cannot be otherwise accounted for is here. There is, however, a weak, general background absorption from λ 5148.2 to λ 5146.1 which may be attributed to C_2 . Other bands of the Swan system are evidently absent.

CH.—Several lines in the $(0, 0)$ band coincide with stellar lines in a manner which is convincingly correlated with quantum number. The strongest unblended lines have an intensity of 1. Certain lines are marked in Figure 3,d.

CN.—Unblended lines of the $(0, 0)$ band have intensities as high as 4 in β Pegasi. There is a small region from λ 3879.2 to λ 3880.7, where all the absorption is due to CN . The $(1, 1)$ head is visible, but weak. A few lines of the $(1, 1)$ band have been found, the maximum intensity being 0. The $(2, 2)$ head is absent. The head of the $(0, 1)$ band coincides with very strong Fe and Sr II lines (see Fig. 3,d). There are not enough convincing coincidences with lines of the $(1, 0)$ band to make identification certain.

CaO.—General background absorption, beginning at λ 4084, may be due to the $(1, 0)$ band. The $(0, 0)$ band is unfavorably situated near the CN bands and the λ 4226 line of Ca I.

CrO.—The bands are well spread out. A rotational analysis is greatly needed. Some broad features in the spectrum of β Pegasi, which cannot be explained otherwise and may be due to CrO , are at $\lambda\lambda$ 6053.74, 6055.22, and 6056.99. The $(0, 1)$ band at λ 6394 may possibly be present.

FeO.—The spectrum consists of widespread groups of lines. Neutral Fe is supposedly more abundant than Ti , especially in β Pegasi, where practically none of the Fe atoms are ionized. The apparent absence of FeO from β Pegasi may indicate that the dissociation potential of FeO is relatively low.

MgF.—The band at λ 3594, observed in the sun, does not seem to be present in β Pegasi.

MgH.—The $(0, 0)$, $(1, 0)$, and $(0, 1)$ bands are present. Many lines of the widespread $(0, 0)$ band are present, several being of intensity 8. The intensity maximum is around $Q(18)$. A few unblended MgH lines are marked in Figure 4,b.

MgO.—Several moderately strong lines from λ 5003 to λ 5006 may be partially ascribed to the $(0, 0)$ band of MgO . There is, however, no striking correspondence with a positive laboratory spectrogram.

¹⁷ H. D. Babcock, *Mt. W. Contr.*, No. 708; *Ap. J.*, **102**, 154, 1945.

¹⁸ *Phys. Rev.*, **68**, 124, 1945.

¹⁹ F. A. Jenkins, *Proc. Nat. Acad.*, **13**, 496, 1927; Jenkins and McKellar, *Phys. Rev.*, **42**, 464, 1932; W. Scheib, *Zs. f. Phys.*, **60**, 74, 1930.

MnO.—The (0, 0) band is not very concentrated, being degraded in either direction. All stellar features in the corresponding region can be accounted for, except for λ 5580.90, which is a little stronger than on a laboratory positive of the *TiO* spectrum. For a rich spectrum like that of *MnO*, one coincidence is not enough to establish identification.

NH.—The ultraviolet limit of available plates for β Pegasi is near the (0, 0) band of *NH* at λ 3360. Very definite absorption appears at the correct location; hence it seems certain that *NH* is present.

PH.—Several close coincidences were observed with lines in the complex (0, 0) band, but the number of chance coincidences which may be expected is very little less than the number observed. Hence we must regard the presence of *PH* in β Pegasi as doubtful. Further details may be found in an earlier paper by the writer.²⁰

ScO.—System V is the strongest in β Pegasi, the (0, 0) band being at λ 6017.29. The (0, 0) band of system III at λ 6036 is also strong in β Pegasi. The (0, 0) band of system II is present.

SiF.—Only a small part near the head of the (0, 0) band of this well-analyzed spectrum appears to be present in β Pegasi. Between λ 4368.24 and λ 4372.72 there are weak contributions by *SiF*, none being unblended. Still, there can be no doubt that *SiF* is responsible for some absorption. It is strongest in β Pegasi and μ Geminorum, intermediate in α Tauri, weak in α Boötis, and very weak in γ Cygni.

SiH.—Next to *TiO* and *MgH*, *SiH* produces the strongest molecular absorption in β Pegasi. Four of the twelve branches in the (0, 0) band are marked in Figure 4, *a*. The *SiH* spectrum is exceptionally well developed in β Pegasi; in fact, it was necessary to extrapolate combination differences in order to predict higher-level lines which were not observed in the laboratory but are nonetheless present in β Pegasi. Full details of the investigation are in an unpublished paper. Even the "satellite" branches have been found in β Pegasi. Doubtless there are more stellar lines due to *SiH*, which cannot be identified until higher-level lines are produced and measured in the laboratory. There are enough unblended lines to make *SiH* an excellent prospect for photometry and temperature determination.

SiN.—The closely packed lines of the (0, 0) band at λ 4117 combine to produce a few weak unblended lines and contribute to others in the spectrum of β Pegasi. Although the head of the (1, 1) band is confused with *SiH* and the strong *Fe I* line at λ 4145 (see Fig. 4, *a*), it appears to be definitely present. Other bands may be present (see Fig. 3, *d*).

SrF.—Both (0, 0) heads fall where stellar absorption is due mainly to *TiO*. The ultraviolet band at λ 3712 which is observed in the sun may possibly account for faint, hazy absorption at λ 3712.44.

SrH.—One of the infrared heads may explain wide absorption at λ 7020.24. The other infrared heads, λ 7347 and λ 7508, do not appear to be responsible for any absorption in β Pegasi. If *SrF* is present, one may expect *SrH*, provided that the dissociation potential for the latter is not too low. The writer intends to make a thorough search in the near future, using the rotational analysis of Watson and Frederickson.²¹

SrO.—Certain bands of the blue and violet systems appear to contribute some weak absorption in the spectrum of β Pegasi. The evidence is not, however, very convincing.

TiO.—The α - and γ -systems are stronger than the β -system. In the α -system, transitions have been found in β Pegasi with vibrational quantum numbers as high as (3, 6). In the α (0, 0) band (see Fig. 4, *b*), lines have been observed with rotational quantum numbers as high as 82. Higher levels are suspected. In the α (0, 1) band, quantum numbers have been observed up to 69, while still higher values are observed in the α (1, 0) band. It is unfortunate that the β -system is not yet analyzed. The γ -system may be the strongest of all, though it is risky to compare intensities in the infrared with those in the green. Like the better-known α -system, the γ -system utterly obliterates large portions of the stellar spectrum. Transitions are observed with vibrational quantum numbers as

²⁰ *Ap. J.*, **94**, 276, 1941.

²¹ *Phys. Rev.*, **39**, 765, 1932.

high as (4, 5). Unclassified bands are present, too. There is some evidence that *TiO* lines vary slightly in intensity from time to time in the spectrum of β Pegasi. The *TiO* spectrum richly deserves more attention in the laboratory as well as in the observatory. The innate tripling of the levels, unfortunately, sets a limit to the accuracy and utility of precise photometry.

VO.—Very few coincidences are observed for the (0, 0) or (0, 1) bands. *VO* must be absent from the spectrum of β Pegasi.

YO.—The system beginning at λ 6132 is the strongest. Other bands of *YO* are superposed upon stronger absorption produced by *ScO*.

ZrO.—The structure of the *ZrO* spectrum is similar to that of the *TiO* spectrum, but *ZrO* is far less abundant in β Pegasi. All three systems of *ZrO* are present, the relative intensities being about the same as for *TiO*; that is, in the order γ , α , β .

IV. DESCRIPTION OF TABLE 5, THE SPECTRUM OF β PEGASI

All lines visible in the spectrum from λ 3360 to λ 8839 were measured on at least one plate. The original ledger was reduced to about 9,900 lines, partly by eliminating lines and features which are clearly due to *TiO* alone in the region beyond λ 4954, where *TiO* begins to dominate the spectrum. The numbers of excluded lines and features are indicated by a numeral in the first column.

ATMOSPHERIC LINES

The *Revised Rowland Table of Solar Wave Lengths*²² and Mr. Babcock's forthcoming list of infrared solar wave lengths give a complete list of telluric lines. Hence it was not considered necessary to include in Table 5 the wave lengths and intensities of the many telluric lines which do not appear to interfere with the true stellar lines. Sometimes the number of telluric lines omitted is indicated by a numeral in the first column. Because they often occur singly, however, much space was saved by not indicating these omissions at all. The telluric lines which had to be included because involved with true stellar lines are identified by the following symbols:

Atm *O*, an abbreviation for lines known to belong to the *O*₂ spectrum, *not* atomic oxygen;

Atm *wv*, an abbreviation for lines known to belong to the *H*₂*O* spectrum;²³

Atm, atmospheric lines whose origin is not definitely known, some of which may be due to *O*₂ or *H*₂*O*.

The shift in wave length due to radial velocity was at times sufficient to displace the stellar lines by 0.5–0.6 Å. Hence, a few lines masked by telluric lines in the solar spectrum have been revealed in β Pegasi; for example, the *Fe* I line at λ 6885.77.

First column.—Wave lengths (in Å.), reduced to the star, are given throughout the spectrum to 0.01 Å, though the accuracy of measurement does not warrant this practice in all regions. Symbols in this column have the following meanings:

- * = Region of unresolved absorption,
- : = Wave length may be in error by more than 0.05 Å,
- N* = See note at end of table.

Second column.—Visually estimated intensities on the following scale:

- 3 = *Very* faint line, barely visible on only one plate,
- 2 = *Very* faint line, seen on one plate,
- 1 = *Very* faint line, seen on several plates,
- 0–2 = Faint lines,
- 3–9 = Lines of moderate intensity,

²² C. E. St. John and others, *Carnegie Institution of Washington Pub. No. 396; Papers of the Mount Wilson Observatory*, Vol. 3, 1928.

²³ W. Baumann and R. Mecke, *Zs. f. Phys.*, **81**, 445, 1933.

- 10–20 = Moderately strong lines,
 21–35 = Strong lines,
 > 35 = Very strong lines.

Other symbols in second column have the following meanings:

- Em* = Apparent emission line,
BdRv = Shortward edge of a band which is degraded longward,
BdVr = Longward edge of a band which is degraded shortward,
w = Line is unusually wide, but not wider than 0.5A,
W = Line is wider than 0.5A,
 ? = Reality of line is questionable, or intensity is greatly in doubt.

Third column.—The symbols in this column represent a compromise in an effort to give the fullest possible information about every identification in the least space. The principal contributor is named first, except where contributors are judged equally effective, in which case they are given in the order of wave length. Underlining indicates that a contributor is evidently responsible for more than 80 per cent of the observed absorption. A dash indicates that an unknown contributor is responsible for part of the observed absorption. The position of the dash indicates whether the unknown is acting on the shortward side (dash at left) or longward side (dash at right) of the line.

Neutral elements are indicated simply by chemical symbols, while ionized elements are indicated by the usual Roman numerals.

The Arabic numbers following the chemical symbol correspond to the serial number of a multiplet in the *RMT* to which a given line belongs. For lines not in the *RMT*, sources are indicated by letters which have the following meaning:

- M* = Line in *MIT* table or original research papers;
Ms = Line found only in original, unabridged manuscript for the *RMT*;
p = Predicted line;
Sun = Stellar line coincides with an unidentified solar line of intensity > -3 in the disk spectrum or a line which is strengthened in the sun-spot spectrum. A question mark here usually implies doubt about equivalence of solar and stellar lines, not doubt about solar identification. If “(Sun)” follows a chemical symbol, it is meant that the line in question has been observed in the sun;
u = unclassified line or molecular band.

Molecular contributors are indicated by appropriate chemical symbols followed by vibrational quantum numbers when known. (These quantum numbers have been omitted for all but the first line or head of an abundant molecule with many lines, such as *TiO*.) Following the vibrational quantum numbers, when a rotational analysis is available, is a capital letter indicating the type of branch (P, Q, or R). The symbol “S” is used for lines in satellite branches.

For the *TiO* spectrum an extra symbol had to be inserted before the vibrational quantum numbers in order to distinguish between the α -, β -, and γ -systems. For the *ScO* and *YO* systems, Roman numerals serve a similar purpose. For other molecules it was unnecessary to indicate the electronic transition because all absorption observed in β Pegasi involves only the ground state and a single well-known upper level.²⁴

The American Association of University Women provided the funds for commencing this study in 1939. Some of the work was completed while the author was on the staffs of the Princeton and Mount Wilson observatories. To Dr. Adams the writer is especially grateful for permission to use the finest available plates for β Pegasi. Finally, the writer is grateful to Dr. Merrill and Mr. Babcock for helpful discussions of both the astrophysical and the editorial aspects of this contribution.

²⁴ Further information about molecular spectra may be found in *Identification of Molecular Spectra*, by Pearse and Gaydon (London: Chapman & Hall, 1941).

TABLE 5
 SPECTRUM OF β PEGASI

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
3400			26.64	6	Fe 82	54.70	2	Ti(Sun)?	79.86	2	VII 6 FeII 4?
00.02	3	GdII 22	26.96:	6	Fe 26	54.90	5	GdII 7	80.03	3	Co 67
00.18	4	V(Sun) Ti(Sun)	27.12	12	Fe 81	55.25:	20w	Co 6	80.14	4	M1 123
00.41	6	V 46	27.48	-1	V(Ms)	55.61	4	Cr 51	80.35:	1	Cr 141?
00.66	1	Sun	28.00	4	Sun	55.77	3	Ti 46			ZrII 58?
00.93	3	-Sun	28.25	12w	Fe 81	56.00	8w	Sun	80.54	7	Ti 84
01.52	12	Fe 26	28.40	6	Sun	56.40	4	TiII 99	80.90	5	TiII 22
01.87	1	V(Sun)?	28.58	-1		56.66	5	Ti 134	81.14	5	ZrII 46 Pd 2?
02.06	1	Co 123	28.89:	6	-Ti 168	56.92	8	Co 5	81.30	4	Cr 77 GdII 22
02.22	2	Fe 614	29.19	3	Sc 21	57.28:	0	Ti 46	81.56	4	Fe 132
02.47:	6w	TiII 53 V 46	29.50	2	Sc 21	57.42:	0	Ti 46	81.84	6	GdII 22
02.90	2	NH(0,0)P	29.77	6	Fe 244p	57.55	5	ZrII 20	82.18	3	Fe (Sun)
03.09	-1		30.15	-1		57.81:	0		82.43	2	
03.32w	6	CrII 3 Ti 86	30.31	5	Zr 17	57.99:	2	Ti 46	82.63:	C	Co(Ms)-
03.67:	-1	-ZrII 59	30.55	6	ZrII 11 Ti(M)	58.22:	3	Al 2	82.85:	3	MnII 3
03.89:	-1		30.78	3	Ru 3	58.46	35w	M1 19	83.04:	10	Ti 22 Fe 24
04.25	15	-Fe 25,301,83	30.91	4	Zr(M) Ti 120	58.96	3	ZrII 58	83.44	9	Co 23 ZrII 33
04.55:	3	Pd 2	31.09	-1	Ti 135p	59.19:	1		83.79	14	M1 6 Co 5
04.78	3	Fe 300	31.31	3	Cr 53 Sc 21	59.44	4	Fe 297 Ti u	84.15	2	CrII 2
05.09	20w	Co 23 TiII 63	31.57	12	Co 6	59.74	2	Fe (Sun)	84.42	5	
		Ti 86V46Fe300	31.82	3	Fe 376,676	59.98	10w	Fe501 MnII 1-	84.67	2	VII 6-
05.63	1		32.05	4	Fe 377	60.34	6	MnII 3-	84.84	6	Fe 185
05.84	2	Fe 299	32.28	1	Cr 53	60.51	3	Re 2?	85.02	3	Fe138 CoII 44
06.14	3	NH(0,0)P	32.41	2	ZrII 58	60.75	5	Co 36 Pd 2	85.34	10	Fe 78
06.43	4	NH(0,0)P Fe676	32.56	1	Sun?	60.97	4	DyII (Sun)	85.69	6	Ti 84 Co 68
06.58	1	Ru 4	33.06	9	Co 23	61.20	1	Co 162	85.81	15	VII 6 M1 17
06.81	6	Fe 85 V 46	33.58	25w	M1 19 Cr 52	61.47:	12	TiII 6	86.30:	-1	
07.19	4	TiII 1	33.89	1	ZrII 58	61.69:	17	M1 17	86.55	7	Fe 79
07.50	8	Fe 83 Fe 81p	34.10	4	Cr 52	61.93:	3	GdII 23	86.90:	3	
07.80	3	DyII (Sun)	34.44	6	DyII (Sun)	*		Rh 3	87.04:	-1	V 81
08.09	2	ZrII 72	34.90	8	Rh 2	62.33:	4	Fe 79	87.25	4	
08.51:	1	Sun?	35.23	-1		62.55:	1		87.64	10	Ca 10 Co 65
08.74	5	CrII 3 SmII u	35.47w	7	N1 53 Sc 21	62.80	13	Co 23 Fe 373	87.80:	0	Ti 119p?
09.13	12	Co 23 V(Sun)	35.67	1	Cr 52	63.04	5	ZrII 90	88.01	2	FeII 4
09.30	5	Ru(M) GdII 21	35.82	1	Cr 53	63.23:	8	Ti 85-	88.2		BdRv?
09.56	15w	N1 5	36.03	5	Fe 614	63.36:	2	Fe 48	88.28	4	-N1 121-
09.85	5	TiII 1-	36.20	6	Cr 52	63.50:	0	Co 42	88.45	4	-Cr 109-
10.20	4	Fe 735 NH(0,0)P	36.73	7	Ru 4	63.66:	0		88.68	3	MnII 3
10.54	2	Fe 244p	37.02:	3	Fe 539 Ir 3?	63.98	5	GdII 40			*
10.88	6	Fe 25	37.23:	21w	N1 3	64.13:	2	Fe (Sun)	89.44	11	Co 36
11.08	1	Fe 299 Cr u	37.65	0	Fe 187 Co 162	64.40	4	Yb(M)	89.75	8	TiII 6 Fe 442
11.56	3	Fe 301	37.78	2	V(Sun)?	64.63	8	-Re 2?	90.06:	3	
11.67	1	Ti(M)	37.98	4	Fe 614-				90.54:	30	Fe 6
12.17:	0	Sun	38.25	6	ZrII 1	65.24	5	Cr 51	90.78:	7	Co 20
12.31	12	Co 25	38.71	2	Co 87	65.49:	3	-TiII 99	91.04	5w	TiII 6
12.63	12	Co 6	39.02	4	Fe 299 MnII 1	65.83:	35w	Fe 6	91.32	15	Co 6
13.15	6	Fe 85	39.34	7	Ti 120 Al 2	66.31	4	Fe185 MnII 12	91.79:	0	
13.45	10	N1 5	39.56	3		66.52	9	Fe 52	91.97	5	Co 159 GdII 6
13.65	0	NH(0,0)P	40.00	3	GdII 7	66.89	6	Fe (Sun)	92.23:	3	
13.92	6w	N1 17	40.62:	40	Fe 6	67.24	5	Ti 84 GdII 22	92.42:	3	
14.63:	10	Zr 17 ZrII 73	41.05:	30	Fe 6	67.51	15	N1 3	92.98:	35w	N1 18 VII 6
14.80:	10	N1 19	41.48:	3	Cr 52 TmII u	67.86	-1	VII 17 SmII 54	93.30	7	Ti 22 Fe 48
15.28	1		42.03	6w	N1 104 MnII 3	68.13:	6		93.60:	1	Fe 297
15.53	8	Fe 83 Co 5	42.37	4	Fe 134	68.45	5	Ca 10	93.77	5w	
15.97	1	1(Ms)	42.68	6	Fe 26	68.84	6	Fe 242	93.98	0	
16.46:	6	-V(Sun)	42.96	15	Co 6 Fe 499, 776	69.03	5	Fe614 GdII 40	94.15	4	Fe 137
16.95	2	TiII 53	43.42:	0	TiII 99	69.37	9	Fe 375	94.47	7	DyII (Sun) GdII 7
		GdII 22	43.66	9	Co 22 Al 2	69.54:	11	N1 83	94.73	1	N1 154?
17.22	15w	Co 23 Fe 26	43.9	20w	Fe 6	69.84	5	Fe 242	95.01	0	Cr 109
17.61	12	Co 19 Fe 81	44.37	9w	Ti 120 TiII 6	70.16	8	-YII 40?	95.30	8	Fe 238
18.16	4	Fe 577	44.73	-1	Sun?	70.39:	1w	Cr 77 Cr 77	95.68	10	Co 22 Ti 84
18.50	6	Fe 81	44.87	3	Al 2 Ti 46	70.68:	4w	Rh3 Cr 77	95.92	8	Ti 22
18.71:	3	GdII 7	45.15	8	Fe 81	70.89:	1		96.2	9	ZrII 1 Fe186?
18.86	1	Sun	45.60	4	Cr 51 DyII (Sun)	71.28	15w	Zr15 Fe82,130	96.4	3	N1 118
19.10	4	ZrII 2 Fe 576	46.3	30w	N1 20	71.44	5	-Cr 77	96.69	20	Co 19
19.29	2	Sun?	46.70:	0		72.12	6		97.11	15w	Fe 78 ZrII 10
19.57:	0		46.96	6	Fe 26	72.30	4		97.56	5w	MnII 3-
19.70	10	Fe 377	47.34	15w	Zr16Fe82Cr52	72.56	25w	N1 20	97.91	30	Fe 6 ZrII 58
20.11	1	NH(0,0)P	47.76	3	Cr 52	72.68:	5	-Co 160	98.20	8	N1 2p Fe 326p
20.30	2	NH(0,0)P	47.99	4	Fe(Sun)?	73.26	5	GdII 7-	98.7	4w	Rh (M)? DyII?
20.43	2	NH(0,0)P	48.21	5	Ti 46	73.50	6	Fe 26	98.93	9	Ru 4
20.74	12	N1 9 Co 102	48.40	5	-Fe 444	73.66	1	Cr 77	99.09	8	Ti 84
21.01	-1	Co 119	48.83	8	Fe 242 Fe 372	74.00	13w	Co 4, 23	99.56	4	ZrII 9
21.25	6	Pd 3 CrII 3	49.15	14	Co 22	74.08:	4	MnII 3	99.83	3	VII 5
21.68	4	Sun?	49.48	15w	Co 22	74.46:	6	-Co 24	99.95:	0	
21.92	1		49.88	4	Ti 46	74.73:	5	Ca 10 Rh 2			3500
22.12	1	Sun	50.08:	0		75.43	35w	Fe 6 Fe 78	00.14	4	ZrII 123
22.44	8	GdII 2 Fe 444	50.33:	8	Fe 82	75.87	5	Fe 186, 373	00.34	11	TiII 6
		N1 105	50.45:	5	GdII 22	76.42:	2	Ti 85	00.55	3	Fe 238
22.68	10w	TiII 63 CrII 3	50.76	4	Ti 46	76.69:	30w	Fe 6	00.88	25	N1 6
		Fe 85	51.09	0	Sun?	77.0	4	Fe 139	01.1	-1	
22.87	3	Co 42 N1 122	51.29:	3	GdII 22	77.20	12w	TiII 6-	01.34	6	Zr 14
23.20	2	Ti 120	51.45:	0	Mn(Sun)?	77.58	4	VII 58	01.56	5	
23.40	1		51.63	4	Fe 139	77.88	6	Fe 82-	01.84	0	CoII 2-
23.68	30	N1 20	51.91	8	Fe 81	78.08:	0		02.26	25w	Co 81
23.94	2	GdII 7	52.27	15	Fe 25	78.31	5	ZrII 2	02.61	30w	N1 3 Co 6
24.27	8	Fe 81	52.67	3	Al 2	78.55:	5	Co 120	02.90	6	
24.55	1	Co103 GdII 22	52.92	25w	N1 17	78.7	3	Co 67	03.28	0	
24.83	3	ZrII 2	53.54	30w	Co 22	78.8	4	Fe 137	03.46	0	FeII 4
25.04	4	Fe 541 V u	53.88:	0		78.92:	5	Ti 84	03.75	13	Ti 22
25.34	0		54.15	4	GdII 7 N1II 1	79.0	1	ZrII 20	04.03	-2	
25.60	2	TmII 3	54.35	6	DyII (Sun)-	79.23	2	N1 105	04.44	9	VII 6 Fe 371
25.93	6	GdII 23 Cr168	54.57	3	ZrII 59	79.55	5	CrII 6?	04.69:	5	-Co 135
26.34	12	Fe 25,82,135				79.68	1	Fe 443			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
3500			30.20	4	Zr 52	49.53	5	ZrII 84Rh(M)?	76.34	29w	ScII 5
04.87	12	TIII 88 Fe131	30.38	7	Fe 326	49.88	9	Fe 48	76.74	8	Fe 613a
05.06	7	Fe 498	30.56	11	Ti 22 Ni 121	50.06	1		76.87	8	ZrII 9
05.23	0		30.78	8	VII 5	50.23	6	DyII(Sun)	77.23	18	Ni 3 Co 41
05.46	6	ZrII 90	30.96	2		50.47	8	ZrII 12	77.45	0	CeII 51
05.67	8	ZrII 1 V 81	31.20	1		50.59	23	Co 4	77.86	14w	Mn 8
05.93	2	TIII 88	31.44	9	Fe 182 VII 4	50.82	-2		*		Co 117 Ti
06.20	-1	Fe 327	31.70	11	DyII (Sun)	51.09	7	Fe 321p	78.75	60w	Cr 4
06.32	20w	Co 21	31.85	10	Mn 18	51.48	14	Ni 5	*		Co 41
06.64	14	Ti 22	32.07	24w	Mn 18 Mn 18	51.60	14	Co 87	80.94	25w	ScII 3
06.83	4	DyII V 81	32.36	4		51.94	8	ZrII 1	81.24	25w	Fe 23
07.16	6	Sun	32.59	8	Fe (Sun)	52.10	6	Fe 499	81.88	7	-Fe 497
07.30	3	Rh 2	32.76	-1		52.42	4	Fe 182	82.26	10w	Fe 612-
07.48	8	Ti u-	32.87	-2	Cr u?	52.64		BdRv	82.66	14	Fe 328 Fe 181
07.72	23	Ni 3	33.01	9	Fe 326	52.72	16	Y 8 Co 6	82.91	0	
07.95	4	-CeII 51-	33.21	11	Zr 14 Fe 326	52.85	5	Fe321 TIII 15p?	83.34	7	Fe 574
08.09	1	Cr u	33.39	20	Co 5	53.02	1	Co 67	83.70	4	V 45
08.32	6		33.69	7	V 53 V 53	53.26	3w	V 53 Co 137	83.93	3	
08.51	11	Fe239 Fe 442	33.85	3	TIII 98	53.49	23	Ni 16	84.18	1	-SmII 12
08.89	2	Cr u?	34.06	2	CeII 44	53.78	8w	Fe 810 Ti(M)	84.36	-1	Cr u?
09.13	10	Fe 326	34.28	-2		54.08	2	ZrII 83	84.52	8	YII 9
09.34	11	Zr 15	34.48	6	-Fe 811	54.12	14	Fe 23	84.67	8	Fe 294
09.83	23w	Co 22 TIII 88	34.71	3	-Co 118	54.30	0		84.79	17	Co 6 Fe 322
			34.94	12	Fe 48 DyII(Sun)	54.48	10	Fe 325 LuII 7?	84.95	4	Fe 611, 395
			35.19	-2	Zr 59	54.66	8	Fe 154p?			GdII 7 DyII(Sun)
10.34	40w	Ni 18	35.35	12	TIII 98 Cb 4	54.92	17	Fe 326	85.16	19	Co 21 Fe 438
10.63	0		35.56	3	TmII?	55.15	1	V 53	85.32	20	Fe 23
10.82	8	TIII 88	35.74	13	ScII 11	55.41	5	Fe (Sun)p?	85.74	30	Fe 23 Co 100
11.23	1	SmII 12?	36.03	7	DyII (Sun)	55.76	8	-Cr u?	86.11	8	Fe 611 Co 87
11.62	11	Ti 22	36.22	-2	Fe (M)?	56.16	7	Ti u	86.29	8	Zr 12
11.70	1	Fe 258	36.58	13	Fe 326	56.56	6	ZrII 9	86.55	11	Mn 8
11.84	3	CrII 2	36.78	3		56.68	6	Fe 325	86.73	2	Fe 325
12.09	4	Fe 327	36.95	6	ZrII 10	56.86	15	Fe 327 VII 5	87.00	9	Fe 25
12.23	4	Fe 326 GdII 38	37.24	8	Ni 153 Cr 50	57.44	0		87.21	14	Co 35 Fe 325
12.40	0	Co?	37.48	8	Fe 239	58.05	0	-Fe 572p?	87.45	10	Fe 134 NdII u
12.66	24w	ZrII 57 Co 21	37.70	8	Fe 239 Co 68	58.3	8?		87.76	12	Fe u Y u
13.04	15w	-Fe 48	37.91	9	Fe 327	58.55	30w	ScII 3 Fe 24	87.96	28	Ni 16 ZrII 10
13.48	25	Co 5	38.25	3	VII 4	58.79	17	Co 20	88.32	6	ZrII 10
13.91	35w	Ni 17	38.32	2	Fe 775	59.13	6	SmII u?	88.60	14	Fe 325
			38.53	11	Fe 137			Fe (Sun)	88.92	8	Fe 322
14.64	5	Fe 183	38.76	2	Fe 811	59.49	12	-Fe 498	89.14	30	Fe 23
15.03	30w	Ni 19	38.94	3	-ZrII?	59.75	4	-Cr 89	89.46	7	Fe 295
15.6	5		39.09	3	CeII 118-	59.90	8		89.69	27	ScII 3 VII 4
15.9	2		39.34	5	Ru 4	60.10	4	Fe 321p	90.10	3	Fe 440
16.21	3	Ni 123	39.57	1		60.28	5	Co 64	90.47	30	ScII 3
16.40	6	Fe 442	39.75	-2		60.58	7	VII 4	91.01	3	Fe 573
16.56	5	Fe 326	39.88	2		60.68	7	Co 64	91.20	1	
16.86	4	Ti 167	40.12	9	Fe 329	60.88	21	Co 21	91.34	4	Fe 321
16.94	6	Pd 1	40.32	0		61.19	-1	ZrII 82?	91.40	4	-Fe 568
17.30	11	VII 6	40.58	2	V 45-	61.37	0	Co?	91.72	0	Co 134
17.66	4		40.72	25	Fe 23	61.57	16	TIII 15	92.03	10	VII 4
17.89	-1	GdII 88?	40.97	1	CbII 4?	61.76	25	Ni 2	92.45	4	Fe 237
18.05	2	Cr?	41.09	12	Fe 326	62.11	4	Co 115	92.58	4	SmII 39 NdII u
18.34	21	Co 36	41.22	4	Fe 47p	62.25	2		92.76	0	GdII 69
18.66	6	Fe 327	41.45	2		62.50	2		92.92	10	V 8
18.86	8	Fe 78	41.62	0		62.65	2	Fe 237?	*		VII 4
19.22	1	Ti 2?	41.78	0		62.91	11	Co 64	93.48	60w	Cr 4
19.42	-1		42.05	13	Fe 326 Ni 119	63.15	10	DyII	*		Fe 182p?
19.61	20	Zr 13	42.25	11	Fe 128	63.50	9	-V u	94.05	7	-Fe 154p
19.85	30w	Ni 5 Ti 22	42.55	7w	Fe 321p	63.58	3	Fe 325p	94.63	10	Fe 322
20.06	25	Co 4 VII 5	42.7	-2	V 45	63.77	1	VII 4?	94.88	25	Co 4
20.24	4	TIII 98	42.96	7	Co 19	64.10	16	Fe 48 Co 159	95.09	8	Mn 8 DyII (Sun)
20.50	2	CeII 55	43.13	0		64.52	10	Fe 183p Ti(M)	95.28	5	Fe 322
20.82	5	Fe 238 ZrII 19	43.26	7	Co 64	64.96	30	Co 19	95.43	2	
21.01	2		43.36	6	Fe 183	65.57	25w	Fe 321, 328	95.70	4	Fe 322
21.20	15w	Fe 24	43.47	6	V 53	66.10	8?	Zr 15	95.88	5	Fe 181
21.58	18	Co 20	43.68	6	Fe 734	66.39	40w	Ni 36	96.05	13	TIII 15
21.82	9	Fe 78	43.98	10	Rh 6-	66.80	0	SmII u	96.19	15	Fe181 Rh1 Ru4
22.08	-1		44.27	2		67.04	10	Fe 325	96.50	2w	Co 118 TIII 76
22.27	7	Fe 326	44.62	9	Fe 259	67.37	9	Fe 183	97.03	5	Fe 569
22.86	4	Fe 330 Co 159	44.97	1	CoII 1?	67.70	25	ScII 3	97.12	6	Rh 5
23.07	10	Ni 34	45.17	9	VII 5 Ni 76	68.24	1	SmII 47	97.4	-2	
23.17	3	Fe 673p	45.32	6	V 53	68.46	11w	Fe 321 Co 61	97.70	37	Ni 18
23.45	28w	Ni 16 Co 21	45.51	1		68.90	12w	Fe 673 Fe 294	98.00	3	
23.79	4?	Co 66	45.63	10	Fe 321	69.40	28	Co 55 Mn 18	98.14	1	-CeII 116?
24.01	3?	Fe 239	45.80	10	Fe 536 GdII 2	69.76	4	Mn 18	98.23	1	
24.54	40w	Ni 18	46.00	2		70.04	30w	Fe 24 Mn 18	98.71	16	Ti 59 Fe 674
		ZrII 90 Co 63 Fe 329	46.38	1	Fe 183	70.24	20w	Fe 326	98.95	6	Fe 568 Fe 322
25.98	19	Fe 6 Fe 240	46.69	4	Co 41	70.66	3	Fe 154p-	99.15	4	Fe (Sun)
26.20	19	Fe 24 Fe 327	46.82	5	-DyII(Sun)	70.99	2		99.39	3	Cr 89
26.48	8	Fe 131	47.02	7	Ti 133	71.21	13	Fe 46	99.62	6	Fe 809
26.7	5	Fe 326	47.19	7	Fe 321	71.83	40w	Ni 5 Fe 321	99.77	4	
26.85	30w	Co 4	47.35	2		72.55	37w	ScII 3 ZrII 1	99.99	5	Sun
27.0	3	Cr 274?	47.67	11	Zr 13	73.06	9	ZrII 9			
27.29	0		47.77	11	Mn 18	73.40	9	Fe 673	3600		
27.44	0	ZrII 103-	48.03	13	Mn 18 Fe 496	73.72	12	TIII 15 Cr 75	00.41	2	
27.63	3	Sun	48.22	18	Mn 18 Ni 3, 20	73.84	12	Fe 181 Fe 611	00.75	18	YII 9
27.80	6	Fe 326	48.44	2	Co 41	74.00	8	Cr 74	00.94	-2	GdII 69
28.00	25	Ni 6	48.56	-1		74.21	9	Ti 247	01.19	14	Zr 13
28.30	7		48.75	-1	Cr 76-	74.39	9	Cr u Fe 181p?	01.40	2	Fe 127p
28.59	5	Co 1	48.99	11	YII 9	74.77	7	Cr 75	01.67	8	Cr 74
29.01	30w	Co 5	49.21	3	Mn (M)?	74.96	19	Co 21 Cr 74	01.92	13	YII 9
29.50	6	Fe 537	49.36	6	GdII 7	75.30	30	Co 4	02.10	17	Co 4 Fe 322
29.62	7	Ni 76				75.80	14	Zr 12 Cb 4	02.31	19	Ni 3
29.80	25w	Co 22				76.98	10	Ni 120			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
3600			29.26: 2			52.11	-1		74.41	2	Fe (Sun)
02.52	11	Cr 74 Fe 322	29.45	1	DyII (M)	52.27: 0		Fe 494p	74.75	11	ZrII 9 Fe 369
		Fe 391	29.73	9	Mn 8	52.55	22	Co 4	74.97: 0		
02.74	5	Fe 390p Fe370p?	30.02	5	ZrII 10	52.88	0		75.27	7	Ca 28 ScII 10
02.97: -1			30.32	11	Fe 323 DyII(Sun)	53.06	2	-CeII 38	75.46	2	Fe229p VII4
03.17	17	Fe 295	30.50	-2		53.46	30w	Ti 19	75.71	14	V 29
03.57	10w	Fe 181 Cr 74	30.72	30	ScII 2 Ca 9	53.75	8	Fe 180	75.97	-2	
03.96	17	Ti 20 Fe 496	31.03: 25w		Ca 9 Fe 322	53.93	9	Cr 47	76.27		BdRv?
04.28	18	Ti 21 SmII 47	31.33: 12		Co 4	54.22	-1		76.34	13	Fe 228 Cr 89
		Co 97 Cr 49, 89	31.54: 25		Fe 23 CrII 12	54.46	5	Co 63	76.58	6	Co 145 DyII(M)
05.33	60w	Cr 4 Co 20	31.75	8	CrII 12-	54.62	21	Ti 18 Fe 77	76.72	4	V 115
		Fe 294	32.04	13	Fe 496	54.84	0		76.87	4	-Fe 589-
06.37: 3		Fe 233p	32.34	-2		54.99	5	Fe (M)?	77.08	8	V (M)
06.74	24w	Fe 294 Ti 20	32.56	7	Fe 437	55.20	-1		77.30	5	Fe 773
07.11	2	Ti (M)?	32.84	5	Cr 49 Co 147	55.37	3	Fe 131p	77.46	6	Fe 125
07.36: 0		ZrII 85	32.96: 5		Fe 135	55.46	8	Fe 369	77.64	8	Fe 291
07.54	14	Mn 8	33.10	20	YII 2	55.64	4	Fe (M)	77.84	3	Co 116 CrII 12?
07.81: 0			33.28: 1		Co 116	55.85	2	CeII 51	78.00	3	Co 20
08.14	10	Fe 325, 438	33.49	8w	ZrII 102 Ti u	56.15: 0		GdII 1	78.23	6	Ca 28
08.47	11	Mn 8 PrII (M)?	33.87	9	Fe 440 V (M)	56.24	16	Cr 46 Fe u	78.31: 1		
08.83: 10		Fe 23 TmII 3	34.16	9	Zr (M)	56.32	1	Fe 323	78.60: -1		
		Cr 49	34.31	11	Fe 389SmII 19	56.71	2	V 115	78.84	10	Fe 131 ZrII 101
09.32	25	Ni 16	34.43: 6		-Fe 323p	56.95	11	Co 21	79.05: 9		Fe 124 Cr 45
09.55: 3		Pd 2 SmII 30	34.69	10	Fe u Pd 1	57.14	8	Fe 130	79.33	5	Fe 228
09.78	0	NdII u Co 147	34.94	19	Ni 33	57.47	8	V(M)? Fe(M)?	79.56: 3		Fe 393, 400
10.12	18w	Fe 321 Ti 58	35.23	19	Ti 20 Cr 3	57.74: 0		V(M)? Fe(Sun)?	79.95	45w	Fe 5 Cr 48
10.48	30	Ni 18	35.47: 30w		Ti 19	57.96	12	Rh 1	80.41	2	Sun
10.71	6	Fe 323	35.84	2	Fe 321p	58.11	22	Ti 19	80.65	5	Fe 568
11.04	12	YII 9	36.19	17	Fe 77	58.37	-1		80.79	5	Fe u
11.54	12	Ni 2p	36.47: 9		ZrII 9	58.56	6	Fe 231	80.95	0	Sun
11.71	4	Co 115	36.56: 10		Cr 47	58.89	5		81.24	5	Fe (Sun) Ti 177
11.88	1	ZrII 113	36.88: 9		Fe 493 Co 64	59.22	2	CeII 54	81.43	-2	
12.05	10	Fe 325	37.00	10	Fe 233	59.52	12	Fe 180	81.65	7	Fe 390 Cr 89
12.24	2	Ti u	37.27	10	Fe 180 Co 117	59.75	9	TiII 75	81.87	-1	Fe 951
12.45	2	Rh 1	37.54	0		59.97	2	CeII? Cr u?	82.10: -1		
12.74	23	Ni 6	37.74	3	Fe 229 V u	60.20: 0			82.18	16	Fe 772
12.93	12	Fe 46, 77	37.96	22	Ti 18 Fe 385	60.35	6	Fe 323			Fe 386, 385p
13.12	11	ZrII 1 Fe 324	38.31	13	Fe 294	60.64	20	Ti 18			
13.43	5	ZrII 8, 45	38.59: -2			61.20	7	Zr 12	83.08	40	Fe 5 V 29
13.58	10	Ti p	38.75	2	SmII u	61.36	11	SmII Fe 179	83.47	6	Pb 1
13.7	6	Zr 33 Cr 89	39.02	3	V 83			Ru 3	83.62	4	Fe 130, 671
13.86	30w	ScII 2	39.29	1	Y (M)?	61.66	-1		84.11	11	Fe 292
14.11: 6		Co 64	39.44	8	Co 64	61.96	18	Ni 16	84.30	0	V 114
14.31: 1			39.58	9	Pb 1 Rh (M)?	62.23	10	TiII 75	84.48	5	Co 99
14.55	5	Fe u	39.80	11	Cr 47	62.70	2	SmII u?	84.89: -2		
14.75	9	ZrII 9	40.10	0	-GdII 23?	62.84	5	Cr 46	85.18: 18		TiII 14
15.02	0	Fe 154p	40.25: 3		DyII(Sun)	63.01	0		85.26	13	PrII (M)?
15.19	2	Fe 589	40.39	15	Fe 295 Cr 47	63.21	10	Cr 46 Fe 459	85.57	10	-Cr 44-
15.41	5	Co 66	40.63	2		63.42	8	Fe 229, 231	85.77	-2	NdII u?
15.65	21	Cr 3 Fe 46	41.04	4	Cr 47 V 115	63.64	10	Zr 12 V 114	85.99	14	Fe 385 Ti 117
15.94	2	Fe (Sun)?	41.33	10	TiII 52	64.10	35	Ni 4	86.28	6	Fe 131 V 70
16.15	5	Fe 589p	41.46	9	Cr 47	64.52: 5		Fe 391	86.80	10	Cr 44
16.33	6	Fe 132	41.64	13	Ni 6	64.61: 13w		YII 9 GdII u	87.07	10	-Fe 75
16.58	9	Fe u	41.84	11	Cr 47 Co 99			Fe 390	87.48	40w	Fe21Ti19Cr45
17.09	5w	Fe 535 Ti	42.19	-1		65.16	5	V 115 NdII u	87.75: 6		-GdII 20-
17.35: 5		Fe u	42.42: 0		Ni 75	65.43	1	Cr 48	88.07	16	V 29 Cr 45
17.53: 4		Fe 325p	42.65: 30		Ti 19	65.70	0		88.42	25	Ni 5 EuII 1
17.79	10	Fe 496	42.8	20	ScII 2	65.99	6	Cr 48	88.83	4	Fe 179 Sun
18.00	6	Co 36 Fe 181p	43.18	8	Co 99 CrII 1	66.24	9	Fe 179, 389 Rh3	89.04	5	Fe (Sun)? Fe178p?
18.34	9	Fe 295 Fe 324p	43.65: 8		Fe 385	66.55	14	ScII 2	89.36: 7		Fe 391p
18.78	35w	Fe 23	43.74	9	Fe 233	66.80	2	-Fe 393p	89.47	10	Fe 369, 386
18.96	7	Fe 77p	43.83: 8		Fe 46p	66.97	6	Fe 46 ZrII 8	89.68	1	Ti 222
19.35: 30w		Ni 35 Mn 8	44.15	-2		67.26	6	Fe 570	89.93	21	Ti 18
19.77	4	Fe 180	44.40	17	Ca 9				90.28	16	V 29
20.04	10	Ti(Sun)? Co(Mn)?	44.58	1	Fe 235p	67.73	5	V 114	90.46	4	Fe 497, 570
20.24	4	Fe 324	44.74	11	Ca 9 Ti u	67.99	7	Fe 438, 569	90.73	8	Fe 807 Co 86
20.46	6	Co 116			CrII 1			CeII 40	91.07: 1		
20.93	13	Y 8	45.03	9	Fe323, 495 Ca9	68.20	3	Fe 568	91.30: 2w		Fe (Sun)
21.20	6	SmII 12Fe574p	45.29: 13		ScII 2	68.46	5	ZrII 9	91.81: 0		
21.45	13	Fe 294	45.40: 12		DyII(Sun) LaII 14	68.65	1	-Co?-	92.20: 18		V 29
21.72	5	Fe 808	45.60	2	Cr 48 V 137	68.95	20	Ti 18	92.33	17	Rh 1
22.00	12	Fe295 Fe233p	45.83	9	Fe 496	69.23	21	Ni 2 Fe 437	92.65	11	Fe (Sun)
22.52	4	SmII 6	45.99	0		69.52	6	Fe 291	92.80	2	Mn 7
23.16	14	Fe 180	46.21	22	Ti 18 GdII 2	69.69	1	CrII 1p?	93.07	11w	Fe 439 Co 97
23.46	8	Fe 233, 438	46.61	2	Sun	69.86	0	Mn 7	93.48	7	Co 95
23.79: 10		Mn 8 Fe 323	46.8	0		70.06	11	Fe 369 Co 64	93.76	7	Fe 46p
23.85: 8		Zr 12	47.05	3	Co 118			Fe 435	93.96	20	Ni15Fe394SmII 2
24.11	12	Ca 9	47.64	20	Co 4	70.45	30w	Ni 4 Mn 7	94.20	14	YbII 1
24.33	8	Fe 133 Co 41	47.90: 30w		Fe 23	70.82	7	Fe 133SmII 11	94.43	8	Ti 117
24.78	25	Ni 2				70.91: 0			94.62	-2	V 114
24.95	10	Co 21	48.55	4	Cr 47	71.23	8	GdII 2 ZrII 45	94.82	9	DyII(Sun)
25.14	10	Fe 323	48.75: 0					V 70	95.06	10	Fe 229, 534a
25.21	-1	Ti u	48.98	7	Cr 47 V 115	71.54	4	Fe 570	95.33	6	V 114
25.51	6	Sun	49.31	30	Fe 5	71.67	24	Ti 19	95.52	4	Fe 225, 707
25.74	6		49.50	12	Fe291 SmII 47	71.98	-2		95.62N	0	Sun
26.09	21	Ti 20	49.70: -2		Fe 391p?	72.16	1		95.86	20	V 29
26.55	0	-Rh (M)	49.84: 0		Sun	72.31	2	DyII(Sun)	96.02: 1		Fe 128
26.73	2	Fe (Sun)p?	50.03	7	Fe 394	72.42	2	V 115	96.29	3	Ni 74p?
27.02	6	SmII 30 Fe308	50.29	10	Fe 180	72.71	9	Fe 180	96.58	6	Mn 24
27.37	2	Fe 395	50.54	5	Sun	73.12	6	Fe (Sun)-	96.88	6	Ti 177 Ni 172
27.81	22	Co 19	50.99: 5		-Fe 571p?	73.42	7	V 114 Ca 28	97.46	17w	ZrII 7 Fe 389
28.10: 8		Fe 77	51.08	6	Fe 322	73.67	0		97.72	2	GdII 4
28.67	11w	YII 9	51.25	2	Co 85	73.89: 1			97.82	8	Cb 3
28.90: 5		Fe 458p?	51.46	11	Fe 295	74.05	17	Ni 15p Fe(M)	98.02	3	Ti? V (M)?
			51.81	21	ScII 2	74.20	16	Ni 32p	98.18	8	ZrII 71 Ti 222

50

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
3800			25.89: 40w		Fe 20 NdII 33	53.14: 2		CeII 39 Cr69?	77.78: 3		CN(0,0)P
01.30	0	GdII u?	26.39	1w	Cr 70	53.46	4	Fe 429	78.00: 33w		Fe 20
01.51: 1		CeII 172	26.81	4	Fe 283 V 44	53.71	9	Ti 176	78.60: 45w		Fe 4 Fe 175
01.66	5	Fe 367	27.84: 35		Fe 45	54.00	0	CN(1,1)? V u?	78.86: 2		—CN(0,0)P—
01.81	5	Fe 367	28.19	6	Ti 189	54.21	7	Cr 69 SmII u?	79.20	4w	CN(0,0)P Cr138
01.97: 2		Fe 704	28.56	20	V 9 Fe 287	54.34	4	Fe 567 CeII 61	79.54	4	CN(0,0)P
02.14: 0			28.85	2	V 67	54.54: 0			79.64	1	CN(0,0)P
02.27	6	Fe 666	29.38	33	Mg 3	54.86: -1		CN(1,1)?	80.03	2w	CN(0,0)P
02.50: 0			29.72	10	Mn 6 Fe 221	55.35	19	V 7 Fe 283	80.35: 2		CN(0,0)P
02.91	10	Cb 3 V 67	30.08: 0		—Cr u	55.57	1	Cr 69	80.66: 2		CN(0,0)P
03.23	3	Fe 122p	30.29	3	SmII 10	55.85	18	V 9 Fe 567	80.76	8	SmII 10
03.47	17	V 28	30.47	9	Sun	56.37	50w	Fe 4			CN(0,0)P
03.89: 12		V 10	30.74: 6		Fe 224?			CN(1,1)?	81.05	-2	CN(0,0)P
04.12	4	—DyII (M)?	30.83: 6		Fe 284	57.64	6	Cr 69	81.22	0	Cr 138
04.58	3	—V 97—	31.02	11	Cr 24	57.89	3	Ti (Sun)	81.40	18	Ti 15
04.80	8	Cr 139	31.70	27	Ni 31	58.15: 7		Ti 176	81.63: -2		CN(0,0)P
05.08	4	—Ti (M)?	32.30: 55w		Mg 3	58.32	30	Ni 32	81.86	21	Co 18
05.34	12	Fe 608 NdII 19	32.57	2		58.66	0	V (M)	82.14: 8		Ti 175
05.46	13	—Ti (M)—	32.86	16	Ni 1 YII 7	58.90	4	Cr 138	82.33	8	Ti 176
05.77	2	Co (M)	33.09	4	SeII 1	59.21	8	Fe 175	82.47	2	Fe (M)?
06.01	0	Ti (M)?	33.31	5	Fe 221	59.91	60w	Fe 4			CN(0,0)P?
06.21	5	Fe 751	33.50	5	Cr 11	60.34: 1			82.71	0	CN(0,0)P
06.52	8	Cr 24 Ti (M)	33.68: 1		Ti u	60.51: 0			82.92	12	Ti 176
06.72: 14		Mn 6 Fe 607	33.86	5	Mn 6			CN(1,1)	83.17: 0		CN(0,0)P
07.15	30	Ni 33	34.25	40w	Fe 20 Mn 6			CN(0,0)	83.30	18	Cr 23 Fe 663
07.52	15	Fe 73 V 28	35.54	5	V 44	61.15	11	Co 33	83.38: 1		CN(0,0)head
07.74: 1		Ti (M)	35.97: 10		Zr 8	61.34	3	Fe 283, 663	83.63	3	Cr 138
07.90	2	Cr 139	36.06: 9		V 44 TiII 12	61.59	1	Fe 663	83.85: -1		
08.09	14	Co 17	36.32	5	Fe 664	62.03	0	SmII 10	84.08	3	Ti 175
08.27	2	Fe 489	36.51	5	DyII(Sun)NdII u	62.22	13	V 8	84.36	10	Fe 282
08.51	14	V 9	36.74	6	ZrII 16 Ti u	62.54	4	Cr (M)?	84.61	11	Co 32 Fe 565
08.73	9	Fe 222	36.93	5	GdII 20—	62.83	11	Ti 175	85.20: 21		Cr 23 Co 31
09.04	6	Fe 367	37.14	7	Fe 222	63.08: 0		Ni 181	85.49	12	Fe 124 Zr 7
09.20	1		37.90	-1		63.37	12w	—V (M)?	85.84: 15		Ni 1p
09.61	20	Mn 6 V 28	38.29	38	Mg 3	63.72	6	Fe 280	86.33: 60w		Fe 4
09.88: -1		SmII (M)?	38.68: -2			63.88	7	Zr 8 V 66	86.76: 10		Cr 23
10.02: -2			38.98	10	V 44 NdII 28	64.10	12	Mo 1 V u	87.07: 28w		Fe 20
10.15: -1			39.24	8	Fe 529	64.32	3	Zr 10 V 64	87.85: 0		NdII 31?
10.29	2	Sun	39.34: 1		V (M)	64.49	0		88.02	4	Ti 175
10.45	3		39.77	10	Mn 6	64.67	0	CN(1,1)P	88.39	0	Fe 565
10.64	5	—Mn (M)?	40.03	0		64.87	19	V 7	88.51	30	Fe 45
10.73	10	Fe 665	40.45	30	Fe 20			CN(0,0)R,	88.81	4	Fe 488
11.04	8	Fe 223, 287	40.76	16	V 9			(1,1)P	89.06	5	H 2
			41.06	28w	Fe 45 Mn 6	65.55: 30		Fe 20	89.25	4	V (M)
11.28	13	Ni 15	41.32	3	DyII(Sun) Cr69	66.01	6	Ti (M)	89.66	12	Ni 15
11.40: 3		Ti 185	41.44	4	Co 32	66.46	10	Ti 176	89.94	17	Ti 15 CeII 50
11.61: -1			41.82	2	Ti (M)?	66.78: 2		VII 11	90.16: 17		V 8
11.80: 3		Fe 701p?	41.89	16	V 8			CN(0,0)R	90.29: 8w		Zr 8 Fe 567
11.88	6	Fe 287	42.05	17	Co 33 Cr 70	66.96	-1	CN(1,1)P	90.55	2	NdII u TiII 1
12.06	0	SmII 10	42.35	1	FriI (M)?	67.22	8	Fe 488	90.89: 14		Fe 280 NdII u
12.20	1	YII 61?	42.59	1	Ti (M)?	67.61	12	V 7	91.07: 12		V u HoII u?
12.44	2	Co 40	42.73	1		67.75: 1		Ti 176	91.24: -1		V 43
12.80	-1		42.91	4	Fe 222p Fe 221	67.95	7	Fe 221	91.38	12	Zr 11
12.98	32	Fe 22	43.04	10	ZrII 7	68.22	1	Fe 430	91.76	0	BaII 4
13.26	0	Ti 189	43.26	8	Fe 528	68.40	10	Ti 1,5	91.93	6	Fe 733
13.50	25w	V 9 TiII 12	43.50	4	V (M)	68.58: -2		CN(1,1)P	92.30: -1		
13.93	11	Fe 854 GdII 2	43.72	4	Co 84	68.77	-1	CN(1,1)P	92.45	0	V (M)?
14.25	-2		43.87: -1			69.06	4	NdII 34	92.70: 0		
14.52	21	Fe 22	43.98	8	Mn 6	69.28	6	Ti 175	92.87	16	Fe 283
14.78	3	Fe(Sun)Ti 189	44.14: -1			69.59	13	Fe 284 Ti(M)	93.09: 0		Co 114
15.49	15	V 28 Cr 71	44.45	20	V 7	69.86	3	DyII (Sun)	93.37	14	Fe 430 Fe 364
15.86	35w	Fe 45	44.57	2	GdII 2	70.13	3	Ti (M)	93.72: 0		
16.35	10	Fe 73 Co 62	44.92	5	V 44	70.26	4	Cr 11	93.91: 7		Fe 175
16.40	2	Co 62	45.17	9	Fe 124	70.51	6	Ca 26 Co 129	94.03	25	Cr 23 Fe 663
16.72	6	Mn 6 GdII 1	45.47	19w	Co 34	70.67: -2		CN(0,0)R	94.49: -2		Fe 566
16.90	4	Co 86	45.70	0	Fe 771	70.85	0	CN(1,1)P	94.64	8	GdII 1 NdII 29
17.35: 1		—TiII u	45.92	-1		71.08	4	V 66	94.98	16	Co 18
17.60	5	ZrII 18 Ti 189	46.03	3	Fe 703	71.37: 0		CN(1,1)	95.25	8	Ti 176
		Fe 701	46.24: -1			71.66	2	LaII 13	95.67	40w	Fe 4
17.84	11	V 10 Cr 40	46.42	7	Fe 804 Ti u	71.73	7	Fe 429	95.89: 0		
18.25	20	V 9	46.63: 0			71.79	-2	SmII 18	96.15: 2		V 43
18.47: -1		Cr 40	46.79	6	Fe 664	72.11	12	DyII (Sun)	96.21	12	ErII (M)?
18.67: 0		Sun?	46.98	7	Zr 10 Fe 176	72.40	5	Rh (M)?	96.52	5	Zr 9
18.97	6	—Ru (M)?	47.34	17	V 7	72.53	36w	Fe 20	96.77	1	CeII 188?
19.63	12w	Cr 70 RuII 1				72.74: 2		CN(0,0)R	96.98: 4		SmII 5
19.96	9	V 28	48.02	6	TmII 2	72.95: 2		Fe 284	97.04: 3		—V 126
20.19: 18w		—V 44	48.27	10	—Fe 224p	73.12	30w	Co 18 Ti 176	97.29	2	Ti 175
20.40	35w	Fe 20				73.61	-2		97.44	5	Fe 429
20.80	3w		48.54	3	NdII u CeII 36	73.76	7	Fe 175	97.62	3	Ti
21.11	9	—Fe 608	48.74	6	—SmII u	73.96	21	Co 18 Fe 120	97.98	30w	Fe 20
21.49	15	V 28	48.99	7	Cr 69	74.17	5	TbII (M)	98.21: -1		
21.83	3	Fe 222	49.25	3	Zr 6			Ti (M)?	98.49	20	Ti 13 Co 58
22.01	15	V 9 Ti 189	49.34	3	V u Cr 138	74.51	4	Sun	99.02: 6		Fe 175
22.23: -2		Rh 8	49.52	10	Cr 24	74.73	-1	CN(0,0)P?	99.16: 4		TbII (M) VII 33
22.41	5	Zr 10	49.95: 37		Fe 20	75.07	21	V 7	99.74	45w	Fe 4 Ti 15
22.62	-3		50.40: -1		Ru (M)?	75.25	15	Ti 15, 175			
22.88	18	V 28	50.58: 38w		Fe 22 Co 17	75.5	3w	—SmII 17—			
23.20	10	Mn 6	51.17	2	V 44	75.83	10	V 7 Ca 26			
23.49	17	Mn 6 Cr 24	51.75	6w	NdII 35 Co 128	76.02: 13		Fe 22	00.22	4	NdII u
23.70	-2	ZrII 31?	52.08: 2		V (M)?	76.12: 18		V 8	00.55	13	TiII 34 Zr 6
23.90	10	Mn 6	52.21	11	Cr 2	76.76: 2		Os (M)?	00.95: 19		Ti 15
24.09	0w	Fe 224	52.48: 7		GdII 2—	76.84: 14		Co 17, 62	01.22: 0		Ru (M)?
24.45: 55w		Fe 4	52.60	10	Fe 73 Cr 11	76.95: 14		Sun	01.66: -1		
24.92: 2		Cb 2 FeII 29	52.81: -1		FriI u	77.27: 0w		CN(0,0)P	01.85	1	NdII u
25.27: 4w		Zr (M) Cr 70	53.04	6	Ti176DyII(Sun)	77.64: 2		Ti 175—	02.26	19	V 7

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
3900			29.88	15w	Ti 13	68.76	Em	CaII 1	98.63	40	Ti 12
02.51: 3	V 43	GdII 19	30.29	42	Fe 4	70.4	Em?	H 1?	98.98: 11	ZrII 16	
02.90: 25	Fe45Cr23Mn		30.45: 3	BaII 5		72.18	12	Ni 29	99.34	13w	Ti 188 CeII 57
03.18: 18	Cr 23		30.86	6w		72.58	0w	Ca 41	99.60: -1	Cr u?	
03.44: 1	SmII u		31.28	3w	-V 90?	72.93	0w	Fe 803	99.87	5	
03.88	11	Fe 429	31.80	6w		73.82	30w	Ni 31 Ca 6			
04.08: -2			33.32	Em	CaII 1	74.74	16	Co 18 Fe 72	4000		
04.23: -2	V(M)?		33.52N	45	CaII 1	75.24	2w	Fe 153	00.07	4	Fe 360 V(M)?
04.51: -2	V(M)?		33.95	Em	CaII 1	75.84	-1	Fe 977?	00.23	6	Fe 556
04.80	16	Ti 56			Ti 15 ZrII 7	76.26	0	SmII 9	00.48	17	DyII(Sun) Fe 426
05.06: -3			35.94	Em?		76.44	1	SmII 33			
05.24	-1		38.03	Em?		76.66	8	Cr 38	00.88: 0		
05.53	15	Si 3	40.89	22	Fe 20 Co 18	76.84	3	NdII 21	00.95: 0		
05.88	6	NdII u	41.24	0	-Fe 562			TbII(M)	01.2	6w	Sun GdII 49?
06.32: 20	Co 17		41.60	12	Cr 25 NdII 27	77.30	1w	Zr 46	01.5	4w	
06.51	32	Fe 4	41.75	22	Co 17	77.76	19	Fe 72	01.65	16	Fe 72
06.74	8	Fe664 V42,45	42.44	6	Fe 364	78.26	-1	Zr(M)?	02.0	-1	
06.91: -2			42.78: 1	CeII 57?		78.48	0	Fe 361	02.26	19	V 7
07.12	13	BaII 5--	42.95: 0			78.68	18	Co 17 Cr 67	02.49	12	Ti 188
07.49	19	Sc 8 Fe 284	43.36	6	Fe 72	79.02	-1		02.66: 3	Fe 320	
07.70: 2	Fe(M)?		43.69: 5	V 42		79.19	0	SmII 51	02.91	6	VII 9
07.93	10	Fe 280	44.02	35w	Al 1	79.53	25	Co 3	03.12	3	
08.15	1		44.34	7		79.66	1	Fe 561	03.44: 2	V(M)?	
08.43	9	CeII 65 PrII 11	44.69	9	DyII(Sun)	79.79	0	Cr 67	03.53: 2	V(M)?	
08.78	18	Cr 23			Fe 361	79.98	-1		03.77	11	Ti 188 Fe 728
09.01: -1			44.92	0	Fe 430	80.49	0		04.02	8	NdII u
09.22	1		45.10	1	Fe 280	80.63	5	Fe 153	04.3	-1	
09.66	2	Fe 565	45.34	11	Co 29	80.85	2	Ti 186	04.68: 2		
09.83	3	Fe 364	45.71	0		81.10	0	Fe 22	04.85: 10	Fe 601--	
09.93	28w	Co 3 V 7,63	45.98	2	Cr 134?	81.22	9	Cr 67	04.99: 10	Fe 486, 557	
10.06: -2			46.20	7	Ni 1p	81.5	4		05.26	37	Fe 43
10.50	0	Fe 562p?	46.46	0		81.78	35	Ti 12 Fe 278	05.47: 8	Fe 219p	
10.80	11	V 42	46.63	1	Co 60	82.06	10	TiIII PrII 28	05.68	9	VII 32
10.92	2		47.05	7	Fe 561--	82.51	35	Ti 11 VII 6	05.95	9	Ti 187
11.18	7	Ti 175 NdII u	47.37	3	Fe 153	82.86	-1		06.14	1	Ni u?
11.40	2	Ti 176	47.57	3	Fe 361, 426--	83.12	4		06.30	10	Fe 603
11.75: 4	Fe 664		47.78	22	Ti 14	83.35	4	Fe 485	06.58	9	Fe 488
11.84	16	Sc 8	48.09	5	Fe562 SmII 9	83.62	3	DyII(Sun)	06.75	5	Fe 320
11.96	6	Cr u	48.31	0	Fe 561p?	83.92	20	Cr 38 Fe 277	06.99	0	Sun
12.21	12	V 42,43	48.72	24	Ti 13 Fe 604	84.34	11	Cr 38 Ti 188	07.26	14	Fe 277 Ti 187
12.39: 0	CaII 60		48.91	11	Ca 6	84.72	6	-ZrII 7	07.44	7	NdII u
12.57	3	Ti 175	49.11	4	LaII 41	84.93	3	Fe 561p?	07.74	-1	
12.84	24	Ni 15 V 42	49.46	0w	PrII 16	85.22	3	-Ti u	08.00	16	Ti 187
13.44	12	TiII 34	49.94	15	Fe 72	85.37: 15	-Fe 661		08.13	4	VII 32
13.64	12	Fe 120	50.35	11	VII 6	85.58	4	Ti 188	08.77	16	W 6? PrII 28
13.92: -2			50.56: -2			85.92	0		08.93	35	Ti 12
14.34	22	Ti 15 Fe 567	50.92	-1		86.18	12	Fe 655	09.2	-2	
14.51: -3			51.15	10	NdII 19 Fe661	86.32: 0	Fe 560p?		09.66	35	Ti 11 Fe 72
14.74	17	Ti 14	51.58	0	VII 16	86.75	14	Mg 17	10.15	3	Fe 915
15.06: -2			51.80	-2		87.13	16	Co 16 GdII 19	10.41: 2		
15.18: -2	V(M)?		51.99	1	VII 10 GdII 1	87.39	1		10.56: 2	Sun	
15.45: -2			52.18: 3	NdII 23		87.60	11	TiII 11	10.74	2	Fe 219, 320
15.59: -1	DyII (M)?		52.30: 11	Co 16		87.80	0		10.92	-1	
15.88	16	Ti 15 Cr 136	52.38: 2	Cr 136		87.98	11	Yb 2	11.09	13	Co 2
16.24	14	Cr 23	52.61	5	Fe 278				11.38	9	Fe 218
16.38: -2			52.70	4	Fe 362	88.50	10	LaII 40	11.50: 12	Ti 10	
16.71	4	Fe 606	52.92	15	Co 28	88.66	2	Zr 46	11.72	7	Fe 153
17.03	-1		53.16	3	Fe 430 Cr 136	88.81	3	V 89	11.93	-1	Fe 424p
17.20	22w	Fe 80	53.52	2	PrII 9	89.04	5	ScII 8	12.24	10	NdII 10
17.57: 2	Cr 137		53.85	3	Fe 362	89.28: 3	Zr 6		12.40	16	TiII 11
17.91: 0			54.56	0	Ni(Sun)?	89.56: 15	Ti 81		12.74	12	Ti 188 NdII u
18.07: 0			54.71	2	Fe 606	89.73	30	Ti 12	13.07	-2	
18.34	12w	Fe 124 Fe 364	55.14	0		89.88: 7	Fe 768		13.24	5	Ti 186p
18.63	6	Fe 430	55.33	4	Fe 562	90.26: 12w	Ti 188 Co 58		13.60	14	Ti 187 Fe 557
19.17	22	Cr 23	55.64	-1			Fe 527		13.87	22	Fe 486 Co 58
19.58: -1			55.95	3	Fe 488	90.57	12	V 89	14.26	5	Fe 426, 427
19.79: 1	Ti 150		56.34	30w	Ti 13	90.86: -1			14.51	16	ScII 8 Fe 802
20.27	35w	Fe 4	56.68	9	Fe 278	91.12	14	Cr 38 ZrII 30	14.70: 2	Cr 268	
21.02	15	Cr 23	57.05	13	Ca 6 Fe 562	91.50: 1	Co 173?		14.90: 5	-CeII 157--	
21.42	22	Ti 14	57.39: -1			91.89	17	Co 17 Cr 38	15.07	-1	
21.81: 4	Zr 8		57.60: -1	Fe 564				NdII 19	15.37	11	Ti 185
21.90: 4	V 42		57.69: -1	GdII 19		91.99	5	Co 3	15.60	5	Sun
22.09: -1	Fe 153p		57.93	15	Co 18	92.08: -1	Cr 38		15.90: -1		
22.42	10	V 42	58.21	30w	Ti 13 ZrII 16	92.37	4	Fe 604	16.26	8	Ti 186
22.70: 0	Fe 429p		58.88	0	Ni 7	92.60	0		16.44	8	Fe 560
22.94: 40w	Fe 4		59.24	-1		92.82	13	V 89 Cr 67	16.51: 0	Fe 277	
	Mn u?		59.48	-2		93.02	-1		16.76: -2		
24.56: 25	Ti 13		60.44	3		93.28	6w	SmII 4 GdII 1	16.95: 4	Ti 208	
24.87	-1		60.88: 0			93.59	-1	Sun	17.12	15	Fe 279 Fe 527
25.23	10	V 8 Fe 567	61.19	8	Fe 361--	93.76: 3w	Ti 188 CeII 12		17.48	5	Sun
25.46: -1	PrII 11?		61.50	40w	Al 1	94.06	14w	Fe 526 GdII 49	17.76	10	Ti 185
25.65	6	Fe 364	62.04	0	Sun			Cr 67	18.09	32	Mn 5
25.95	11	Fe 364 Fe 562	62.87	29	Ti 12	94.57	18	Co 17	18.24: 12	Fe 560	
26.30	0	Ti 292	63.26	-1		94.68	10	Ti 188 NdII u	18.81	6	-NdII 19--
26.56: -1			63.43	0	Fe 654p?	94.82: 7	PrII 11 V(M)?		19.05	8	Fe 219
27.11: 0	NdII(M)?		63.69	12	Cr 38	95.31	24w	Co 49	19.28	14	Co 16
27.56: -1			63.98	-3		95.74	9	LaII 27	19.81	3	
27.93	42w	Fe 4	64.29	23	Ti 12	95.99	12	Fe 279	20.02	4	SmII 16 Fe 556p
28.31: -1	SmII 17?		64.59	0	Fe 361--	96.27	5w	-Fe 561p--	20.40	32	Sc 7
28.63	12	Cr 23				96.61	21w	Sc 7	20.89	22	Co 16
29.10	1	Fe 280	66.10	10	Fe 45	96.97: 7	Fe 945		21.08	2	
29.25	1	LaII 27?	66.42	5		97.12: 7	VII 9		21.33	6	NdII 36
29.52	-1	Zr 7	66.6	Em?		97.42	18	Fe 278	21.60	8	Fe 180, 557
			68.12	Em	CaII 1	97.88: 18	Co 32		21.81	18	Ti 185 Fe 278
			68.28	40	CaII 1	98.01: 14	Fe 276		22.19	9w	Sun? Ru(M)?

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4000			47.20: 12	K 3	SmII 16	69.02: 8	Fe 557		93.60	10	Ni 1
22.42	4	Fe 173	47.32: 12	Fe 117		69.27	10	NdII 20	93.98	3	
22.71	5	Fe 556, 554	47.64	12	Y 6	69.61	3	Sun	94.24	4	Zr(M) Tm u?
22.98	7	NdII u	47.82	20	Sc 7	69.93	4w	Mo(M)? Mo(M)?	94.43	4	Sun
23.19	5	SmII 4	48.35	1	ErII(M)?	70.29	18	Mn 5	94.62	2	
23.37	7	VII 32 Co 59	48.70: 35	Mn 5	ZrII 43	70.76	18	--Fe 558--	94.96	19	Ca25 PrII(M)?
23.70	26w	Sc 7	49.07: -1			71.16	8	--Ti 254	95.26: 0	Sun	
24.05	12w	Zr 46 Fe 277	49.36: 18w	Fe 218 Ti 185		71.80: 42w	Fe 43		95.49	18	V 41
24.41: 3	ZrII 54		49.43: 2	GdII 50		72.47: 5	Fe 698		95.71	9w	
24.52	25	Ti 12	49.81	9	GdII u Cr 251	72.70	12	Zr 46	95.98	15	Fe 217 SiH u
24.70	10	Fe 560	50.02	4	Cr 36	72.95: 5	Ni 197--		96.22	25	Fe 18p
24.96: 11	Zr 46		50.31	7	ZrII 43	73.14	8	DyII(M)	96.48: 9	SiH u SiH(O,O)R	
25.16	15	TiII 11	50.50	10	Zr (M)		GdII 34		96.72	10	SiH u SiH(O,O)R
25.43	2	Cr 37	50.65	5	Fe(Sun)?	73.48	7	CeII 4	96.97: 11	Fe 173p--	
25.83: 4	Sun			DyII(Sun)?		73.76	11	Fe558 GdII 44	97.18: 9	Fe558 SiH(O,O)R	
26.16: 8	Cr 37		50.96	8	V 121	74.36	10	Ti 254	97.45	5	--SiH(O,O)R
26.41: 5	Mn u		51.15	9	NdII 66	74.76	12	Fe 524	97.65	5	Cr 97
26.52: 13	Ti 185		51.35	12	V 121	74.88	20	Ni 28	97.81	0	Ru 9
27.03	20	Co 3	51.94	14w	Fe 700 CrII 19	75.12	8	NdII 62	97.98	8	Cr 97
27.18	12	Zr 46 Cr 37	52.29	4	Fe 700	75.27	10	NdII 19	98.17	14	Fe 558 Cr 97
27.45	3	Ti u Sun	52.46	12	Fe 563	75.54	4	Ti(p)Mo(M)?		SiH(O,O)R	
27.66	2	Sun	52.69	10	Fe 524 Fe 557p	75.66: 1	VII 14		98.40	8	--PrII(M)?--
27.94: 0	Sun		52.95	7	Ti 208	75.85	9	CeII 206	98.56	14	Ca 25
28.36	10	TiII87CeII47	53.25	9	Fe(Sun)?		SmII 51		98.94: 5	GdII 49	
28.74	1	Fe(Sun)		GdII u			CeII 57		99.16	20w	Ti207 SiH(O,O)R
28.95	7	Zr(M)	53.47	8	--CeII 36	76.13	14	Co 16	99.42: 5		
29.26: -1			53.63	8	GdII 5?	76.38	21	Ti 9	99.79	27	V 27
29.63	11	Fe 556, 563	53.85	13	Fe 485	76.52: 9	Fe 218		99.97: 2	--Fe 698p	
				TiII 87--		76.64	11	Fe 558			
29.87	9	Sun	54.17	7	Fe 557	76.83	12	Fe 557			
30.07: 3	Zr 46		54.54	30	Sc 6 Co 2	77.11	9	Ti 207 Cr 66			
30.18	10	Fe 72	54.81: 9	Fe 698 Fe 698		77.36	16	Y 7 LaII 41			
30.71: 50w	Mn 2		55.03: 20	Ti80Zr46Fe218		77.66	40w	ErII 1			
31.28: 7	Fe 486 ZrII 42		55.55	25	Mn 5	77.94: 20	DyII(M)				
31.51: 0			55.65: 5	Zr(M)?		78.35	16	Fe217 GdII 15			
31.72	12w	Ti 185 LaII 40	55.86: -1				CeII 19				
			56.00	2	Fe 914	78.51	19	Ti 80			
31.95	8	Fe 655	56.20	8	TiII 11	78.73	5	Gd 5?			
32.22: 0	Ru(M)		56.4	4w		78.83: 2	--AlH(1,0)P,Q?				
32.47: 6	Fe 320		56.57	8	Fe 320 PrII 26	78.92	3	SiH(O,O)S?			
32.62	15	Fe 44	56.89: 0			78.93	16	Mn 5			
33.08	45w	Mn 2	57.13: 6	V 121		79.42	18	Mn 5			
33.44: 3	Ti p?		57.20	16	Co 3	79.77	25	Co 1 Ti 207			
33.64: 3	Sun		57.32: 9	Fe 277 Ni 89			Fe 359				
33.90	6	Ti 208 Cr 36	57.53	8	Mg 16	80.22	10	Fe 558 Cr 66			
34.06: -1	Cr(M)?		57.65: 5	Ti 254		80.4	-1	CeII 36			
34.51	45w	Mn 2	57.84: 8	Pb 1 Cr 251		80.59	11	Ru 7 Cr 156			
34.89	5	Ti 208	57.97	4	Mn 29?	80.90: 9	Fe 557				
35.15	3	SmII 33--	58.20	20	Co 16 Fe 558	81.00: 1	PrII 14				
				Ti 254		81.24	14	Zr 46 CeII 4			
35.78	40w	Mn 5	58.59	5	Co 58	81.44	2	AlH(1,0)Q?			
36.35	6	Fe 279p	58.80	16	Fe 120 Cr 251	81.75	2	Cr 66			
36.57: 1			58.96	22	Mn 5 Co 1	81.82: 1					
36.75	6	VII 9	59.33	15	Co 2 Mn 29	81.98: 1	AlH(1,0)P?				
37.10: 4	Sun		59.48: 0			82.13	5	Fe 698			
37.32: 9w	Cr 36 GdII 49		59.73	9	Fe 767	82.40	30	Sc 6 Ti 80			
37.62	3	--Fe 118	59.99	5	NdII 63	82.59: 16	Co 16				
37.88	3	GdII 49	60.11: 5	Ti 254p Zr(M)?		82.95	21	Mn 5			
38.12	5	NdII 31	60.28	17	Ti 80	83.27	8w	CeII 60--			
38.26: -1	Ni 150p?		60.56	2		85.64	35w	Y 6 Mn 5			
38.43	8		60.79	6	Fe(Sun)?	84.49w	11	Fe 698			
38.57: 0	Fe 600?		61.09	14	NdII 10 Fe(Sun)	84.60	0				
38.78	7	Sun Mn(M)?	61.04: -1			85.00	11	Fe 358			
39.06	5	--Cr 251?	61.52	9	Zr(M)	85.29	16	Fe 559			
39.28: 3	--Cr 251?		61.73	8	Mn 29	85.38: 0	AlH(1,0)P?				
39.54	2	VII 32	61.99	6	Fe(Sun)	85.57	2	GdII 50			
39.81	11	Y 5	62.23: -2	CeII 34		85.81	4	NdII 16			
39.90	4	Fe 276	62.45	18	Fe 359	85.99	2	Fe1073 Sc(M)?			
40.09	2	Fe(Sun)?	62.77: 9	PrII 26		86.31	17	Co 58			
40.30	9	Ti 185	62.86: 6			86.67	16	LaII 10 Sc(M)?			
40.63	8	Fe 655	63.21: 14w	Co 18		86.92	3	SiH(O,O)S			
40.79	16	--NdII 30--	63.63: 45w	Fe 43 Mn 5		87.11	9	Fe 694			
41.05	4	NdII(Sun)?	63.82: 9	V 121		87.32	2	CeII 59--			
41.37	25	Mn 5	64.19	20	Ti 80 Zr 46	87.46: 1					
41.67	3	SmII 22	64.50	19	Fe44 SmII 24, 35	87.63	5	ErII(M)? Zr(M)?			
41.91: 0	Fe 602 Cr 36		65.10	21	Ti 80	87.78	-1				
42.24	5	Cr 36	65.38	8	Fe 698	87.96	2	AlH(1,0)Q?			
42.42	-2		65.59	3	Ti 207	88.29	16	Co 2			
42.64	14w	SmII 4 CeII 140	65.72: 0	Cr 279		88.55	6	Fe 906			
42.88: 7	SmII 9 LaII 66		66.15	3	Sun	88.85	0				
43.03: 0	NdII(M)?		66.37	17	Co 30	88.99	0				
43.29: 0			66.58	8	Fe 424	89.19	11	Fe 422			
43.60	10	Zr 32	66.70: 3	SmII 28		90.04	8	Fe 700 Mn(M)?			
43.74	7	Ti 208	66.96	15	Fe 358 Cr 66	90.30	7	Cr 66 Fe 44p			
43.93	11	Fe 276, 557	67.29	18	Fe 217	90.56	25	V41 ZrII 29			
		Fe 558p?	67.38: 1	LaII 26		90.98	10	Fe 695			
44.15	10	K 3	67.5	0	Fe 422p		SiH(O,O)S				
44.37: 1			67.73	4	V(M)	91.55	9	Fe 357			
44.60	14	Fe 359 Zr 46	67.98	17	Mn 5 Fe 559	91.94	10	V 52			
44.80: 6	--PrII 8--		68.08: 7	--Ti 207		92.26: 7	SmII 1				
45.07: 7			68.34	4	SmII 42?	92.37	20	Co 29 V 52			
45.40: 16w	Co 31 ZrII 30			Ru(M)?		92.68	25	V 27 Ca 25			
45.85: 50w	Fe 43		68.53	7	Co 58	92.78: 6	Co 59				
			68.68: 3	Ti 254		93.14	3	Sc(M)?			
			68.88	7	--CeII 82?	93.48	9	V 52			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4100			34.17	12	SiH(0,0)Q	52.16	25	Fe 18	69.51	8	SiH(0,0)S
13.51	21	V 52 SiH(0,0)R	34.35	60w	Fe 3 V 27	52.33	9	Sc 20			SmII 24
13.85	9	—SmII 16	34.66	16	Fe 357	52.56	19	Ob 1 SiH u	69.82	11	SiH(0,0)P
14.12	10	SiH(0,0)R	34.90	7	SiH(0,0)Q			Zr(M)?			Fe 693
14.48	15	Fe 357	35.04	0	Mn(Sun)?	52.77:	-1	LaII 78	70.00	6	SiH(0,0)P
		SiH(0,0)S	35.28	11	Rh(M)? NdII u	52.96:	8	SiH(0,0)S	70.21	3	Cr 278
14.92	4	Fe 695			SiH(0,0)S	53.05	9	—Cr 35—	70.52	4	SiH(1,1)R?
15.19	25	V 27	35.44	2	Sun	53.34	15	V 26	70.68	6	SiH(1,1)R?
15.50	10	—SiH(0,0)R	35.71	15	SiH(0,0)Q Zr50	53.62	5	Zr(Sun)	70.91	15	Fe 482
15.68:	2	SiH(0,0)R	35.90	0	SiH(0,0)P?	53.86	20	Cr 35 Fe 695	71.03	16	Ti 206
15.82	8	SiH(0,0)R	36.08	6	V(Sun)	54.10	6	Fe 694	71.29	2	V(M) SiH u
16.23	7	SiH(0,0)S	36.36	17	V 26	54.51	18	Fe 355	71.45:	3	Zr(M)?
16.50	25	V 27	36.52	16	SiH(0,0)Q	54.82	16	Fe 694 SiH u	71.63	11w	SiH(0,0)P
16.69	20	V 27			Fe 694	55.00:	6	SiH(0,0)S	72.06	35	Ga 1 Fe 649
16.94	6	Fe 558	36.72	0		55.20:	2	SmII 8			SiH(0,0)P, S
17.12	-1		36.90	4	Ti 221	55.40:	-1		72.75	35	Fe 19
17.26	0	Fe 484 CeII 77	37.06	19	Ob 1 Fe 726	55.56	10	—SiH u	*		SiN(2,2)R?
17.50	10	SiH(0,0)R	37.28	7	Ti 253	55.69	9	SiH(0,0)P	73.30	7	Fe 355
17.87	20w	Fe 700	37.41	14	SiH(0,0)Q	55.88	8	V(Sun)	73.51	10	TiII 21
		SiH(0,0)R	37.67	7	CeII 2	56.07	11	NdII 10			SiH(1,1)R
18.13	9	CeII 11 7 112	37.93	7	—Fe 320p	56.26	11	NdII 14	73.59:	3	—SiH(1,1)R
18.53	25w	Fe 801 PrII 8	38.12	-2	V(Sun)?			ZrII 29	73.94	24	Fe 19
		V 41	38.34	8	SiH(0,0)Q	56.45	3	Fe 693	74.11	18	Y 6 Ti 55
18.81	30	Co 28 Fe 559	38.44	8	SiH(0,0)Q	56.64	10	SiH u Fe 419	*		SiN(2,2)R?
19.08	8	SiH u	38.63	0		56.80	16	Fe 354	74.46:	8	Ti 220 Fe 799
19.43	15	SiH(0,0)R V41	38.84	6	Fe 117	56.99	5	SiH u Mn(M)?	74.78:	0	SiH(1,1)R
19.64:	7	SiH(0,0)S	39.05	2	—Sun	57.22	8	SiH(0,0)S	74.92	37	Fe 19
19.86	3	SiH(0,0)R	39.23	4	V(M)	57.75	18	Fe 695	75.30	7w	SiH(0,0)P, S
		Zr(M)?	39.41	12	SiH(0,0)Q			SiH(0,0)P	75.62	17	Fe354 SiH(0,0)P
20.01	1	Ti 253			Ti 221	*		SiH u			NdII 39
20.21	13	SiH(0,0)R	39.53	4	SiH(0,0)Q	58.07	3	Ti(M)?	75.93:	6	Cr 106 Fe 694p
		Fe 423	39.70	16w	Ob 1	58.38	14	Sun	76.03:	4	SiH(1,1)R
20.57	16	SiH u V 41	39.92	22	Fe 18	58.81	17	Fe 695	76.31	7	SiH u
		Cr 65	40.24	6	—Sc 20			SiH(0,0)P	76.57	11	Fe 695
20.85	1w	CeII 112	40.46	14	Fe 694, 695	59.00	0	—CeII 246?	76.79	12	V 6
		SiH(0,0)R			SiH(0,0)Q	59.17	6	Sun	77.05	6	Fe 690 V(Sun)
21.36	30w	Co 28	40.72	7	SiH(0,0)Q	59.40:	6	SiH(0,0)Q	77.34	11	Ti 163 NdII 10
		SiH(0,0)R	40.90	0	Pd(M)?	59.50:	6	SiH(0,0)S	77.59	32	Fe 18 Co 2
21.63	9	Ti u	41.10	6	Mn(M)?	59.67	19	V 25 Ti 206			VII 14
21.83	11	Fe 356 Cr 108	41.26	9	PrII 10	59.92	6	SiH(0,0)P	77.89	4	Cr(M)?
21.93:	2	Co 2p	41.51	4	DyII(M)?	60.18:	4				SiH(0,0)S
		SiH(0,0)R	41.81	14	SiH(0,0)Q	60.36	10	V(Sun)?	78.05	5	Fe(Sun)?
22.15	6	Cr 65 Ti 296?			Fe 422 LaII 40	60.55	6	Fe 419			SmII 16, 50
22.32:	0	SiH(0,0)R	42.01	13	SiH(0,0)Q	60.78:	1	Sun	78.22	5	SiH(0,0)S
22.52	10	Fe 356	42.26:	6	SiH(0,0)Q	60.87:	4	SiH u	78.41	5	SiH(0,0)P
22.72	11	SiH(0,0)R	42.42	5	SiH(0,0)Q	61.08:	6	Fe 689			VII 25
23.02	5	SiH(0,0)R, S	42.60	12	SiH(0,0)Q V26	61.20:	8	ZrII 42	78.58:	5	SiH(0,0)P
23.22	10	LaII 41 V 112	42.85	18	Y 5 SiH(0,0)Q	61.38:	3h		78.86:	2	—FeII 28—
23.50	23	—V 27	43.11w	10	SiH(0,0)Q, P	61.51	14	TiII 21 Fe422	78.99:	1	SiN(2,2)P?
23.81	20w	Fe 217, 422			Ti 253 PrII 4	61.82:	5	SiH(0,0)Qp?	79.27	5	Cr 179, 250
		Ob 1 CeII 60			DyII(Sun)	61.84	7	SrII 3			Co 144?
24.06	11	V 52	43.41	14	Fe 523	62.05	12	SiH(0,0)P	79.39	25	V 25
24.53:	3	Sun			SiH(0,0)Q	62.28	9	SiH(0,0)P	79.54	8	NdII 19
24.75:	1	Ln 3?	43.59	8	SiH(0,0)Q	62.55:	2		79.85	6	Ti206 ZrII 99
24.90	7	VII 14	43.89	35w	Fe 43	62.69	4	GdII 17	80.01:	0	SiN(2,2)P?
25.19:	0		44.14	6	SiH(0,0)Q Ru 7	62.96	1	Fe 476a p	80.45	12	Ti 206
25.36	9	SiH(0,0)R	44.33	7	SiH u	63.05	11	Ho(M)?	80.86	7	Ti(M)
25.55:	4		44.51	8	SiH u			GdII 44	81.13	10	SiH(0,0)S, P
25.64	3	Fe 1103	44.68	7	SiH(0,0)Q	63.45	4		81.39	6	SiH(1,1)R
25.87	8	Fe 354	44.88:	8	SiH(0,0)Q	63.66	16	Ob 1 Cr 35	81.47:	6	SiH(0,0)P Sun
26.16	13	Fe 695 Cr 65	45.02	12	Ti(Sun)			Fe 274, 699	81.76	18	Fe 354
26.51	11	Cr 35	45.26	14	SiH(0,0)Q			TiII 105?	82.00	3	
26.87	8	Fe 354 Cr u?			Fe 274	64.04:	0		82.22	3	
27.08:	-1	Ti 114p	45.70	12	SiH(0,0)Q	64.15	15	PrII 8 Ti 163	82.36	8	Fe 476a
27.34	2	CeII 4 Cr 35	45.77:	1	Ru(M)?	64.33	11	SiH u	82.57	11	V 24
27.58	15	Fe 357 Ti 296	46.04	11	Fe 422	64.39:	2	SiH(0,0)S	82.78	4	Fe 694
		Cr 65	46.21	13	SiH(0,0)Q	64.67	21	Ob 1 Ni 28?	83.01	3	Fe 697
27.78	8	Fe 558, 727	46.36	6	SiH(0,0)P	64.79	10	SiH(0,0)P	83.27	9	Ti 220 Zr 51
28.06	24	V27 SiH(0,0)R, Q	46.68	18	SiH(0,0)Q			Fe 418	83.41:	3	VII 37
28.32	17	Y 5			Cr 107	65.06	10	SiN(1,1)P-head	83.68:	4	SiH(u)
28.61:	-2	SiH(0,0)P	46.98	1	Sun	65.20	15	SiH(0,0)P Sc20	83.74	5	SmII 4
28.90	14	SiH(0,0)Q	47.20	6	SiH(0,0)Q	65.39	10	SiH(1,1)R	83.98	5	Sun SiH(1,1)Q
		V 112	47.35	5	Zr(Sun)?			SiH u	84.26	12	GdII 15
29.18	12	Cr 97 Fe 698	47.54	2	Mn 37? Sun	65.58	4	CeII 10 Cr 305			SiH(0,0)P Fe274
		Ti u?	47.69	33	Fe 42	65.76	1		84.42	5	—Ob(M)? Ni 89
29.42	7	DyII(Sun)	47.97	-1		65.97:	8	SiH(1,1)R	84.60	4	SiH(0,0)P
29.71	22	EuII 1						BaII 4	84.90	17	Fe 355
		SiH(0,0)Q	48.19	7	SiH(0,0)S	66.08:	7		85.14	3	SiH u
30.03	16	Fe 44, 486	48.41	7	SiH(0,0)Q	66.32	12	Ti 163 SiH u	85.35	8	—Cr 106—
30.36	10	—GdII 19, 49	48.60	6	SiH(0,0)Qp?			Zr 45	85.69	3	SiN(2,2)P?
30.60:	17	Co 16 BaII 4	48.86	14	V26 SiH(0,0)Q	66.67	7	SiH(1,1)R	86.10	25	Ti 129
		SiH(0,0)Q	49.18	9	ZrII 41	66.98	4	SiH(0,0)S	86.32:	1	SiH u Cr 249
31.14	15	SiH(0,0)R	49.38	15	Fe 694	67.26	19w	Mg 15 SiH(0,0)P	86.58	5	CeII 1 Zr(M)?
31.32	13	SiH(0,0)Q			SiH(0,0)P	67.52	14	Y 7 SiH(1,1)R	86.81	8	Sun Dy(M)?
31.74	3w	SiH(0,0)S	49.78	30	Fe 3p SmII 8	67.69	-1		87.04	18	Fe 152
32.06	50w	Fe 43 V 27	49.92	4	—CeII 158, 189?	67.88	5	Fe 599	87.24	0	Co 93
32.48	5w	SiH(0,0)S Sun	50.26	12	Fe 695 SiH u	67.95:	5	Fe(Sun)?	87.42	6	SiH(1,1)Q
32.73	8	SiH(0,0)Q	50.42	6	Co 15	68.10	16	Ob 1	87.57	8	Zr 45 Fe 694
32.93	16	Fe 357	50.55	5	Ti 253	68.43	13	SiH(0,0)P	87.81	19	Fe 152
33.12	3	SiH(0,0)S	50.72	7	SiH(0,0)S	68.58	6	—Fe 689	88.12	3	SmII 50
33.40	5	NdII 19	50.94	11	Ti206 ZrII 42	68.85:	3		88.31	0	Mo(M)?
		SiH(0,0)Q	51.08	11	SiH u	68.98:	8	Fe 694—	88.72	14	Ti220 SiH(1,1)Q
33.60	-1	Fe(Sun)?	51.24	5	SiH u	69.08	10	Fe 18p	88.98:	7	SiH u
33.80	14	CeII 4 Fe 698	51.34	1		69.30	8	Ti 163 V(M)?	89.08	3	
			51.90	16w	SiH(0,0)P				89.43	3	SiH u
					CeII 2 LaII 40						

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4100			08.56	12	Fe 689, 696 SiH(1,1)Q	34.15	3	—NdII 20	53.45:	7	GdII 46 CeII 77 Fe 690p?
89.51	11	Co 2 PrII 8 Fe 940	08.74	-1	ZrII 41	34.55	18	V 6	53.60:	3	GdII 17 Zr(M)?
89.85	15	V 24	08.99	11	Cr 248	34.63:	6	SmII 42—	53.93	8	—Fe 905p?—
89.91	0	SiH(2,2)P?	09.37	5	—Cr 155	34.89	1	Bo(O,O)P	54.34	60W	Cr 1
90.24	9	SiH(1,1)Q TiII 21	09.70:	2	V 24	35.17	8	Mn 23	54.73	4	
90.71	33	Co 1	09.85	17	SiH(1,1)Q	35.30	11	Mn 23	54.93	6	Fe 419, 477
90.88	10	—VII 37—	10.20:	3	Fe 152 SmII 8	35.76	9	V 111 VII 5	55.09:	5	AlH(O,O)R
91.08	3	GdII 34	10.37	16	SiH(1,1)Q	36.09:	3	Zr(M)?	55.29	0	CH(O,O)R—
91.24	3	Cr 35 SiH u	10.85	4	SiH(1,1)Q	36.31	-1	CH(O,O)R	55.50	8	Fe 416 Cr 105
91.49	16	Fe 152 V 24	11.15	3	Rh(M)?	36.59	9	V(Ms) ZrII 110?	55.81	8	CeII 81—
91.72	13	Fe 355 Cr 35	11.31	8	Cr 133 NdII 57	36.77	5W	SmII 53 Fe 906—	56.02	8	Ti 252 Zr(M)
91.90:	5	SiH(1,1)Q—	11.56:	-2	DyI(M) Ti 279	37.10	25	Fe 19	56.21	4	Fe 690
92.10:	9	Zr(M)	11.76:	9	ZrII 15 GdII 15	37.45	2	Bo(O,O)R,Q	56.36	11W	SmII 37 DyII(Sun)
92.42	5	Pt(M)?	11.93:	7	Ru 6	37.69	8	SmII 8 Cr 132	56.61	1	Cr 131
92.51	0	Fe(M)? Sum	12.08:	9W	Cr(M)? Zr(M)?	37.88:	7	—Ti 284—	56.83	4	Sun
92.73	4	SiH u	12.68	6W	Pd 7 SiH(1,1)Q	38.03:	9	Fe 689, 696	57.09	3	AlH(O,O)R CeII 123?
92.90	-2	Co 94—	12.96	7	Cr 155	38.38	9	LaII 41	57.16	3	—Cr 131 V(M)
93.09	3	—CeII 79—	13.16	6	Cr 155	38.60	1	Fe 849p	57.32	6	SiH(1,1)P
93.27	2		13.40:	0	SiH(1,1)Q	38.80	12	Fe 693 GdII u	57.63	9	Mn 23
93.40	5	SiH(1,1)Q	13.61	9	Fe 355	38.95	8	Cr 131	57.75	2	
93.62	7	—Cr 248	14.11:	-2	CH(O,1)P	39.32	15	Zr45 SiH(1,1)P	58.02	7	ZrII 15
93.82	4	—CeII 85	*	3	CN(O,1)P	39.84	30	Fe 18, 273	58.32	35	Fe 3
94.07	4	SiH u	15.03:	6	GdII 32	40.05:	8	V(M)?	58.56	12	Fe 351 Ti 252
94.29	2		15.53	45W	SmII 1 Fe 274, 419	40.34:	9	Zr 45 Fe 764	58.96	6	Fe 419
94.48	3	—Fe 274p?	15.98:	4	Fe 273	40.44:	9	Ca 38	59.08:	2	Sun Cr 131
94.62	4	SiH u	16.22	43W	Fe 3	40.72	8	Cr 105, 178	59.31	15	V 6
94.84	8	Dy(M)?	16.61	0	SiH(1,1)P-head	40.94:	4		59.49	2	AlH(O,O)Q
95.06	9		16.87	2	SiH(1,1)P	41.13	20W	AlH(O,O)R-head	59.70	5	AlH(O,O)Q
95.31	12	SiH(1,1)Q Fe 693	17.03	2	—SiH(3,3)P?—	41.27:	2	Zr 45 Fe 351	60.03	11	Fe 689
95.63	9	Fe 478	17.22	7	GdII 49	41.52:	5	GdII 117? V(M)?	60.12	10	Fe 476a
95.92	1	CN(O,1)R, CN(1,2)P	17.56	14	Fe 693 LaII 78 CeII 19 Cr 132	41.65	14	Zr45 AlH(O,O)R	60.47	30	Fe 152
96.20	8	Fe 693	17.8	5W		42.00	4	Zr(M)	60.80	10W	Ti 251 Os 1
96.30:	3	CeII 123	17.91	6		42.16	7	TmII 5	60.97	8	—AlH(O,O)Q
96.57	7	SiH u LaII 41 Fe 418	18.21	5W		42.42	7	AlH(O,O)R	61.19	2	CeII 19 CH(O,O)R
96.64:	8	—Fe(Sun)?	18.43	4	Zr(M)	42.60	5	Fe 273 CH(O,O)R	61.32	6	Cr 96
96.88	0	CN(1,2)P Ru(M)?	18.70	10	V 24	42.74	8	Fe 649	61.61	13	Ti 252 Cr u
97.11	16	Fe18p GdII 117?	18.86	2	SiH(1,1)Q	43.01	7	AlH(O,O)R	61.83	1W	PrII 23—
97.23	11	SiH(1,1)Q Cr 249	19.20	4	Fe(Sun)	43.30	2	CH(O,O)R	62.10	12W	GdII 44 V(M)
97.64	15W	SiH u GdII u Ru(M)	19.40	15	Fe 419p SiH(1,1)Q Fe 800	43.38	4	Fe 906 CH(O,O)R	62.41	9	AlH(O,O)Q Cr154
98.10	15W	Fe(Sun)	19.52	5	V 24	43.62	10	AlH(O,O)R	62.70	7	SmII 57
98.36	25	Fe 152 Co 2	20.05	6W	—VII 25—	43.84	5	Fe 994—	62.81	5	
98.64	16	V 24 Fe 693	20.32	11	Fe 482 NdII 32	43.97	5	AlH(O,O)R	63.15	15	Ti 162 Cr 247
98.90:	3	Ru(M)	20.64	6	SmII 15, 50	44.26	1	SiH(1,1)P	63.37	8	AlH(O,O)Q
99.09	15	Fe 522	20.86	3	SiH(1,1)Q	44.49	3		63.67	-1	
99.34	11W	SiH(1,1)Q	21.18	5	SiH(1,1)Q Dy(M)	44.70	5	SmII 27	63.89	2	PrII(M)?
99.69	3	SiH(1,1)Q	21.28:	3		44.94	2		64.20	9	Fe 692
99.98	30	Fe 3	21.60	3W	Cr 155, 248—	45.15:	8	AlH(O,O)R	64.43	7	AlH(O,O)Q
4200			22.10	1		45.26	17	Fe 352	64.72-	4	Fe 993
00.15	13	V(M) Cr u	22.22	15	Fe 152	45.49	4	Ti(Sun)?	65.10	9	SmII 15 V u
00.51	6	M 89	22.33	6	SiH(1,1)Q	45.61:	2		65.23:	10	Ti252 Fe993, 994
00.78	8	Ti 220	22.62	4	CeII 36	45.87:	1		65.69	8	AlH(O,O)Q Ti162
00.89	9	Fe 689 V 6	22.72	3	Cr 132	46.04	9	Fe 906 Sun	65.90	7	Mn 23 V?
01.22:	1		22.92:	4	SiH(1,1)Q	46.47	7	—AlH(O,O)R	66.17	9	—Ti 252
01.45	8	Zr 45	23.01:	4	PrII 4	46.82	37W	SoII 7	66.53	5	AlH(O,O)P
01.67:	4W	SiH(1,1)Q,Q	23.23	6	Sum	46.95:	2	NdII 14?	66.72	5	NdII 58
02.02	30W	Fe 42	23.73	7	SiH(1,1)P	47.38	22W	NdII 8 Fe693	66.96	10	Fe 273
02.34	3	VII 25	24.01:	2		47.68	5W	PrII(M)?—	67.10	8	AlH(O,O)Q
02.74	4	Fe 476a, 521	24.19	11	Fe 689	47.94	9	AlH(O,O)R	67.33	4	—CH(O,O)R
02.96:	7	SiH(1,1)Q	24.54	9	Fe 689 VII 25	48.23	8	Fe 482	67.56	2	V p?
03.11:	3	SiH(1,1)Q	24.99:	7		48.39	17	Fe 19p	67.80	9	Fe 482
03.54	25	Fe 19 Ti 220	26.67:	200W	Ca 2	48.59	17	Fe 19p	68.02	14	Zr 45 Co 1
03.97	13	Fe 355 Fe 850	28.54:	6	Bo(O,O)R	48.68	6	CeII 1	68.24	3	
04.23	7	VII 25	28.93:	4	Bo(O,O)R	48.90:	-1	CH(O,O)R	68.40:	2	—Co 127
04.49	3	SiH(1,1)Q	29.49:	9	Fe 416, 649	49.11	7	Ti 252	68.67	20W	V88 AlH(O,O)Q Fe649
04.71	8	VII 1 SiH(1,1)Q	29.75	17	Fe 41	49.25:	-3		68.92	3	Ti 252
04.94	12	GdII 46—	29.96	9	Co 1	49.43:	-1	CH(O,O)R	69.13	4	—AlH(O,O)P
05.06	15	BaII 1	30.24	6W	Bo(O,O)R,Q—	49.54	11	AlH(O,O)R	69.24:	0	
05.31	6		30.54	9W	Cr 132 Fe 478	50.13	24	Fe 152	69.39:	3	Ti p?
05.57	6	Fe 689 NdII 19	30.77	-1		50.24:	2?	—AlH(O,O)R?	69.53:	1	LaII 76
05.72:	1		30.99	7	N1136 LaII 83	50.50	3		69.73	5	—V(Sun)
05.94:	2W	—Ru(M)?	31.17	5	—VII 25?—	50.82	45W	Fe 42	69.91	2	Fe 690p Cr 154
06.20:	4W	SiH(1,1)Q CH(O,1)R	31.40:	-1		51.03:	3		70.13	6	Ti 251
06.70	35W	Fe 3	31.57	4	Fe 647 ZrII 99	51.31	9	AlH(O,O)R	70.30	2	V(Sun)
06.94:	0	SiH(1,1)Q	31.70	3	—CeII u?	51.62	8	Ti 162	70.64	9	NdII 12 AlH(O,O)Q
07.12	8	Fe 352	31.99	6	Sum	51.76	9	Ti 251 GdII 15	70.78:	2	CeII 21—
07.29:	2		32.41	20W	V 111 NdII 8	51.84	7	—Fe 216p?	71.16	24W	Fe 152
07.90	6		32.73	30	Fe 3	52.06:	2		71.34:	2	V 88
08.26	9W	SiH(1,1)Q Cr 249?	32.92	5	V 111 Cr 132	52.31:	33W	Co 1	71.51:	6	Fe 42
			33.16	3	FeII 27	52.42	2	NdII(M)?	71.78	45W	Fe 42
			33.32	1	Bo(O,O)R,Q	52.65	2	SiH(1,1)P	72.12:	2	Y(M)?
			33.62	19	Fe 152	52.76	5	—V?	72.40:	17	Ti 44
			34.01	29	Co 1 V 6	52.88:	-2		72.55:	8	Sum AlH(O,O)Q
						53.14	6	AlH(O,O)R SiH(1,1)P	72.78	5	NdII 11

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4200			95.77	32	Ti 44 Cr 64	18.85	23	Ca 5	41.47	5	Cr 64 TiO
74.2	3w	—GdII(M)?—	96.10	10	V120 GdII 46	18.93	8	SmII 27	41.80:	6	Sum
74.80	70w	Cr 1 Ti 44			LaII 53?	19.05:	3	Zr(M)?	42.10:	7	NdII 8
75.27	3		96.33:	0	Cr 176	19.35	1	TiO	42.21:	13	GdII 15
75.66	8	LaII 40 Fe215	96.74	13w	ZrII 98 Sm3?	19.46	6	Fe 214p	42.39	2	
75.91	3	—Cr 240			CeII 2	19.62	6	Cr 96	42.48	2	
76.07:	-1		96.91	0	CH(O,O)Q	19.86:	2		42.81	6	V 103
76.20	0		97.04	6	Cr 64				43.03	6	—Zr(M)—
76.42	10	Ti 148	97.32	2	CH(O,O)Q—				43.16	14	—Cr 64—
76.68	12w	Ti252 Fe976	97.51	1	Sun	20.48	8	AlH(O,O)P	43.28:	2	Fe 645
76.94	8	V 88	97.74	12	Cr 247 Ru 5			Fe 691?	43.46:	1	Zr(Sun?) CH(O,O)P
77.25	5				V120 PrII 7	20.73	14	ScII 15	43.71	12	Fe 517 Ti 204
77.58	4	ZrII 40	98.02	10	AlH(O,O)P	20.94	8	TiII 41	43.90:	0	
77.58:	8	AlH(O,O)Q,P			Fe520 V120	21.18	9	Cr83 GdII 46	44.00:	-2	CH(O,O)P
77.68:	6	Fe 172	98.20:	1	Fe 476a p	21.40	2	Sun Zr(M)?	44.29	13	TiII 20
78.17	1	FeII 32?—	98.68	32	Ti 44 Ni 28	21.63	10	Ti 235	44.51	29	Cr 22
78.24	9	Fe691 Ti291	99.00	16	Ca 5	21.77	7	Fe(Sun)	44.70	-2	
78.45	3		99.26	17	Fe152 Ti148	22.11	5	—GdII 15	44.89	2	Sun
78.53	3	TbII(M)?—	99.64	18	Ti43Fe416Cr96	22.50	7	LaII 25	45.07:	4	Cr 198 TiO
78.79	8	Ti 252				22.79	2	Sun Fe 215p?	45.13	3	AlH(O,O)P
78.96	7		4300			22.98	5	TiO	45.25	2	TiO
79.27	-1?		00.05	12	TiII 41	23.26	6	SmII 8	45.48	1	
79.46	3	Fe 993	00.58	25	Ti 44	23.42	16	Ti 79 Cr u	45.59	1	Zr(M)?
		CH(O,O)R	01.11	30	Ti 44	23.86	2	—Zr(M)?	45.84	9	—SmII 7—
79.67	7	SmII 27	01.60	4	—Zr(M)?—	24.33	0		46.09	9	Ti 234
		CH(O,O)R	01.93	11	TiII 41	24.60	2	Sun	46.30	1	TiO?
79.88	8	Fe351 ScII15	02.17	8w	Fe520 PrII 32	25.03	30w	ScII 15 Fe 70	46.53	14w	Ti 204 Fe 598
80.05	6	Ti 252			W 7?			Cr104 Ti235			GdI 4
		CH(O,O)R	02.54	25	Ca 5	25.42	5	Zr(M)	46.81	8	Cr 104
80.48	12	GdII 15	02.93	10w	Zr 45 Ti 79	25.77	50w	Fe 42	47.23	24	Fe 2
		AlH(O,O)Q	03.22	12	Co 1	26.36	24	Ti 43	47.51	3	PrII 30? CH(O,O)P
80.59:	6	AlH(O,O)P	03.57	12	—NdII 10—	26.74	8	—Fe 413	47.83	16	SmII 37 Zr(M)
		Fe 598	03.80:	3	CH(O,O)Q	26.99	12	Ti 43	48.02:	-1	Zr(M) CH(O,O)P
80.78	6	SmII 46	03.91:	3	CH(O,O)Q	27.10	11	Fe761 GdII 15	48.31	3	—CH(O,O)P
81.02:	5	SmII u	04.18	2	Fe 647p?—	27.24	-1		48.63	1	
81.09:	6	Mn 23	04.40:	0	CH(O,O)Q	27.52	0	TiO?	48.77:	0	Y(M)
81.37	20	Ti 44	04.52	4	Fe 414	27.76:	1		48.94	11	Fe 414 Zr(M)?
81.58	0	Fe 171p			CH(O,O)Q	27.92	12	Fe 597	49.27	0	
81.71	0	Sun	04.66	4	Zr(M)	28.14	2		49.48	0	
82.00	2	CH(O,O)Q?—	04.84	3	Sun GdII u	28.55	1	TiO Sun	49.74	6	CeII 59 TiO
82.20	5	Zr 45	05.13	6	Fe 272p	29.02	10w	SmII 15	49.96	1	
82.40	18	Fe 71	05.43	8	Fe476 SrII 3			CH(O,O)P—	50.17	2	TiO?
82.67	14w	Ti162 NdII13			Cr 96	29.42:	2	TiO	50.45	6	SmII 46
83.02	23	Ca 5	05.70	8	ScII 15 PrII 8	29.54	7	Zr(Sun) Fe70p?	50.68	0	
83.42:	1w	Fe 215p?	05.90	35	Ti 44	29.76:	-1		50.80	11	—TiII 94 V(M)
83.82	13	AlH(O,O)P,Q	06.20	18	V 5	30.01	25	V 5	51.05	22	Cr 22
		Fe 19p	06.56	3	—Fe 691p	30.24	2	TiII 94	51.27	10	—NdII 10
84.04	10	V 88 Mn 23	06.70	2	CeII 1	30.42	1	CeII(M)?	51.55	10	Fe 413
84.35	4	—Fe 417			CH(O,O)Q	30.69	15	TiII 41	51.79	38w	Cr 22 Mg 14
84.47	4	—NdII 10	06.93:	7	Ti 43			GdII 46	52.09	6	SmII u
84.69	5	Cr 96 Ni 86	07.18	18	V 5	30.95	5	Fe 597	52.42	2w	—CH(O,O)P
84.98	13	Ti 148	07.86:	50w	Fe 42 Ca 5	31.36	6	Fe ?	52.82	39w	Fe 71 V 5
85.34:	3	CeII 11?			TiII 41	31.66	10	Ni 52	53.12	3	AlH(1,1)R
85.47:	9	Fe 597 SmII 27			—CH(O,O)Q—	31.76	9	—AlH(O,O)P	53.33:	8w	V(M) AlH(1,1)R
85.79	18	Co 1	08.51:	10	Ti 79	31.94:	1	TiO	53.56:	0	TiO
86.02	30	Ti 44	08.64	6	DyII(Sun)	32.11	0	TiO	53.86	5	AlH(1,1)R—
86.43	7	Fe 414	09.04	11	Fe849 SmII 15	32.42	3	V(M)?—	54.08	8	Ti204 AlH(1,1)R
86.62	3	SmII 42	09.39	13	Fe 414	32.56	3	Cr 176	54.37:	-1	LaII 58 TiO
86.86	5	Fe(M)?	09.60	10	YII 5	32.81	29	V 5	54.57	12	ScII 14 Mg 13
86.99:	4	LaII 75 Fe976	09.81	25	V 5	32.91:	1		54.92	6	AlH(1,1)R V 105
87.06:	4	AlH(O,O)P	10.12	0	CH(O,O)Q	33.20	2	TiO ZrII 132?	55.05:	11	—Ca 37 AlH(1,1)R
87.41	24	Ti 44	10.37	9	Ti 79	33.74	11	LaII 24	55.28	6	Ti 174
87.52	11		10.87	6	—AlH(O,O)P—	35.91	8	PrII 8	55.66	2	CH(O,O)P TiO
87.71	4	Ti45p PrII(Ms)?	10.96	1	GdII 15	34.08	12	V(Sun)?	55.94	24	V 5
87.89	12	TiII 20 Ni178	11.17	2	CH(O,O)Q—			SmII 27	56.38:	3	CH(O,O)P
88.17	14	Ti 43,79 Fe273	11.42	1	CH(O,O)Q	34.38	-1		56.57	2	CH(O,O)P
88.37	2	—Cr(M)?—	11.64	10	Ti 205	34.55	0				AlH(1,1)R
88.67	4	Rh(M)?	12.06:	0	CH(O,O)Q	34.65	0	CH(O,O)P			CH(O,O)P
88.79:	3w	Ti 45p V?	12.29:	2	CH(O,O)Q Sum	34.84	17	Ti 43	56.77	6	Cr 130
89.06	29w	Ti 44 Fe 214	12.53	7	Mn 23	35.06	-2		56.91:	0	
89.34	17	Ca 5	12.86	16	TiII 41	35.26	2	Sun TiO?	57.42	4	V(M)
89.77	45w	Cr 1	13.08	2	CH(O,O)Q—	35.64	0		57.61	2	TiO
90.19:	8	TiII 41 Zr(M)	13.33:	-1		35.87	0	Fe 991	57.87	2	TiO?
90.34:	7	Fe 416	13.51:	-1	CH(O,O)Q	36.09	0	TiO	58.15	10	NdII 10
90.58	2	AlH(O,O)P	13.68:	-2	CH(O,O)Q	36.25	2	CeII 89			AlH(1,1)R
90.92	23	Ti 44 Fe 351	13.84	2	Gd 4? V(Ms)?	36.44	2		58.49	10	Fe 412
91.14	11	—Ti 45,147	14.08	21	ScII 15	36.63	-1	TiO	58.70	14	YII 5
91.30:	1	V?	14.34	15	Ti 45	37.05	29	Fe 41	58.85:	1	AlH(1,1)R
91.46	39	Fe 3,41	14.50	7	NdII 9	37.22	5		59.05:	0	TiO
91.77	1		14.78	30	Ti 43	37.56	23	Cr 22	59.23	0	TiO?
91.84	12	V 120 Ti 251	14.95:	15	TiII 41	37.79	4	—CeII 82—	59.5	-3	CH(O,O)P
92.13	8	Fe 70p SmII 32	15.09	22	Fe 71	37.91	16	TiII 20	59.63	23	Cr 22
92.29	10	Fe 70	15.42	8	AlH(O,O)P	38.23	15	Fe70 AlH(O,O)P	59.93	1	Tm 1? Cr 198?
92.68	10	Ti 79	15.61:	-1		38.46	7	Ti 204	60.13	0w	TiO CH(O,O)P
93.09:	2	CH(O,O)Q	15.82:	2	V(Ms)?	38.73	12w	NdII 68 Cr198	60.48	9	Ti 204 AlH(1,1)Q
93.22	3	V p? Mo 7?	16.02	4	—GdII 43			Fe 117p	60.75	10	—Zr 31
93.54	4	Cr 96	16.33:	0		59.12	2	Co 1p TiO	61.09	12	Co 1 AlH(1,1)Q
93.80	6w	—Sun—	16.51	-1		59.44	29	Cr 22	61.17	4	
94.14	37w	Fe 41 TiII 20	16.79	9	TiII 94	59.72	27	Cr 22	61.38	2	V(M) TiO?
94.56:	1	—W 6?	17.07:	3	—TiOa(6,1)?	40.02	3	Ti 174	61.62	2	TiO
94.78	11	Zr 45 ScII 15	17.33	9	ZrII 40	40.16	6	Cr 64—	61.94	9	Zr(M) Co 1
95.01	3	CH(O,O)Q	17.93	2	TiO ?	40.49	30w	H 1	62.02	8	SmII 45 AlH(1,1)Q
		Fe 598	18.17	-1		41.03	29w	V 5 Zr 61	62.48	4	—Sun
95.21	1	CH(O,O)Q	18.40	1	TiO	41.36	12	TiII 32	62.69	1	—Sun
								GdII 14	62.93:	2	—Cr 82

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4300			86.06	0	TiO?	99.52	8	TiII 61	33.75	9	Fe 825 TiO
63.11	6	Cr 103	86.40	3w	TiO N1 168	99.64	0	Zr(M)?	33.96	24	Ti 113, 161
63.31	3	CH(O,O)P			Tm 1?	10.01	6	Ru 5 TiO			SmII 41?
		AlH(1,1)Q	86.72:	2		10.26	7	—Cr 129	34.33	12	SmII 36 TiO
63.51	18	V 23	86.83:	8	CeII 57 TiII 104	10.52	6	Ni 88	34.57	8	V(Sun) TiO
63.81	-1		87.00	-1	CH(O,O)P	10.74	1	Ti(Sun)p?	34.96	20	Ca 4
64.00:	2	CH(O,O)P	87.19	7	V 40	11.07	15	NdII 8 TiII 115	35.15	30	Fe 2
64.17:	8	V(M)	87.37	2	Cr 84			Cr 129	35.45:	2	
64.49	1	TiO?	87.51	6	Cr 103	11.53	-1		35.56:	12	BuII 4
64.65	7	CeII 135	87.66	2	GdII 15 TiO	11.68	2	TiO	35.64:	25	Ca 4
		LaII 53	87.89	9	Fe 476	11.89	11	Mn u? TiII 61	36.13	23	V 21
65.03	5	AlH(1,1)P,Q	88.04	10	Ti 219	12.20	25w	V(Sun) Cr 22	36.34	8	Mn 22
65.5	0w	TiO	88.40	8	Fe 830	12.42	13	Fe 69p Ti 54	36.60	13	Ti 160 Ti 267
65.70	2	—V 79	88.52	4	Ti(M)	12.67	1w	TiO	36.93	10	Fe 516 Ni 86
65.90	8	Fe 415	88.62:	-1		13.01	3	Zr 61	37.28	2	TiO
66.06	1	TiO	88.80	0	—CH(O,O)P	13.15	2	TiO	37.58	3	Ni 168 TiO
66.37	12w	NdII 12 Zr 61	89.25	32	Fe 2	13.70	9	V(Sun) PrII 26	37.83	26	V 21
66.69	1w	TiO CH(O,O)P	89.58:	-1	CH(O,O)P	13.92	7	Cr 234 TiO	38.01:	5	Sr 6 TiO
67.06	5w	—AlH(1,1)R—	89.99	32	V 22	14.14	8	Zr(M) Ti(M)	38.18	8	Ti 218 PrII(M)?
67.25	1	TiO	90.25	0	TiO	14.48	8	Fe 643 ZrII 79?	38.30	7	Fe 828 GdII 44
67.58	13	Fe 414 TiII 104	90.43	6	Fe 413			V(M)?	38.65:	2	TiO
67.88:	15	Fe 41	90.62	2	Sun	14.84:	10w	—Mn 22	38.97	6	V(Sun)?
68.00	23	V 5 SmII 37	90.88:	8	SmII 15	15.14	40	Fe 41	39.23	2w	TiO
68.24	6	Cr 130	90.96	18	Fe 414 TiII 61	15.55	16	ScII 14	39.61	5	—Fe 515
		SIF(O,O)R-head	91.08:	8		15.81	0	TiO	39.73:	0	TiO
68.33	6	PrII 5	91.31:	-1	Fe(Sun)?	16.47	24	V 22 Ti 161	39.86	9	Fe 116
68.62	9	NdII u	91.74	25w	Cr 22	16.66:	6	—V(M)	40.08	-1	TiO
68.94	6	Ti 245	92.07	15	V 23	16.86	6	FeII 27—	40.34	12	Ti 159
69.26	1	TiO SIF(O,O)R	92.30	1	Cr 130 TiO	17.29	13	Ti 161	40.43:	8	Fe 829 ZrII 79
69.50	6	AlH(1,1)Q	92.59	4	Fe 973 TiO	17.40:	3	Co 150 Ti(M)?	40.52:	0	TiO
		SIF(O,O)R	92.90:	-1	TiO?	17.70	16	TiII 40	40.84	7	Fe 992
69.66	1	Ti260 CH(O,O)P	93.14	6	—V(M) TiO	18.34	14	TiII 51 Fe412	40.99	7	Fe 645
		SIF(O,O)R	93.30	8	Sun AlH(1,1)P	18.58:	0	Fe 899p?	41.26	9	Ti 160
69.76	14	Fe518 GdII 15	93.55	3	Cr 102—	18.74	4	CeII 2 TiO	41.66	30	V 21
69.86:	3	CH(O,O)P	93.86	15	V 40 Ti 244	19.02	7	GdII 15	41.95	0	Co(M)?
		SIF(O,O)R	94.04	11	TiII 51	19.29	2	TiO	42.34:	28	Fe 68
70.11:	0w	TiO SIF(O,O)R	94.49	8	Zr(M)	19.92	21	V 21	42.81	10	—Fe 69
70.39	1w	TiO? SIF(O,O)R	94.85	25w	Ti 78 Cr 130	20.22	1	—Sun	43.02	3	ZrII 88
70.85:	0	Mn 17	95.04	20	TiII 19	20.47	14	Zr 61 Ca 1?	43.19	12	Fe 350 Ti(M)?
70.95	8	ZrII 79	95.25	30w	V 22	20.60:	9	SmII 32 ScII 14	43.32	2	V(M)
71.06:	3	CH(O,O)P TiO	95.51	7w	Fe 991, 992	21.13	7	SmII 37—	43.79	19	TiII 19
71.25	27	Cr 22			Cr 129	21.31:	1	Co 150	44.21	27	V 21 Ti 218
71.58	3		95.84	12	TiII 61	21.54	27	V 22 Ti(Sun)	44.54:	10	TiII 31
71.76	0	TiO? Sun?	96.15:	0	TiO	21.75	10	Ti 218	44.67	1	CeII u?
72.18	8	—Ru 13?	96.35	0	TiO	21.93	5	TiII 93	45.00	1	TiO
		SIF(O,O)P	96.58	0	TiO	22.21	2	V(M)	45.48	26	Fe 2
72.57	10	AlH(1,1)Q	96.83	4	—CH(O,O)P?	22.43:	2		45.84	2	TiO
		SIF(O,O)P	97.26	8	Cr 129	22.52	15	Fe 350 V 79	46.21	0	TiO
72.72	1	CH(O,O)P TiO	97.51:	-3	CH(O,O)P	22.84	20	Ti 78 Fe 646	46.39	11	NdII 49
		SIF(O,O)P	97.70	3		23.19	18	V 40 Fe 412	46.84	9	Fe 828
72.98	6	Fe 473	98.03	17	—YII 5—	23.61	2	CeII 21 TiO	47.12	12	Fe 69
73.24	17	Cr 22	98.28	6	—YII 61	23.85	7	Fe 830	47.37	-1	TiO
73.53	9	Fe 214, 413	98.56:	-1	TiO			TiO(4,0)R-head	47.71	24w	Fe 68
		SmII 42	98.78	5	CeII 81 TiO	24.06	2	Cr 82 V(Ma)?	47.93:	-1	
73.82	7	V(M)	98.98	1	AlH(1,1)P	24.31	13	SmII 45 Cr129	48.14:	0w	TiO
74.16	10	Cr 104	99.20	6	CeII 81			Ti 243	48.53:	0w	TiO
74.44	15	ScII 14	99.43	5	V(M) TiO	24.56	9	V(M)	48.90	1	Zr(M) TiO
74.80:	5	TiII 83	99.68	0	Ni 196—	25.07:	3	TiO Cr 52	49.13	15	Ti 160
		DyII(Sun)Rh(M)?	99.77	15	TiII 51	25.43	23	Ca 4	49.33:	4	Ru(M)?
74.94	25w	YII 15 NdII 15, 8				25.72	9	V(M) Fe 798?	49.61	8	V 62—
75.35	9w	Cr 103 Ti 219	44.00			25.82	13	Ti 78	49.75	10	DyII(Sun)
		V 140	00.14:	3	TiO	26.01	29	V 22 Ti 161	49.93	9w	Ti 159 PrII 4
75.6	0	CH(O,O)P	00.38	16	ScII 14	26.30	0	TiO	50.28	8	Fe 476 Zr(M)
75.93	45w	Fe 2	00.68	27	V 22 Ti(M)	26.40	0	TiO	50.47	12	TiII 19
76.77	8	Fe 471, 904	00.82	8	NdII 10 Ni149?	26.66	0	TiO?	50.74	3	CeII 2 TiO
		Cr 304	01.03	1	Sun TiO	26.81	3	TiO	50.87	14	Ti 160
77.10	2	TiO CH(O,O)P	01.28	5	Fe 828	27.10	35	Ti128 Ti 78p	51.23:	0	TiO
77.32	2	Fe 990	01.48	13w	Ni 86 Fe 350	27.34	35	Fe 2	51.57	13	Mn 22 NdII 50
77.52	5	Cr 83	01.83	5	TiO GdI(M)?	27.58:	5	SmII(M)?	52.00	14	V 87 NdII 6?
77.81	8	Fe 645	02.24	2w	TiO	27.68:	1	Cr 129	52.09	2	
78.23	6	SmII 53	02.56	1	TiO	27.84	8	—TiII 61p?	52.33	-1	TiO
78.49:	0	TiO	02.70	1	TiO	28.50	22	V 21 Cr 129	52.58	0	Sun
78.78	-1		02.95	6	Zr(M) TiO	28.88:	-2		52.71:	11	SmII 26 V(M)
79.23	38w	V 22	03.14	6	Sun TiO	29.13:	10	Zr(M)—	52.99	8	Mn 22
79.77	8	Cr 130 ZrII 88	03.33	9	ZrII 79 SmII 22	29.27:	7	PrII 2, 4	53.31	21	Ti 113
80.06	3	—CH(O,O)P—	03.49	1	Cr u			CeII 19	53.70	15	Ti 160
80.33	1	—Mg 12?	03.66	7	V(M) PrII 34	29.87	26	V 22	53.78	1	
80.51	8	Cr 130 Sun	03.97	3w	TiO	29.93:	8	LaII 38	54.10	0	—TiO—
80.78	2	TiO CH(O,O)P	04.34	30	Ti78 Ti218, 219			—Ti 267?—	54.36	12	Fe 350 PrII 5?
80.96	5		04.73	38	Fe 41	30.19	7	Fe 472	54.60:	0	SmII 49?
81.11	8	Cr 64 V 23	05.04:	38	V23Fe2pTi161	30.37	12	Ti 113	54.76	31	Ca 4
81.74	0w	TiO	05.32	1w	AlH(1,1)P TiO	30.62	23	Fe 68	54.98	9	Mn 28 Fe 974
82.15	5	CeII 2	05.68	14	Ti 78	31.00	2	TiO	55.32	24	Ti 113
82.43:	0		06.09	14w	V 40 PrII 4	31.31	13	Ti218 ScII 14	55.87	25	Ca 4 Mn 28
82.74	9	Fe 798a Cr 64	06.64	37	V 22	31.45:	3	Zr 61	56.34	11	Fe 516 Zr(M)?
83.55:	60w	Fe 41	07.07	5		31.55:	0	TiO	56.60	11	Ca 4
84.18	11	V(Ma)?	07.25:	2	Fe(Sun)p	31.82	3	TiO	56.80	0	TiO
84.71	38	V 5, 22	07.65	40	V 22 TiII 51	32.09	10	TiII 51	57.01	6	Mn 28 TiO
85.00	28	Cr 22	08.19	32	V 22	32.23:	2	TiO Cr 81	57.45	33	V 21 Ti 113
85.23:	3	Fe 415	08.53	35	V 22 Fe 68	32.56	8	Fe 797 Ti(M)	57.75	8	V 101
		CH(O,O)P 18?	08.85	16w	—PrII 4—	32.88	10w	V(Sun) TiO	58.05	5	Fe 992 TiO
85.35:	3	FeII 27? Ru(M)?	09.18	15w	TiII 61 Fe 645	33.21	10	Fe 830	58.24	9	Mn 28
85.66	11	NdII 50 Ru(M)?	09.34	10	DyII(Sun) SmII(M)?	33.37:	3	Fe 412p	58.51	10	SmII 7 Cr 127
85.84	-1	CeII 57?				33.55	6	Ti 267	59.11	26	Fe 68 Ni 86
									59.33	11w	Cr 63 TiO

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4400			83.32	1	GdII 62	06.10	3	TiO V(M)	30.08:	0	TiO
59.75	29	V 21 Cr 127	83.43	1		06.30	8	Ni 133	30.33	6w	TiO
60.02	-1	TiO	83.75	5	Sun TiO	06.57	10	V(M) Ca 24	30.70	17	Cr 33 Cr 33
60.30	38	V 21	83.86	5	CaII 3 TiO	06.78	5	TiII 30 Cr288	30.90	9	-Co 150
60.76	6	Cr 63			Co 150	06.96	-1	GdII 13	31.15	27	Fe 39
61.05	4	Mn 28 V(M)	84.20	11	Fe 828	07.13	12	Zr 31-	31.62	13	Fe 555,847,992
61.12:	14w	CaII 10 Fe 471	84.49	8	Co 27	07.41	5	Ca 24			Ti 112p?
61.37	2	Fe 725 TiO	84.76	0	TiO	07.80:	6	Ca 24 TiO	31.86	0	TiO
61.66	42w	Fe 2	84.83	4	TiO	08.02	8	Ti(Sun) Zr(M)	32.10	5	Ti(Sun) TiO?
62.07	32	Ti 8 Mn 28	85.06	7	Ti 184 TiO	08.27	8	Ti(Sun) FeII 38	32.31:	-1	TiO
		Fe 471,825,902	85.30:	0		08.49	0	LaII 52?	32.58	2	TiO
62.38	13	V 87 Ni 86	85.43:	0	ZrII 79	08.70	4		32.75	2	TiO
62.74	6	Cr 127 TiO	85.66	10	Fe 830	08.86	0	TiO	32.98	4	Sun
62.99:	8	NdII 50 TiO	85.94	5		09.11	1	TiO	33.24	37w	Ti 42
63.16:	5	Fe 471p? TiO	86.30	1w	TiO Zr(M)	09.26	7	V 110 Fe514,937	33.96	19	TiII 50
63.39	10	Ti 160	86.71	2	TiO Co(M)?			TiO	34.1	3	PrII 20
63.51	11	Ti 160	86.91	11	CaII 57 TiO	09.41	7	-Ca 24	34.55	2	TiO?
63.81	0	TiO	87.22	3	TiO Y 14	09.73	7	Ti(Sun)	34.78	30	Ti 42
64.16	4	Zr(M) TiO	87.46:	4	Y 14 Cr 63	10.10	6	Ru(M)?	35.14	10	Cr 33
64.44	9	TiII 40	87.74	3	TiO	10.21	6	PrII 20-	35.57	35w	Ti 42
64.70N	12	Mn 22 Fe 472	87.81	3	PrII 3	10.38	1	TiO	35.99	43w	Ti 42 Ti 42
		Cr 127	88.06	11	-Cr 293-	10.52	2	TiO	36.50	7	Fe 896
64.92	4	Cr 127 TiO	88.24	10	Ti 184p	10.84	8w	Fe 823p TiO	36.72	-1	TiO
65.05	7	NdII 5	88.61:	1	TiO	11.16	5	Ti u	36.99:	5	TiO
65.34	5	Cr 127	88.88	13	V 86,110	11.32	6	In 1	37.16:	6	TiO
65.57	6	V(M) NdII 13			Fe 213,827	11.41	8	V(M) TiO	37.63	8	V 82 Fe 594
65.81	13	Ti 146	89.08	16	Ti 146	11.63	-1	TiO	37.83:	5	Ti(Sun)?
66.13	3	Cr 127 Fe 901	89.44	2	Cr u TiO	11.86	15	SmII 14 Cr 150			GdII(M)?
		TiO	89.74	32	Fe 2			TiO	37.90:	9	SmII 45
66.57	34	Fe 2p Fe 350	90.05:	10	Mn 22 Fe 469	12.24	8	Ca 24	38.25	4w	TiO
66.89	9	Co 150 TiO	90.18:	1		12.73	27	Ti 42	38.55	3	
67.32	8	SmII 53	90.49	6	-TiO-	12.98	1	TiO Ni 163	38.76	13	Fe 115
67.64	7	Cr127 CaII 17?	90.74	12	Fe 973,974	13.32	7	Zr(Ms)?	39.06	4	Ti u
67.84:	0	TiO	91.13	7	V 62 TiO	13.53	1	TiO Y 15	39.32	1	TiO?
67.98:	8	V 87	91.39	4	TiO FeII 37	13.66	10	TiII2 Fe 213p	39.53	1	TiO
68.12:	1	TiO	91.63	8	Mn(M) Cr 95	13.92	0	TiO	39.75	12	Cr 33 CaII 108
68.50	17	TiII 31	91.82	4	Cr 83 TiO	14.17	10	V 110 Fe 514	39.99	7	V 100
68.73	10	PrII 20 V 102	92.27	8	Cr 197 TiO	14.44	11	Sun	40.18	3	TiO
69.15	9	TiII 18	92.50	10	Ti 184	14.66	1	TiO	40.48	22	Ti 8 Cr 33
69.36	10	Fe 830	92.64	5	TiO Fe 969	14.89	-1	TiO	40.68	9	Cr 150
69.56	4	Co 150 TiO	92.91	0	TiO	15.09	10	SmII u	40.83	9	Ti 112
69.68	11	V 87	93.41	7	Fe 796-	15.36	7	FeII 37 Zr(M)	41.04	11	Cr 33
70.11	7	Mn 22	93.51	10	TiII 18			Cr 128	41.27	7	NdII 58
70.48	9	-Ni 86-	93.77	2w	TiO	15.58	10	Ti 184 V 100	41.51	4	Cr149 FeII 38
70.84	12	I 40	94.15	8	Na 15 TiO	15.81	2	Zr(M) TiO	41.75:	-1	Na 147-
71.23	14	146 CaII 8	94.56:	27	Fe 68	16.07	2	TiO	42.02	6	SmII(M)?
71.35:	5	TiO	95.00	8	Ti u	16.33	10	Nd(M)? TiO	42.19	9	Zr 49
71.68	22	Fe 2p	95.39	9	Fe 319,970	16.65	5	TiO	42.41	4	Fe 894
72.02	-1	TiO			TiII 40p	16.81:	0	TiO	42.62	11	Cr149 NdII u
72.13:	1	TiO	95.53:	4	Fe 827	17.10	9	TiO Co 150	42.74:	0	Fe 827
72.40	5	SmII(M)? TiO	95.98:	6	Fe 825 V 110	17.51	14w	Fe 472 PrII 2	43.18	3w	TiO
72.48:	5	Fe 39p TiO	96.20	33	Ti 8 Ti 146	18.02	29	Ti 42	43.75	9	Cr 100 Co 142
72.74	12	Fe 595,900	96.51:	9	PrII 4,25	18.33	10	TiII 18? TiO	43.97	15	TiII 60 SmII 32
		Mn 22	96.86	29	Cr 10 V 86	18.64	17	Cr34,100 Ti 112	44.67	32	Ti 42 Cr 33
72.98	7	SmII 26 TiO	97.04	1	TiO	19.4	3w	TiO	44.92:	0	CaII 123?
73.22	-1	TiO	97.30	0		19.60	10	SmII 49 Gd(M)?	45.12	10	TiII 30
73.52	4	-TiO-	97.40	0	V(M)	19.96	9	Ni 51	45.37	15	Cr 33 V 109
73.83	7	Sun Cr 63	97.67	11	Na 15 Ti 184	20.20	9	V(M) FeII 37	45.95	25	Cr 10
74.02	11	V 110	97.87:	4	CaII 19	20.49	3	TiO	46.42	0	AlH(O,1)R
74.53:	0	TiO	98.10:	-1	Ru(M)?	20.75	-1	TiO?	46.85	6	TiO AlH(O,1)R
74.78	20w	Ti 113,184	98.30	6	GdII 31 TiO	21.08	6	TiO	47.00	16	Fe 39
		V 101	98.52:	-1	TiO	21.28	4	GdII 44 TiO	47.28	5	Ni 146
75.17	3	Cr63p? Ti184p?	98.69	7	-Cr 81	21.98	5	Cr 173 Zr(M)			AlH(O,1)R
		TiO	98.87	10	Mn 22			TiO	47.53:	-1	
75.27	8	-Cr 95	99.12	4	Sun	22.19:	0	TiO	47.82	13	Fe 755 Ti 270
75.51	8	Ti 184	99.46	9	SmII 23	22.35	3	LaII 8,74	48.11	6	Ti 270
75.87:	0	V(M)	99.71	3w	TiO V(Ms)?	22.53	13	-EuII 4 TiO			AlH(O,1)R
76.03	19	Fe 350				22.79	30	Ti 42	48.38:	-1	
76.57	5	Ti184p? TiO	4500			23.05	12	SmII 3 CaII 2	48.76	28	Ti 42
		TiO	00.06	-1	TiO	23.39	10	Fe 829	49.24	0	TiO
77.04	4	Cr 63 TiO	00.29	13	Cr150 TiII 18p	23.53	1	TiO	49.45:	5	FeII 38
77.43	5	Y 14 TiO	00.58	5	-TiO-	23.89	10	SmII 41 TiO	49.60	15	TiII 82 V u
77.62	1	TiO	00.75	4	-TiO-	24.20	11	V 89			AlH(O,1)R
78.00	8	Sun Fe 69	00.90:	1	TiO	24.36	-1	TiO	49.75	11w	TiII 39p
78.34	7w	Co 150 TiO	01.05	6w	-Cr 81	24.69	9	TiII 60			Mn(M)?
78.66	6	-SmII u-	01.25	16	TiII 31 V 86	24.90	4	BaII 3? TiO?	50.09	2	TiO
78.81	2	GdII 15 TiO	01.77	2	Cr 81 TiO	25.16	16	Fe 826 V 110	50.42	1	
79.04	0	Fe 899p?	01.92	14w	V 62 TiO	25.82	13	Fe 319 Sn(M)?	50.74	9	Zr(M)? Sun? TiO
79.20	2	TiO	02.20	11	Mn 22	26.07	6	LaII 50 Cr 196	51.25	7	TiO Ni 256?
79.38	5	Mn(M)?	02.58	10	-Fe 796			TiO	51.62	3	TiO Fe 972
79.66	17	Ti 146	02.85	5	TiO	26.44	29w	Cr 33 Ti127,184	51.85	4	V 82
		Fe 828,848	03.19	1	TiO			Fe 471	52.28:	5	TiII 30p
80.00	7	V(M) Sun	03.42	0		26.90	15	Ca 36	52.46	35w	Ti 42
80.12:	11	Fe 515	03.58	-1	TiO?	27.33	45w	Ti 42 Cr 33,82	52.62:	7	SmII 23
80.24:	5	-Cr 197	03.74	12	Ti 184			Y 14	53.00	11	Zr 31 V 133
80.57	12	Ti 146 Ni 211	04.20:	3	TiO	27.76	7	Y 14 Fe 641	53.40	4	TiO
80.79:	1	Sun	04.43:	1	TiO	27.92	3	TiO Co 158?	54.05	37	BaII 1
81.00	4	GdII 44 TiO	04.79	9	Fe 555 TiO	28.26	1w	TiO	54.47	15	Fe 319 Ru(M)?
81.27	15	Ti 146 TiII 1	05.00	4	Ca 24	28.46	5	CaII 1			SmII(M)?
81.58	6	-Fe 827	05.32	2w	TiO Cr 151	28.60	35	Fe 68	54.79	3	Cr 173
82.20	40w	Fe 2 Fe 68	05.71	6	Ti 184 TiO	28.80	10	Rh(M)? TiO	55.06	10	-Ti 266-
82.51	-1	Zr(M)?			NdII 3	29.22	1	TiO V 95	55.49	25	Ti 42
82.70	14	Ti 113 Fe 828	05.91	4	TiOa(6,2)Ro-head	29.49	10	TiII 82	55.86	2	FeII 37
82.85	5	Cr 197 TiO			Y 14	29.60	8	V 99 Fe 987	56.12	12	Fe 410,820,974
						29.83	10	Cr 33			Cr 173

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4500			81.99: 3		TiO ALH(0,1)Q?	06.48: 10		CeII 6 SmII 1	31.86	9	TiO
56.73: 3		NdII 12	82.27 2		Zr(M)	06.75 11		Cb(M)? TiO	32.12	8	Cr 171 TiO
56.94 7		Fe 638	82.50 7		CeII 7?GdII 65	07.05: 0		ALH(0,1)P?	32.41 8		TiO
57.35 3w					TiO	07.33 15		Sr 2	32.66: 7		TiO
57.82 6		Ti 270	82.77: 0		FeII 37	07.65 13		Fe 554,969	32.82 30		Fe 39
58.11: 10		Fe 894,974	82.90 4		Fe 348	08.03 5		TiO	33.23 5		--Cr 186
		Ti 262,263	83.12 2		Sun?	08.26 0		TiO	33.54 7		TiO
		GdII 44	83.41 9		TiII 39	08.55 2		TiO	33.72 7		Fe 41C TiO
58.44 4		LaII 39	83.80 13w		V 109 FeII 38	08.84 2w		TiO	33.99 12		Zr 5
58.62 4					Cr 125?	09.13: 5		TiO NdII 3?	34.18 11		TiO
58.96: 1		TiO	84.06 8		--Cr 172	09.32: 9		TiII 39p	34.74 9		Ti(M)? Sun
59.22: 3w		TiO LaII 53	*		TiOa(3,0)Rhead?	09.62 8		V 61 TiO	34.85 7		TiO
59.57: 1		TiO	84.43 9		Ru 6	09.90 9		--TiO--	35.18 25		V 4
59.92 12		Ti 112 Ni 115			TiOa(3,0)Rb-head?	10.13 10		--TiO--	35.55 13		Fe 319p Ti 261
60.06 5		Fe 823 TiO	84.78 13w		Fe 820 Cr 125	10.58: 2		TiO	35.84 10		Fe 346
60.27: 1		CeII 8			Fe 822	10.91 10		V 39	36.07 7w		TiO
60.41 6		SmII(M)?	85.08: 0		Cr 212?	11.14: 6w		Fe 319? TiO	36.34 8		TiII 38
60.70 9		V 109	85.29: 0		TiO	11.28 22		Fe 826 Fe 17p	36.80: 14w		TiO
60.94 2		CeII 2	85.89 25		Ca 23 Ca 23	11.72 4		TiO	37.16 8		Cr 32 Ti 261
61.17 6		--Cr 34?	86.40 50		V 4	11.94 2		Cr u	37.50 11		Fe 554
61.40 6		PrII 23 TiO	86.94 13		TiOa(3,0)Rhead	12.06 4		TiO	37.79 10		Cr 32 TiO
61.72 0		TiO			Cu(M)?	12.26 5		Sun DyI(M)?	37.98 9		ZrOa(0,0)R-head
61.83 -1		ALH(0,1)R	87.08: 10		TiO Fe 795			TiO			TiO Fe 822
62.16 0		TiO	87.53 3		TiO	12.51 8		TiO NdII 3?	*		ZrOa(0,0)R
62.35 4		CeII 1	87.70 1		TiO	13.00 1		Y(M)?	38.44 8		TiO ZrOa(0,0)R
62.63 26		Ti 7	87.94 4		TiO	13.23 10		Fe 554	38.68 7		TiO ZrOa(0,0)R
62.99 0w		TiO	88.16: 3		TiO	13.36 28		Cr 21	39.01N 8		TiO ZrOa(0,0)R
63.20 8			88.44: 2		TiO	13.68 2		TiO	39.36 15		Ti 145
63.59 9		Ti 266	88.72 13		--Co 15--	13.92 9		TiO ZrII 67?	39.66 14		Ti 145
63.76 21w		TiII 50	89.00: 2		TiO	14.20 12		Fe 638	39.98 26w		Ti 145 V 39
64.19 9		Ti 112	89.32 9		--TiO--	14.55: 6w		TiO Ni 99	40.36 6		ZrOa(0,0)P?
64.54: 1		VII 56	*		TiO	14.98: 6w		TiO	40.72 20w		V 39
64.73 9		Fe 823 TiO	89.95 14		TiII 50	15.44: 6		SmII 49			ZrOa(0,0)R-head
			90.34 4w		TiO	15.69 11		SmII 22	*		ZrOa(0,0)R,P
65.35: 5		Fe 641 TiO	90.54 5		Zr(M)	16.13 29		Cr 21	41.38: 8		ZrOa(0,0)R TiO
65.54 31		Cr 21 Co 150	90.84 10w		TiO	16.57: 0w		TiO	41.67 6		ZrOa(0,0)R,P
65.85 2		CeII 21 TiO	91.12 2			16.94: 3		TiO			TiO
66.20 7		SmII 32	91.41 27		Cr 21	17.28 26		Ti 145	42.22 9		SmII 36
66.52 9		Fe 641 Cr 125	91.80 6		SmII 14	17.87: 2w		TiO?			ZrOa(0,0)P
66.84 0			92.05 9		V 95 TiO	18.23: 3w		TiO	42.39: 5		ZrOa(0,0)P TiO
66.97 4		Fe 725			TiO?	18.59: 1		TiO? Sun?	42.60 5		ZrOa(0,0)P TiO
67.38 0		TiO	92.66 28		Fe 39	18.75 12		Fe 409 V 39	42.98: 3		ZrOa(0,0)P TiO
67.62 6		NdII 49	92.83: 2		TiO?	19.27 10		Fe 821	43.19 9		Fe 38p TiO
68.30 9		TiII 60	93.31: 5		TiO	19.51 9		Cr 81 Ti 261	43.45 8		Fe 820 TiO?
68.59 1		PrII 33--	93.52 8		SmII 36	19.78 25		V 4	43.69 11		Y 4
68.79 10		Fe 554 Fe 894	93.92: 4		CeII 6	20.07: 3		ZrOa(0,0)Rhead	44.10 6w		TiO ZrOa(0,0)P
		ALH(0,1)R	94.12 35		V 4	20.40: 2w		ZrOa(0,0)R	44.43 6w		V(M) TiO
		TiO?	94.44 9		TiO Cr u?	20.76: 2		ZrOa(0,0)R	44.74 3w		ZrOa(0,0)P Zr64
69.56 10		Cr 173 Cr 173	94.66 5		TiO Co 176?	20.86: 4		TiO	45.05: 3		TiO? Ru(M)?
69.83 1		NdII 5	94.90 5		TiO Fe 638	21.02: 2		Cr 32	45.18 17w		Ti145
70.02 4		--TiO--	95.08: 4		TiO			ZrOa(0,0)R			ZrOa(0,0)P
70.40 8		V 109	95.35 13		Fe 594 SmII 45	21.31 3w		TiO			LaII 8
70.54 0		PrII(M)	95.62: 7		TiO Cr 286	21.89 14		Cr 32 Cr 32	45.42: 2		TiO?
70.92 5		Ti 266 Cr 173	95.92 3		TiO Ni 101	22.19 2		TiO	45.76 3		TiO
71.11 35w		Mg 1 Cr 125	96.06 11		Fe 820	22.44 8		Cr 233	45.96 8		V 4 ZrOa(0,0)P
71.43 4		Fe 319	96.38 5		TiO Fe 823			ZrOa(0,0)P			PrII 22?
71.71 13w		Cr 32 V 109	96.60 3			22.72 8		Cr 81 TiO	46.17 25		Cr 21
71.97 14		TiII 82	96.84: 5		--Co 177?	23.10 18		Ti 145	46.41 12		V 39 Cr 147
72.26 6		CeII 1	96.98 12		GdII 44 TiO	23.54 5w		TiO	46.65 5		SmII 26
72.58 3		Sun	97.20 3		TiO			ZrOa(0,0)P			ZrOa(0,0)P
72.78: 3		Mn(M)?--	97.37 1		TiO	23.99 3w		TiO	46.76 5		Cr 186
72.90: 4		Zr(M)?--	97.66 9		TiO	24.39 12		V 39	47.00: 2		TiO?
73.08 7		Cb(M)? TiO	97.87 8		GdII 44 TiO	24.61: 1		V 94?	47.46 14		Fe 409
73.34 0		TiO	98.13 11		Fe 554	24.92: 2		CeII 27? TiO			ZrOa(0,0)P
73.67 2		TiO	98.35 12		Fe 17p			ZrOa(0,0)P	47.57: 2		TiO?
73.95 3		TiO	98.46: 2		Cr 172	25.01 12		Fe 554	47.72 3		Fe 722p? TiO
74.22 7		Fe 554 TiO	98.76 3		TiO?	25.31: 2		Cr 171	47.99 3		TiO
74.47 2		TiO Cr 148?	98.97 3		TiO	25.47: 0		ZrO(0,0)P	48.11 10		Cr 32 SmII 1
74.74 11		Fe 115	99.20 5		Ti u	25.90 -1		Cr 244?			ZrOa(0,0)P
74.89 10		LaII 23	99.40 2		TiO?	26.06: 1		ZrOa(0,0)P	48.27: 1		TiO?
75.09 2		TiO Cr 196	99.83 6		Fe(Sun)?	26.17 29		Cr 21	48.51: 2		TiO
75.50 18		Zr 5	4600			26.45 15w		V 39	48.65: 6		Ni 98
75.78 3		Fe 593,970 TiO	00.13 12		Cr 32			TiOa(4,1)Rbhead	48.90 12w		Cr 32 V(M)
76.29 3		FeII 38 TiO	00.27 10		TiII 60p	26 75 4		Fe 410			ZrOa(0,0)P
76.45: 5		ALH(0,1)Q	00.38 5w		TiO Ni 98	*		ZrOa(0,0)P	49.41 9		Cr 32 TiO?
		Ti 262?	00.76 25		Cr 21	27.20 3		TiO Eu 1?	49.78 5		TiO ZrOa(0,0)P
77.19 28		V 4	01.06 20w		Cr 32 GdII 44	27.53 5		TiO	50.02 12		Ti 145
77.69 9		SmII 23			Fe 591	27.72: 4		Zr(M)?	50.27 3		ZrOa(0,0)R?
		ALH(0,1)Q	01.55 5		TiO	28.15 11		CeII 1 TiO	50.50 4		TiO
78.03: 0		--ALH(0,1)Q	02.02 17		Fe 39	28.43 7		Cr 186 TiO	50.73 5		TiO ZrOa(0,0)P
78.31 -1		Cr 246?	02.52: 3		Cr 210 Zr(M)?	28.77: 3w		PrII 1	51.28 25		Cr 21
78.55 12		Ca 23	02.95 29		Fe 39	28.90 12		Co 15 TiO	51.58 6		PrII 6 TiO
78.71 9		V 109	03.33 8		Fe 348p? TiO	29.17: 5		TiO	51.75: 4		TiO ZrOa(0,0)P
78.90 5		TiO	03.80: 2		TiO	29.30 12		Ti 145 TiII 38	52.19 28		Cr 21
79.01: 0			03.90: 10		Fe 410 TiO			FeII 37 Co 156	*		5W TiO
79.22: 3		V 109	04.18 10		SmII(M) TiO?	29.46 7			52.90 7		TiO
79.31: 7		Fe 319,936	04.42 4		Zr 29	29.57: 4		Sun	53.46 15		Fe 17p
79.80 5		Fe 469 TiO	04.65 4		TiO	29.92 9		TiO	53.83 8w		TiO
80.08 28		Cr 10 Co 27	04.98 15w		Ni 98 TiO	30.13 13		Fe 115	54.14 4		Sun
80.42 29		V 4 TiII 60	05.40 1		TiO	30.43 5		TiO	54.32: 2		TiO? Ru(M)?
80.57: 5		Fe 827	05.61 7		TiO	30.63: 3		TiO	54.49 19		Fe 38
81.04 1		Cr 148 GdII 44	05.87 6		TiO LaII 52?	30.87 6		TiO	54.67: 16w		Fe 554,821
81.38 23w		Y 15 Ca 23	06.15 24		V 4	31.11 0		TiO			Cr 186
81.57: 8		Fe 555 Co 150				31.43 10w		Fe 1152? TiO	55.02 2		TiO
									55.19 5w		TiO

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4600			81.49:	0	Fe(Sun)	08.02	12	Cr 186	35.59	2	TiO
55.71	13	Ti261 TiII 38p	81.91	40w	Ti 6	08.20:	5	TiO	35.85	8	Fe 1042
56.04	9	Ti 145	82.33	10	YII 12	08.67	9	TiII 49 TiO	36.04:	-1	
56.18:	5	Cr 147 TiO	82.56	4	Fe 384	08.96:	7	Fe 889 Ti 203	36.21	4	
56.49	32	Ti 6	82.75	8	TiO V(M)?	09.05:	12	Fe 821	36.78	14	Fe 554
56.95	3	FeII 43? TiO	83.26	3	TiO	09.35	0	Sc 22	36.93:	3	TiO
57.21	8	TiII 59	83.52	12	Fe 346 TiO	09.49	7	TiO Ru 14	37.33	8	Cr 145
57.41	4	TiO Co 156?	83.80	3	TiO	09.73	12	Mn 21 NdII 7	37.66	15w	Sc 14 Fe 590
57.62	11	Fe 346 Zr 64	84.01	4	TiO	10.18	39w	Ti75,203 Zr43			Co 57
58.24	7	TiO Fe 591?	84.25	4	TiO			Fe 409	38.37:	0	
58.59	6	TiO	84.50	11	Ti 203 V 94	10.57	7	V 119	38.65:	0	TiO
58.98	6	TiO	84.82:	2	TiO	10.93	3	TiO	39.10	10	Mn 21
59.12:	2	TiO	84.96	6w	TiO	11.26	0	TiO?	39.49	11	Zr 43
59.48	12w	TiO Zr(M)				11.50	5	TiO	40.19:	4	TiO Ni 99?
59.89	10w	TiO--	85.26	11	Ca 51	11.70:	7	--Sc 22	40.30	9	Fe 409 LaII 8
60.24	2	TiO	85.62	0	Zr(M)?	11.90	1	Zr 64	40.51:	4	TiO
60.44	1	Sun?	85.85	12w	Ge 3? Co(M)?	12.07	7	Fe 467 Ni 131	41.02	15	Sc 14 Fe 688
60.71	4	TiO	86.21	7	Ni 98	12.40:	3	TiO	41.52	9	Fe 346
60.93	6	TiO	86.90	12	Ti 203 V 93	12.73	4	TiO	41.64:	3	TiO?
61.28	5	TiO	87.18:	9	SmII 3 TiO	12.80:	1	LaII 38	41.93	9	ScII 5
61.56	7	TiO Fe 1207?	87.30	16	Fe 17p	13.09	5	TiO SmII 49	42.06	7	--Ti 202
61.96	14	Fe 409	87.44:	2	Fe 347	13.23	5	TiO	42.30:	3	
62.24	2	TiO	87.67:	3	Fe 347p?	13.52	3	TiO	42.58:	2	--V 128
62.52	7	LaII 8 TiO	87.80	13	Zr 43	13.79	6	TiO	42.80	11	Ti 233
62.74	4	TiII 38p	88.14	1	TiO	14.13	10	V 119 Fe 591	43.24:	0	TiO
62.94	2	TiO	88.46	15	Zr 5	14.43	12	Ni 98	43.82	15	Sc 14
63.22:	3	Fe 754	88.75	5	TiO			TiO	44.16	-1	PrII(M)?
63.54	12	Cr 186 Co 156	88.98	1	TiO	15.33	28	Ti 6	44.40	4	TiO
63.49:	3	TiO?	89.40:	10	Cr 186	15.62:	4	NdII 49	44.64	7	Fe 17p
63.88	14w	Cr 186	89.47	10	TiII 38p?	15.78	9	Ni 98 TiO	44.85	1	TiO
64.27	1	TiO				15.89	10	V 136? TiO	44.94:	1	PrII 3
64.48	1	TiO	89.60	7	TiO	16.24	4	TiO	45.14	7	Fe 67
64.74	15w	Cr 186 PrII 27	90.13	7	Fe 820	16.49	6w	TiO LaII 52	45.30	7	Cr 61
		Na 12	90.38	8	Fe 17p	16.85	4	TiO	45.72	8	SmII 7
			90.79	12	Ti 76	17.08	6	TiO	45.83	8	Fe 821,1068
65.08	6w	--TiO--	91.35	25	Ti 75 Fe 409	17.27	3w	TiO?			TiO
65.67	1	TiO	91.63	8w	TiO	17.67	13	V 119 Cr 170	46.64	4	V 113
65.89	5	Cr 233	91.72	3	TiO	18.40	13	Cr 186	47.02	2	TiO
66.16	11	Cr 99 V 94	91.99	2w	TiO	19.11	3	Zr 66	47.26	11	Ti 75
66.48	9	Cr 186	92.44	10	Ti 77p LaII 75	19.38:	2	TiO	47.67	8	Ti 253
66.72	4	FeII 37 TiO?	92.64:	8	TiO	19.52:	6	TiII 59	47.94	0	Na 11
66.97:	2	Ni 146	92.97	4	Cr 99	19.86	10	SmII 3 LaII 81	48.11	2	Sun
67.16	9	Cr 99 TiO	93.17	4	Co 156 TiO	20.46	5	TiO	48.50	6	V 113
67.58	38	Ti 6	93.64	28	Ti 6 SmII 14	20.78	1	TiO	48.72	3	LaII 65
68.11	13	Fe 554 TiO	93.94	6	Cr 99	21.01	11	Fe 1071,409 TiO	49.13	5	
68.35	11	Ti 77	94.11	3	TiO	21.28	0	V(M)?	49.32	0	--Zr(M)?
68.58	13	Na 12	94.85	13w	TiO	21.50	6	V 108 TiO	49.62	13	--Co 156
		TiOa(5,2)Ra-head	95.17	11w	TiO Cr 99	22.08	5	TiO Zn 2	49.98	5	Fe 1206--
68.92	5	TiO LaII 76	95.50:	0	TiO	22.26	3	TiO Sr 5	50.38:	3	TiO
69.24	9	V 4 Fe 821	95.92	12w	TiO	22.61	17	Ti 75	50.52:	2	TiO
69.35	16	SmII 7 Cr 186	96.38	4	TiO	22.88	3	V 108	50.70	0	SmII(M)?
69.65	12	SmII 26 TiO	96.58	-1		23.16	18	Ti 75	51.00	7	V 113
70.00	5	TiO	96.77:	0		23.65:	1	TiO	51.12:	3	TiO
			96.95	9	Ti 203	24.07:	1	TiO	51.54	8	V 94
70.48	26w	V 39 ScII 24	97.04	10	Cr 62	24.38	9	--Cr 145	51.84	8	Nall Zr(M)?
70.69:	2	TiO?	97.13	8	TiO?	24.65	5	Ti 203	52.10	4	Ni 165
71.07	5w	TiOa(5,2)Ra-head	97.39	0	TiO	25.06:	1	--CeII 153?	52.37:	0	TiO
71.40	5	TiO				25.27:	4	TiO	52.49	11w	Ni 132 TiO
71.65	5	Mn 21	98.24:	10	ScII 13 TiO	25.65	3	TiO Cr 195	52.89	0	Cr 194?
71.79:	3	TiO LaII 80	98.44:	7	Cr 186 Co 156	26.07	12w	Fe 384 Cr 99	53.18	15	Sc 5 Zr 66
72.07:	8w	PrII 21 TiO	98.78	24	Ti 75 Cr 62,146	26.39	5w	TiO	54.03	20	Mn 16 V 113
72.38	2	Sun? TiO	98.87:	8	Ti 203p	26.75:	3	TiO	54.37	3	Co 156 Ti 202
72.51	2	TiO	99.20	7	Co 27	26.97:	3	TiO	54.67:	2	TiO PrII 4
72.81	13	Fe 40p	99.35	8	Sun? V(M) TiO	27.16	7	Cr 99	54.76:	6	Cr 168 Ni 141
73.19:	21w	Fe 820 TiO	99.61	4	TiO LaII 39?				55.14	5	Cr 124
73.77:	7w	TiO				27.43	15w	Mn 21 Fe 821	55.40:	2w	TiO
74.15:	7	TiO				27.77:	2	TiO	55.76	4	TiO
74.26:	6	TiO Sun	4700			27.91	18	Co 15	56.12	11	Cr 145
74.63	14	SmII 14 Fe 40p	00.17	10	Fe 935	28.08:	3	TiO?	56.25	0	
74.85	14	Y 4	00.45	9	Fe 67p TiO	28.38	3	LaII 22 TiO	56.54	7	Ni 98
75.12	16	Ti 77	00.61	9	Cr 62	28.50	8	Fe 822 GdII 65	56.63:	2	TiO?
75.39	9	TiO	00.86	0	TiO	28.75	8	Sc 14	56.90	1w	TiO MgH(1,0)?
75.65	2	Ni 115 TiO?	01.09	9	Fe 820 Mn 21	29.00	5	Fe 1043a TiO?	57.35	5	V 113 Cr 290
75.89	3	TiO	01.34	5	Ni 101	29.22	11	Sc 14	57.54	10	V 113 Fe 634
76.37	9	TiO Sun?	01.54:	3	Ni 235 TiO?	29.53	3	V 93	57.84	4	Ru 12?
76.46	2	TiO	01.74	2	TiO	29.71	5	Fe 688 Cr 169	58.12	11	Ti 233
76.70	2	TiO	01.91:	0	Cr 170	29.81:	5	TiO	58.76	5	V 51
76.93	12	SmII 3 Ti(Sun)	02.08	4	TiO	30.05	8	Mg 10	58.92	12	Ti 41
77.25	8	TiO	02.24	5	TiO	30.38	2	V 108	59.27	12	Ti 233
77.54	9	Co 15 TiO	02.62:	2	TiO	30.69	9	--Cr 145	59.69	5	--Ti 202 Cr 124
77.91:	7	TiO	02.99	25w	Mg 11	30.98	0	Sun	60.05	2	Fe 384p
78.14:	4w	--Sun	03.35:	2	TiO	31.18	11	Ti 202			
78.48:	1	TiO	03.55	6	Sun	31.50	8	TiO FeII 43	60.97	7	Y 4
78.60	4	TiO	03.80	7	Ni 133 TiO	31.80	5	Ni 163	61.17	7	TiOa(2,0)Ra-head
78.85	11	Fe 821	04.16	5	TiO			DyII(Sun)			Cr 169
79.20	12w	Fe 688 TiO	04.43	11	SmII 1--	32.04	8	Co 15	61.53	12	Mn 21
79.67	8	TiO Ti 77p?	04.98	10	Fe 821	32.34	10	Zr 48	61.63:	7	TiOa(2,0)Rb-head
79.91:	1	TiO	05.47	8	Fe 752	32.60	0	GdII 65			TiO
80.09:	4	TiO CeII 18	06.15	11	V 94 Cr 170	33.41:	9	Ti 202	62.10:	3	
		Zn 2?	06.29:	3	TiO?	33.60	25	Fe 38	62.38	11	Mn 21
80.30	12	Fe 39 Co 15?	06.56	14	V 119 NdII 3	34.11	11	Sc 14	62.65:	7	Ni 71
80.50	10	Fe 346 Cr 186				34.40:	0	GdII 43	62.79	8	TiII 17 Zr 66
80.80	10	TiO Cr 170	07.29	13	Fe 554	34.65	6	Ti 233	63.00	5	TiO
81.05	0	TiO Ni 145p?	07.49	13	Fe 346	34.80:	1	Co 156	63.30	8	TiO
81.36	8w	TiO	07.75	2	TiO Cr 195?	35.08	2	TiO	63.60	8	TiO

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4700			88.16:	4	T10	13.56:	8	T10	41.62:	2	T10?
63.92	13	T10 N1 146	88.32:	6	T10	13.64:	3	T10	41.82:	8	T10 Fe 1070
64.34:	13	T10a(2,0)R-head	88.72	11	Zr 43 Fe 588	13.99	4	T10 Co 158			MgH(1,0)?
		Cr 231	89.06:	4	T10	14.26	12	T10 Cr 144	42.26	9	T10
64.52:	12	T10 T11 48	89.36	20	Cr 31	14.64	4	T10 N1 98	42.63:	0	AlO(0,0)R-head
64.74	10	T10	89.70:	5	Fe 753	15.02	12	Zr 44 T10	42.74	4	AlO(0,0)R
65.04	5	T10	89.83:	12	T1 41	15.14:	4	T10			Fe 1069?
65.22	7	PrII(M)? V(Ms)?	90.03	0	T10	15.70:	17	Zr 43 SmII 14	43.12	13	Fe 687 N1 50
65.49	10	Fe 40	90.31	8	Cr 31			T10	43.37:	8	Co 158?
65.60:	7		90.64	12w	T10	15.87:	5	T10			AlO(0,0)R
65.90	10	Mn 21--	90.97	4	N1 71	16.30	4	T10	43.54	0	AlO(0,0)R
66.14:	4w		91.20	6	Fe 633 T10	16.45	8	T1 40p	43.98	8	T1 217
66.35	10	T1 233 Mn 21	91.51	14	So 5 SmII 7	16.61	6	T10	44.21	8	SmII 26
66.62	11	V 113 Cr 231	91.78	11w	T10 MgH(1,0)?	16.89	0	T10?	44.53:	5	Sun AlO(0,0)R
66.84	9	T10	92.26	9	T1 40p T10?	17.27	16w	T10			MgH(1,0)?
67.12	6	T10 Co 182?	92.49	11	T1 260 Cr 168	17.48:	3	T10?	44.88:	2	AlO(0,0)R--
67.28	8	T10 Cr 231	92.60:	6	T10	17.80:	6	Fe 67 MgH(1,0)	45.10	6	T10 AlO(0,0)R
67.65:	8	T10	92.93:	7	T10 Co 158	18.03	10	T10	45.38:	3	--AlO(0,0)R
67.88:	7	T10 Cr 231	93.10	5	T10 MgH(1,0)?	18.24	6	T10	45.62	6	--Y13Fe588,888
		T10 Co 156	93.46	5	T10 N1 158	18.75	7	T10	46.03	5	T10
68.37	12	Fe 384 Fe 821	93.91	4	T10	19.02	13	T1(Sun) T10	46.30:	9	T10
68.83:	8	T10	93.98:	2	Os(M)?	19.45:	2		46.63:	1	V(Ms)? CeII 17?
68.97:	7	T10	94.26:	6	T10	19.68	15w	T10 MgH(1,0)	*		T10 AlO(0,0)P
69.4	12w	T10	94.39:	9	T10 Fe 115p	19.86	4		47.25	9	T10 Ca 50
69.83:	8	T10 T1 233	94.67	5	T10	20.41	23	T1 126			AlO(0,0)R,P
70.24:	3	T10	95.14	9	T10 MgH(1,0)?	20.72	7	T10	47.47	2	T10a(4,2)R-head
70.58:	2	T10 Cr 124?	95.53	8	T10 MgH(1,0)?	21.09:	2	T10?	47.70:	6w	SmII 53
70.92	5	T10 MgH(1,0)?	95.65	0		21.37	7	T10			AlO(0,0)R,P
71.11	14	T1 41 Co 156	95.86	4	T10	21.59	7	T10 MgH(1,0)	48.11:	5w	T10 AlO(0,0)R,P
71.52	9	T10 Cr 124	96.22	9	Ti 260 MgH(1,0)?	21.88:	0	T10?	48.42	14	
71.67	9	Fe 67	96.39	13	Co 14	22.07	2	T10	48.84	11	Fe 114 V 78?
71.93	0		96.85:	3	T10	22.50	8	T10	49.12:	10	T10 AlO(0,0)R
72.17	7	T10	96.94:	9	V 113	22.04	5	T10 MgH(1,0)	49.66	6	T10 AlO(0,0)R
72.32	12	Zr 43 T10	97.13	6	T10 NdII 60?	22.28	4	T10 YII 22?	50.14:	3	ZrOa(0,1)R-head?
72.83	20w	Fe 38,467 N1162	97.62	6	T1(Sun)p? MgH(1,0)	23.52	19	Mn 16	50.43	8	T10a(4,2)R-head
72.99:	6	T10?	97.78	3	T10	24.02	4	LaII 50 T10	50.95:	4	T10
73.19:	3	T10? MgH(1,0)?	98.00	8	T1 260 V 93	24.22	10	Zr 43 T10	51.47:	40w	V 3
73.41	6w	T10 N1 167	98.30	8	Fe 1042--	24.46	8	T10	51.96	5	T10
73.76	8	T10	98.53	6	T111 17	24.87	5	T10	52.45	8	T10
74.10	10	T10 CeII 17	98.74	11	Fe 38	25.47	14	NdII 3 T1250	52.61	8	Y 13 N1 130 T10
74.53	10	T10 Cr 124	99.11:	2	T10	25.78:	2	MgH(1,0) T10?	53.12	7	T10 AlO(0,0)R?
74.71:	6	T10	99.42	9	Fe888 NdII 2	25.99	5	T10	53.52	8	T10
74.86	7	T10 MgH(1,0)?	99.78	23	V 3 T1 242	26.50	7w	T10 PrII 20?	54.04	6	T10 AlO(0,0)P?
75.39	16w	T10 Cr 230?				26.84	6	T10 MgH(1,0)	54.34:	2	SmII 36
		MgH(1,0)?	4800			27.28:	5	T10			AlO(0,0)R?
75.68	5	T10	00.15:	5w	Fe 384 T10	27.44	30	V 3	54.76:	15w	T1 217 YII 22
76.11:	14	T10 Fe 635	00.66	11w	Fe 1042 MgH(1,0)	27.78	5	T10			T10
76.33	10	V 113 Co 158	01.02	12	--Cr 168	28.04	6	Zr 44	55.16	13	Cr 61 T10
		Fe 1206			GdII 65	28.40	6	--T10--	55.36:	6	T10 Co 14
76.51	8	V 128 T10	01.33	-1	T10 MgH(1,0)	28.62	8	Cr 31 T10			N1 130
76.97	11	T10	01.62	4	T10	29.02	6	N1 131 T10	55.64	9	Fe 687 T10
		T10	01.93	14	T1 40p T1 40p	29.33	13	Cr 31 MgH(1,0)	56.00	15	T1 231
77.79	14	T10 SmII 3	02.30:	3	T10	29.55:	4	SmII 35	56.36:	5	T10
78.06	3	MgH(1,0)?--	02.42:	8w	T10 MgH(1,0)	29.73:	8	T10 MgH(1,0)	56.66:	7	T10 AlO(0,0)P
78.26	9	T1 232	02.92	10	Fe 888, 934 T10	30.26:	3	T10?	57.22	13w	T10 Cr 61
78.56:	5	T10 Cr 124	03.32	4	T10	30.53:	4	T10?			N1 111?
78.80	6	T10	03.62	6w	T10 MgH(1,0)	30.85	8	--T10	57.84	8w	T10 Co 15
79.06	5	T10	03.86:	5	T10	31.15	3	N1 111 T10?	58.26:	14w	T10
79.37	20	So 5	04.04:	4	T10	31.46:	3	Sun?	58.97	16w	T10 NdII 3
79.73:	4		04.40	9	T10a(3,1)R-head	31.65	30	V 3	59.24:	7	T10
79.98	11	T10 Co 158			Fe 794	31.79	2		59.54:	2	SmII(M)?
80.28	8	T10 MgH(1,0)?	04.70	10w	T10a(3,1)R-head	32.04	7	Sr 4,5 T10	59.75:	20	Fe 318 Y 13 T10
80.64	6	T10 MgH(1,0)?			Cr 64	32.43	25	V 3	60.12:	2w	T10?
80.92:	6	T10	05.08	11	T111 92 T10	32.66:	7	T10 Fe 888	60.48:	3w	T10AlO(0,0)P
81.08:	6	T10 Y 14	05.41	11	T1 260	32.97	5w	T10 V 78	60.94:	13	LaII 8 Sun
81.45:	6	Co 57	05.67:	3		33.28	8	T10	61.33	95w	H 1
81.73	14	T1 41	05.88	9	Zr 43 GdII 60	33.73	4	T10	61.87	16	Cr 31 T10
81.84:	9	SmII(M)?	06.25	8	Cr 61	34.07	3	T10	62.27:	5	Sun T10
		MgH(1,0)?	06.50	6	T10	34.27	5	T10 GdII 65?	62.60	7	V(Sun)
82.04	7	T10	06.80	12	T1 40p	34.47	9	Fe 115 T10	63.03	6	T10 Sun
82.59	11	T10 Zr(M)?	07.01:	2	N1 183 T10?	34.84	3	T10	63.40	4	T10
82.98	12	T10	07.27	2	Fe 634,1098	35.22:	5	T10 MgH(1,0)?	63.70	10	Fe 687 T1 217p
83.35	18	T1 41 Mn 16			T10a(3,1)R-head	35.42:	5	T10			AlO(0,0)P?
83.49	9		07.53	10	T10 V 113	35.65:	4	T10	63.93	5	T10 N1 113
83.92	20w	T10	07.69:	6	T10 Fe 688	36.02	10w	T10 T1 241	64.20	10	T1 201
84.22:	0	MgH(1,0)--			T10			Fe 1068	64.73	38w	V 3
84.48	17	V 3	07.55:	7	T10 N1 114	*		T10?	64.98	1	T10 GdII 65?
84.60:	2		08.33:	3	T10 N1 160	36.63	6	T10	65.27:	1	T10
84.78:	3	T10?	09.04	6	T10 LaII 37	36.90	6w	Cr 144 T10	65.64	12	T111 29
84.95	13	Zr 44			Fe 633			MgH(1,0)?	66.04	8	Zr 44
85.12	3	T10? Co 186?	09.49	8	T10 Zr(Sun)	37.16	3	T10			AlO(1,1)R-head
		MgH(1,0)?	09.98:	8w	T10 Fe 793	37.38:	3		66.26	8	N1 111
85.67	3	T10	10.53	7	T10 Zn 2	37.91	7w	T10 Co 15?	66.71	9	NdII(M)?
85.84	3	T10	10.85:	7	T10	38.24	6	T10 Mn 43?	66.97	3	T10
86.08	9	T10	11.07	8	T1 158 Fe 467	38.50	8	Fe 687 T10?	67.30	0	T10 AlO(0,0)P
86.29	8	N1 50 T10	11.38	12	NdII 3 T10	39.18	11	T10 T1 217	67.55	10	Fe 58p
		MgH(1,0)?	11.71	6	T10			MgH(1,0)?	67.87	10	Co 158
86.55	9	V 113 N1 98	11.92	0	Sr 5?	39.55	12	Fe 688 T10	68.27	17	T1 231
86.85	12	Fe 467	12.28	6	T1 260	39.89	4	Y 13	68.51:	3w	T10
87.0	2		12.37:	6	T10	40.27	9	T10 Co 158	68.79:	0	T10
87.14	8	T10	12.68:	2	T10			Fe 1068?	68.98:	1	T10
87.47	2	T10	12.94	10	T1 41	40.48	6	T10	69.18	5	Ru 11 T10
87.65	7	T1 40p	13.10	7	Fe 630	40.87	30	T1 53			AlO(0,0)P
87.88	10	Fe 384 T10?	13.49:	3	Co 158	41.56:	0	T10?	69.52	5	T10 Sun
									70.10	16	T1 231

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
4800			02.02: 15w	T10 T10		38.10: 6	SmII 23 Fe 966		3		T10
70.81 10	Cr 143 N1 131		02.44 2	T10?		38.27 17	T1 289		93.24 14	MgH(O,O)R T10	
71.31 25	Fe 318		03.07 8w	Ru(M)?		38.40: 5	Ru(M)?		93.70 9	MgH(O,O)R T10	
71.66 5	Ta(Sun)? T10?		03.28 25	Cr 31 Fe 318		38.84 18	Fe 318			Fe 1111	
71.94 7	Sun		03.74 7	T1 u? T10		39.22 6	Fe 1065, 1070		94.14 25	Fe 16 T10	
72.15 18	Fe 318		04.14: 3	T10		39.70 30w	Fe 16		94.60 9	T10	
72.46 4	--Sr 4		04.34 17w	V 118 V 118		40.4	3w		95.09 11	T1 216 T10	
72.73 5	T10			V 50		40.95 9w	--T1 173		2	T10	
73.02: 2	T10		04.85: 4	V(M)?		41.30 13	T1 39		96.51 7	MgH(O,O) T10	
73.20: 4	T10 AlO(O,O)P		05.13 8	Fe986 Zr43 T10		41.56 9	T1 200		97.12 45	T1 5 T10	
73.48: 6w	N1 111 T10		05.92 1	T10?		41.90 3	N1 114			T10	
73.98: 6w	T111 114 T10		06.35 9	Sun T10		42.51 27	Cr 9		99.53 45	T1 38	
74.34 8	Sun T10		06.66: 4	Sc(M)? Sun		43.04 13	--T1 52				
	AlO(O,O)P		07.11 7	Co 14		43.46 2	T10		5000		
74.83: 4	N1 98 T10		07.36: 4	T10		43.87 8			00.01 -1	T10	
74.99: 4	T10		07.50 -1	Co(Ma)?		44.32 9	Sun T1 173		00.41 20	T10 MgH(O,O)R	
75.50 43w	V 3		07.71 10	Fe 687		44.78 4	--Sun?		01.06 19	T1 173	
75.09 8	Fe 687		08.04 8	Sun T10		45.15 4				MgH(O,O)R T10	
76.15 6	T10 Fe 631p		08.42: 5	T1 295p		45.55 10w	N1 145? Fe1113?		01.85 15	Fe 965 T10	
	Sr 4		08.65: 9	T10 Fe 115p		45.95 5w	N1 148		02.70 16	T10 Fe 687	
76.46: 5w	T10 Sr 5		09.09 20	T1 39		46.40 16	Fe 687		03.24: 10	T10 MgO?	
	AlO(O,O)P		09.38: 7	Fe 985		46.63: 4			03.60 10	T10 Ni 50 MgO?	
76.79 6	T10		09.72 7	T10		47.05 5	T10		04.23 10	T10 Cr122 MgO?	
77.36: 8w	T10 AlO(O,O)P?		10.02 10	Fe 687		47.53 1			04.78 12	T10 Mn2O? MgO?	
77.72: 10w	T10		10.33 7	Fe 1068		47.98 15	T1 39		05.20 11	T10 MgO?	
78.16 30w	Ca 35 Fe 318		10.55: 8	Fe 1068		48.17 12	T1 200		05.69 15	Fe 984	
78.53: 3w	T10 AlO(O,O)P?		11.07: 1	T10		48.66 10	SmII 49--		06.24 20	Fe 318-- MgO?	
78.91: 5	T10		11.17 7	T111 114 T10		*	Sun		06.76 3	T10	
79.20: 4	T10		11.51 6	T10		49.42 3			07.24 35	T1 38 Co 14	
79.77 7	T10 AlO(O,O)P?		11.76: 7	Fe 984		50.10 14	Fe 687			Fe 966, 1065	
80.01: 4	--Cr 167		12.02 8	N1 111 T10		50.61 7w	T10		08.17 8w	MgH(O,O)R T10	
80.24 12	Co 15 Sun?		12.40 10	Co 14		51.15 4w	T10		08.93 9	MgH u T10	
80.55 12	V 50		13.21 4w	SmII 53 T10		51.59 5w			09.63 30	T1 5	
80.91 8	T1 201		13.63 15	T1 157		52.30 13	N1 113 T10		10.23 0	T111 113	
81.57 55w	V 3		13.97 6	N1 132		52.63 7	Fe 1068, 1111		10.62 0	T10	
82.14: 17w	V 50 Fe 687 T10		14.38 10	T10			T10		10.97 1	N1 144	
82.65 11w	T10 Co 158?		14.89: 2	T10?		53.20 30	Co 14 N1 111		11.27 3	--T10--	
83.10 1	T10		15.21 12	T1 157		53.66 11	T10 Cr 166		11.72 10	--T10--	
83.34: 1	T10		15.79 4	T10		54.08N 13	T10		12.06 45	Fe 16	
83.66 14	Zr 44 YII 22		16.15: 3	T10		54.50 Bdrv	T10a(1,0)Rb-head		12.60 -2	T10	
84.02 6	T10		16.45: 2	T10		54.67 30	T10 Cr 166		12.97 4	T10 MgH u	
	T10		16.80: 2	T10		55.30 35	T10a(1,0)Rb-head		13.33 10	Cr 60 T1 173	
84.88: 8	Cr u		17.00: 1	T10		56.07 2	T10			T10	
85.04 28w	--T1 157		17.25 7	Fe 1066		56.58 5	T10a(1,0)Rb-head		13.72 -1	T111 71	
85.42 8	Fe 966		17.44: 2	T10		57.15 3	T10		14.18 40	T1 5 T1 38	
85.78 10	Cr 30 T10		17.58: -1			57.32 60w	Fe 318 Fe 318		14.82 6	MgH(O,O)R	
85.92: 8	T10 Cr 143		17.74 1	T10?			DyII(Sun) T10		15.29 4	MgH(O,O)R	
86.31 10	Fe 1066 T10		18.03 3	Fe 1070		58.30 10	T10 T1 52p		15.67 -1	T10	
86.78 8	V 50 N1 158		18.38 8	N1 177 T10		58.75 11	T10 GdII 64		16.20: 45w	T1 38--	
87.03 10	Cr 143		18.72: 2	N1 113?		59.15 13	T10 NdII 1		16.92 10	Fe(Sun)?	
87.33: 6	--T10		19.00 50	Fe 318		59.66N 15	T10 Co 14			MgH u T10	
87.70 9	Cr 31 Zr 43		19.28: 1	T10?			T10		17.55 19w	--N1 111	
88.55 12	Cr 31 Fe 1066		19.52: 1	T10		62.10 12w	T10 Sr 4		18.30 17	T10-- N1 162	
89.04 29w	Fe67,749 Fe985		19.85 12	T1 200		3	T10			FeII 42?	
	Re 1?		20.4 6	T10?		64.79 30w	T1 173 Cr 9		18.78 4	--Sun	
89.41: 3w	T10		20.56 45w	Fe 318		65.36 17	T10		19.20 10	Cr 20	
89.80: 5w	T10 Cr 61?		20.98 9w	LaII 7 Cr 143		66.00 25w	Fe 687 T10		20.01 35	T1 38	
90.09 3			21.78 17	T1 200 LaII 7		66.71 25w	T10 Co 14		20.59 3	Sun--	
90.33: 5	T10		22.26 15	Cr 143		67.38 30w	T1 5 T10		20.85 3	Fe 748	
90.76 30	Fe 318		22.84 2w	T10		68.06 3	T10		21.43 4		
91.50 30	Fe 318		23.16 7	T10		68.46 20	T1 173 T10		21.88 10	Cr 8 MgH(O,O)R	
91.88: 7	T1 201 T10		23.88 8	--FeII 42		68.80 10	Fe 687? T10			Fe 629?	
92.08 1	T10		24.29: 2			8	T10		22.37 5	MgH(O,O)R	
92.26: 2	T10		24.52 3	T10		72.97 25	T10 T1 173			Fe 965 T10	
92.48 -1			24.78 15	Fe 114		73.82 23	--T10--		22.92 25	T1 38--	
92.72 Bdrv	--T10a(5,3)Rb-head--		25.36 10w	T1 157 Fe 1065		2	T10		23.50: 8	Fe 1150 T1 199	
92.80: 8	Fe 1070 T10		25.65 16	V 50 N1 141		75.37 20	T10 T1 283			T10	
93.04 9	T1 231 Zr 43 T10		26.16 16	T1 39			Fe 586?		23.77: 3	T10? MgH u	
93.39 7	T10 Y 13		26.54: 2	T10		75.82 10	T10		24.17 6	--T10--	
93.65: 3	T10		*	T10?		76.33 19	T10 N1 49		24.83 32	T1 38	
93.90: 4	T10 T1 201		27.46 8	Fe 792		2	T10		25.58 25	Cr 20 T1 173	
94.20 8	V 118 T10		27.85 7	Fe(Sun) T10		77.66 30	T10 Fe 985		26.43 8w	N1 158 MgH u	
94.77: 4	T10		28.33 14	T1 200			T1 173		26.95 Bdrv		
95.14 6	T10a(5,3)Rb-head		28.89 9	T1 39		78.18 5	T1 173		27.14 12	Fe 1065 Fe 883--	
95.56 4	T10		29.22 2	T10		78.67 32	T10 Fe 966		27.68 12	--T10--	
95.88: 6w	T10 T10		29.65 3	T10		79.17 2	T10		28.15 30w	Fe 791 Cr 122	
96.41 5	Fe 984 T10		30.03: 4	Fe 631p		79.60 8	T10 Fe 883			T10	
96.8 3	T10?		30.28 10	Fe 985 Cr 259		80.10 25w	N1 112 T10		28.86 13	--MgH(O,O)R	
97.0 4w	T10?		30.50 2	T10?		2	T10		29.48 12w	Fe 718 MgH(O,O)R	
97.44 8w	T10		30.71 8	T10		81.78 35	T1 38 T10			T10	
97.81: 3	T10		31.18: 2	T10?		82.39 10	T10 Fe 1067		29.92 8	--Mn 20 T10	
98.15 6	T10		31.48 2	T10?		82.96 20w	T10 NaII Fe1067		31.09 20	Fe 746, 883	
98.43 5	T10		32.06: N13	V 50--		83.83 8	Fe 1066			ScII 23	
99.00 5w	T10		32.23: 5	Sun?		84.21 12	T10 N1 143		31.94 10	Fe 1150 T10	
99.49 9	Co 92 T10?		32.52 2	T10?		84.73 10	T10		32.35: 2	T10	
99.90 20	T1 157 LaII 7		32.84: 4			85.39 25	T10 Fe 984		32.77 4	N1 207	
	BaII 3		33.29 13	Fe1065 SmII(M)?		86.02 5	MgH(O,O)R T10			T10	
4900			34.08 40w	BaII 1 Fe 1068		86.65 20	T10 MgH(O,O)R		34.83 3	MgH u	
00.08: 12	YII 22		34.59 5	T10		87.22 7	T10		35.27 6w	Fe 885 N1 143	
00.2 3	T10		34.95 5	--T10--		87.86 25	T10 Co 14		35.92 28	T1 110 N1 145	
00.59 8	V 118 T1 295		35.37 4w	T10		88.47 3	T10		36.48 20	T1 110	
00.96 8	N1 98 T10		35.83 7	N1 177 T10		89.10 24	T10 T1 173		36.94 3	Fe 465 MgH u?	
01.37: 7	T10		36.33 12	Cr 166		89.54 3	MgH(O,O)u T10		37.38 3		
			*			89.92 6	MgH(O,O)u T10		37.87 5w	T10	
			37.32: 8	N1 114		90.42 20	T10				
			37.73 20	T1 39		91.08 35	T1 38				

TABLE 5-- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
5000			87.05	28	Ti 109 Sc 13	32.87	9	MgH u Ti 230	68.88	60	Fe 1 TiO
38.45	29	Ti 110 Ni 166	87.46	4	YII 20	33.28	15	MgH u Zr 27	3		TiO
39.20	20	—Fe 687	88.13	2	Fe 1066p?			Co 180	70.69	10	TiO MgH(0,0)Q
40.00	35	Ti 5	88.65	18w	MgH(0,0)R MgH u	33.68	21	Fe 1092 MgH u	71.09	10	TiO MgH(0,0)Q,R
40.63N	13	Ti 38	89.13	3	MgH(0,0)R 20	34.25	10	MgH(0,0)Q	71.60	30	Fe 36 TiO
41.10	20	Fe 16 Ni 158	89.35	9		34.53	18	MgH(0,0)Q	*		TiO
41.74	23	Fe 36 Ca 34	90.36	3	MgH(0,0)Q			Cb(M)?	72.69	32	Mg 2 TiO
42.17	8	Ni 131	90.78	18	Fe 1090	35.12	13	MgH(0,0)R	73.83	40	Ti 4 TiO
42.78	8w	MgH(0,0)R			MgH(0,0)Q			FrII 37	74.48	15	TiO(0,0)P-head
			91.28	3	Co 14	36.08	12	MgH u Fe 1036	75.16	15w	TiO MgH(0,0)Q
43.35	3	MgH(0,0)R	91.85	15	Cr 20	36.44	12	MgH u	75.92	20w	TiO Co 92
43.66	30	Ti 38	92.76	10	NdII 48?	37.04	20	Ni 48 MgH u	5		TiO
44.27	27	Fe 318	93.49	12	Cr 20 MgH u	37.43	11	Fe 1090	78.63	16	TiO MgH(0,0)Q
44.86	2		94.22	3w	Ni 164 MgH u	37.93	0	Cr 207	79.02	3	TiO MgH(0,0)Q
45.49	30	Ti 38—	94.88	11	Co92 MgH(0,0)R	38.35	13	MgH(0,0)Q	79.46	15w	TiO
45.99	0		95.28	12	—MgH(0,0)R			V 123	80.34	15w	TiO MgH(0,0)Q
46.48	15	Ti(Sun) Zr 62	95.78	11	MgH(0,0)Q	38.75	16	Cr 19 MgH(0,0)Q	81.09	20	TiO
47.18	13w	MgH u—	96.20	11	MgH(0,0)Q	39.28	30	Fe 383 Ni 129	81.41	15	TiO MgH(0,0)Q,P
47.74	6	Ti(M)	96.88	20w	Ni 111 Sc 13			MgH u	81.95	20	TiO MgH(0,0)Q
48.22	5	Ti 199			Fe 1092	39.60	30	MgH(0,0)R Fe383	82.28	16	TiO MgH(0,0)Q
48.73	16	Cr 20	97.54	5	Sun			Cr 207	82.78	15	TiO
49.17	0	MgH u	98.70	25	Fe 66	40.24	15	MgH(0,0)R	83.18	8	TiO MgH(0,0)Q,R
49.80	32	Fe 114	99.24	23	Sc 13	40.99	3	Sun?	83.68	60	Mg 2 TiO
50.36	3		99.82	5	—Ni 161	41.14	14	MgH u	84.60	16	TiO Cr 201?
50.83	13	—TiO—				41.74	20	Fe 114	84.92	18	TiO Sun
51.72	40	Fe 16 Cr 8	5100			42.40	20w	MgH(0,0)Q			MgH(0,0)Q
52.97	20	Ti 199	00.20	3	SmII(M)?			Fe 1090,1092	85.73	20w	TiO MgH(0,0)Q
53.49	9	—MgH u	00.54	3	MgH u			Cr 60	86.32	5	Ti 183
53.99	9	—Ti 171,294	01.03	18	MgH(0,0)R,Q	42.94	40	Fe 16			MgH(0,0)Q TiO
54.49	12	—Fe 884			Co 13	43.66	12	MgH u Fe 65p?	86.65	16	TiO Ni 205?
55.03	4		01.48	15	MgH(0,0)Q,R	44.05	10	MgH u	87.11	4	TiO MgH(0,0)Q
55.42	10	MgH(0,0)Q TiO	02.33	12w	MgH u MgH u	44.65	10	MgH(0,0)R Cr60	87.62	20	TiO
55.87	5	MgH(0,0)Q			Fe(M)?	45.10	20	Fe 66 MgH(0,0)R	88.05	8	MgH(0,0)P TiO
56.36	5	MgH(0,0)R	03.05	20	Ni 49 SmII(M)?	45.45	22	Ti 109	88.72	30	TiO Ca 49
56.78	15	—MgH(0,0)R	04.02	17	Fe 465 MgH u	45.80	1	MgH u	89.15	4	TiO
57.39	12	—Fe 1067,1150	04.56	16	MgH u SmII(M)?	46.17	10	MgH(0,0)Q	89.68	20	TiO
57.95	4	Fe 967	05.53	28	Cu 2	46.54	8	MgH(0,0)Q Ni162	90.11	8	TiO MgH u
58.44	10w	—Fe 884	06.20	3	MgH(0,0)Q	46.87	5	MgH u Co 170	90.71	18	TiO MgH?
59.09	13w	—MgH u TiO?	06.70	6w	MgH(0,0)Q	47.52	35	Ti 4			TiO MgH(0,0)P
59.65	2	MgH u			MgH(0,0)R	48.22	16	MgH u Fe 1095	91.49	14	Fe383 NdII 45
60.11	35	Fe 1,1095	07.54	40	Fe 16 Fe 36	48.76	15	V 123 Ni 158	91.90	15	TiO Cr 201?
60.91	7				Cr 19			MgH u	92.38	16	Fe 383
61.43	10	MgH(0,0)Q	08.20	5	—MgH u	49.47	11	MgH(0,0)R	93.01	45	Ti 4 TiO V 125
61.82	11	MgH(0,0)Q	08.67	3	MgH u	49.80	9	MgH(0,0)Q Co39	93.97	18	TiO Ti 183
62.16	15	Ti 199	09.03	14	MgH u Cr 60	50.14	12	MgH(0,0)R	94.23	18	TiO
62.66	1		09.56	20	Ti 109 Fe 1089			MgH(0,0)Q	94.92	30	Fe 36 TiO
63.02	4	MgH(0,0)R			MgH u	50.85	30	Fe 16	95.42	25	TiO Fe 1092
63.61	10	MgH(0,0)R—	10.44	40	Fe 1	51.12	0	MgH u	95.96	6	MgH(0,0)P
63.84	10	—TiO—	11.26	9	MgH(0,0)Q	51.88	35	Fe 16 Cr 19			Fe 1091
64.67	38	Ti 5	11.54	10	MgH(0,0)Q	52.22	35	Ti 4	96.55	20	Cr 207 TiO
65.22	10	Fe 883 MgH u?	12.55	15	Cr 9 MgH(0,0)R	52.66	4	MgH u			MgH(0,0)P
65.95	20	Ti 110 Cr 60	13.06	8	MgH(0,0)R Cr 60	53.31	17	MgH(0,0)Q Cu 7	97.29	5	Mn 32 TiO
66.36	-1		13.44	18	Ti 109	53.74	10	MgH(0,0)Q	97.80	23	TiO
66.95	15w	MgH u Fe 1092	14.04	15	—MgH u C ₂ (0,0)R	54.15	15	MgH(0,0)R	98.29	3	MgH(0,0)P
67.67	10	MgH(0,0)Q Cr 60	14.60	13	LaII 36 MgH u			MgH u	98.71	25	Fe 66 Fe 743
68.30	14	Cr 20 Ti 294	15.33	10	Ni 177 Zr(M)	54.67	25w	MgH(0,0)R	99.03	24	TiO
68.80	14	Fe 383	15.97	13w	MgH(0,0)Q			MgH u	99.67	3	MgH u
69.46	17w	Ti 199			MgH u Fe 789	55.09	9	SmII(M)? MgH u			
			16.43	15	MgH(0,0)Q			Ni 206	5200		
70.19	23	So13 MgH(0,0)R	16.69	13	MgH u SmII 56	55.44	-2	MgH u	00.30	30w	TiO Cr 201?
70.90	6		17.04	5		55.81	11	Ni 210 MgH u	01.09	10	Ti 183 TiO
71.44	18	Ti 110	17.87	7	Mn 32?	56.37	3	MgH u	01.61	21	TiO MgH(0,0)P
72.17	9	Fe 1089	18.33	12	MgH(0,0)R	56.69	18	MgH(0,0)Q	02.34	23	Fe 66 TiO
			18.86	20w	MgH(0,0)R			LaII 7	02.92	23w	TiO MgH(0,0)P
72.87	25	Cr 8 Fe 1095			YII 20?	57.01	14	MgH(0,0)Q	03.54	4	MgH(0,0)P TiO
			19.32	5	MgH u	57.60	7	MgH u	04.18	20	TiO MgH(0,0)P
73.37	3	MgH(0,0)Q	19.68	0		57.99	0	Ni 111	04.51	55w	Fe 1 Cr 7
73.97	20w	Zr(M) MgH(0,0)Q	20.04	7	MgH u	58.27	0	MgH u Sun	05.12	4	MgH(0,0)P TiO
74.70	16	—Fe 1094	20.38	20	Ti 288 MgH u	58.65	10	MgH(0,0)R	05.62	10	TiO MgH(0,0)P
75.25	3	—CeII 14 MgH u?	20.87	8	MgH(0,0)Q	59.05	9	Fe 1091	05.96	45w	Cr 7 MgH(0,0)P
75.81N	15	—So 13—	21.30	4	MgH(0,0)Q—	59.24	11	MgH(0,0)R VI23	*		MgH(0,0)P
			21.59	6	Ni 177	59.75	14	MgH(0,0)Q	07.16	12	TiO
			22.13	20	Ca 19 Ti 230	60.29	23w	MgH(0,0)Q—	07.80	10	MgH(0,0)P TiO
76.55	15	—MgH(0,0)R—	22.83	4	MgH u			C ₂ (0,0)P	08.39	55w	Cr 7 Fe 553
77.21	25w	MgH u MgH u	23.67	60w	Fe 16 Cr 20	60.96	4	Zr(M)? MgH u?			MgH(0,0)P TiO
78.14	4	—Zr 82?	24.49	10w	MgH(0,0)R			C ₂ (0,0)P	09.16	20w	MgH(0,0)P
78.95	14	Fe 1092 MgH u			Fe 585p	61.71	9	Cr 60 C ₂ (0,0)P	09.56	3	MgH(0,0)P TiO
79.21	14	Fe 66 MgH(0,0)Q	25.12	13	Fe 1090 Ni 160	62.28	13	Fe 1089 C ₂ (0,0)P	09.78	6	MgH(0,0)P TiO
79.50	40	Fe 16 Ni 60	25.54	9	MgH(0,0)Q	62.75	11	MgH(0,0)Q	10.39	40w	Ti 4 MgH(0,0)P
					MgH(0,0)Q			C ₂ (0,0)P	10.90	4	MgH(0,0)P-head
80.54	12	Ni 143 MgH u			GdII 99	63.11	11	MgH(0,0)Q	11.25	8	Ti 37p
81.06	6	Ni 194 MgH u	26.14	9	Co 170 Fe 1089			MgH(0,0)R	2		TiO
81.58	16	Sc 13	26.91	9	MgH u	63.65	10	LaII 7 MgH(0,0)R	12.30	20	Ti 215 TiO
82.13	5	—MgH u	27.36	35	Fe 16 Ti 230			C ₂ (0,0)R	12.90	20w	Co 170 TiO
82.47	10	MgH(0,0)R Ni130	27.72	30	Fe 1p	64.63	10w	C ₂ (0,0)P Fe1166			Ti 215
82.93	4	MgH(0,0)R	28.07	0	Ni 113	65.17	6	Co 39	13.28	2	TiO
83.36	24	Fe 16	28.52	13	V 123			C ₂ (0,0)P-head	13.78	4w	TiO
83.70	12	Sc 13	29.32N	30w	MgH(0,0)R	65.50	12	Fe 1089	14.45	10	TiO Cr 189
84.12	8	Ni 162—			Ni 159 Fe 965			MgH(0,0)Q	15.25	16	Fe 553 TiO
84.68	10	Sun MgH(0,0)Q	29.93	15	MgH(0,0)R	65.92	3	MgH(0,0)Q			Cr 206
					MgH(0,0)Q	66.34	35	Fe 1	15.88	3	TiO
85.43	40	Ti 109 Sc 13	30.52	20w	MgH(0,0)Q Ni177	66.71	BdRv	TiOa(0,0)P-head	16.29	28	Fe 36
					NdII 75	66.82	20	TiO	16.91	4w	TiO Ni 113?
86.26	0	Sun	31.47	23	Fe 66	67.50	60w	Mg 2 Fe 37 TiO	17.42	13	Fe 553 TiO
86.70	8	MgH u	32.25	14	MgH u			TiO MgH(0,0)Q	*		Fe 880 TiO

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
5200			57.50	12	T10 T1(M)?	10.74	12w	—T10—	57.12	1	
18.11	8	—Cu 7	58.06	0	T10(O,O)R?	11.42	8	Zr 27	57.72	2	T10?
18.47	4	T10	58.73	7	T10(O,O)P	*		T10?	58.10	1	T10
19.04	8	T10 FrII 57?	59.61	15	—T10(O,O)P—	12.83	3	Cr 225	58.75	BdRv	TiOa(4,4)R-head?
		Co 170?	60.34	9	Ca 22	13.24	10	T1 74	58.92	6	T10 Co(M)?
19.32:	6w	—T10 Gd 6?	60.84	5	T10(O,O)P Mn32	13.50	2	T10(O,O)P	59.34	5	T10
19.72	25	T1 4	61.69	17	Ca 22	13.72	1	T1II 81?	60.06	OW	T10
20.16	6	Cu 7 T10 N1114	62.21	18	Ca 22	14.19:	5	T10	60.85:	OW	T10
20.68	6	T10	63.06	2	T10(O,O)R?	14.45:	2	T10	61.47	3	NdII 74
20.98	11	T10 Cr 201	63.30	20	Fe 553 T1 183	15.03	7	Fe 1147 T10?	61.65	10	T1 35 Fe 1143
21.42	10	Fe 1	64.16	35	Cr 18 Ca 22	15.60	1		62.70	8w	Zr 27 Co 198?
21.69	12	T10 Cr 193	64.76	0	—FeII 48	15.98	3	T10?			FeII 48
22.37	13	T10 Cr 206?	65.19	3	T10?	16.60	12w	FeII 49 FeII 48	63.40	2w	T10
22.64	13	Cr 59 T10	65.66	29	Ca 22 Cr 18			T10	63.99:	2	
		T1 183			Co 38	17.15	4	T10 Fe 584	64.34	2	T10
23.22	12	Fe 880 T10	65.93	10	T1 156	17.74	7	T10(O,O)P	64.89	9	Fe 1146
23.62	8	T1 183	66.54	22	Fe 383 Co 83	18.30	3	SoII 22	65.43	9	Fe 786
24.16:	20w	T1 37p T1 183			T1 36p	18.68:	8	Cr 225 Ob(M)?	66.02	0	T10?
		T10 Cr 201			T10	19.37	3	T10	66.66	10	T1 35
24.45:	10	T10 Cr 59,193	68.54	7	Co 172 T10	19.84	10	NdII 75	67.50	9	Fe 1146
24.96	12	Zr 27 T1 183	69.04	3	T10?	20.01	3	Fe 877	68.02:	-1	
		Cr 201 T10	69.53	38	Fe 15	20.62	6	T10(O,O)P	68.43:	-1	
25.55	33	Fe 1 T10	70.35	26	Ca 22 Fe 37	21.15	8	Fe 1165 T10?	69.01:	-1	
26.08	6	T10 Fe 716	71.53	4	T10?	22.03	15	Fe 112	69.57:	15	Co 39 T1 u
26.84:	30w	Fe 383 Cr 193	71.98	0	Cr 225	22.74	3	FrII 35	70.00	10	Fe 1146 T10?
		Cr 59 T10	72.29	6	T10	23.54	0	T10(O,O)P	70.48	2	T10?
27.23	35	Fe 37 T10	73.17	10	Fe 553	23.95	20	T1 36	71.50	45	Fe 15
27.80	2	Cr 58 T10	73.37	12	Fe 114	24.11:	25	Fe 553	72.11	-1	
28.42	16	Fe 1091 T10	74.20	1	CeII 15	25.05	0	T10(O,O)P?	72.64	4	T10
29.14	5	T10	74.61	2	T10(O,O)P	25.62	2	FeII 49—	73.72	7	Fe 1166—
29.59	5	T10	75.23	11	Cr 94 Sun	26.14	5	Fe 407,785	74.40	0	T10?
29.84	11	Fe 553,1090	75.72	10	Cr 94 T10	26.67	3	T10(O,O)P	74.84	0	T10?
30.19	13	Co 39 Cr 58 T10	76.11N	11	Cr 94—			Fe 1147?	75.30	4	Sc 19
30.56	0		76.92	3	T10	27.15:	0		75.75	0	
30.95	11	T1 215 T10	77.40	4	Zr 27	27.98	40	Fe 15	76.06	0	
31.41	13	Fe 787 T10	78.09	6	T10	28.56	40w	Fe 37 Cr 94	76.60	20	T1 3p
31.95	13	—T10	78.70	4		29.15	7	Cr 94	77.03	-2	LaII 95?
32.91	26	Fe 383	79.44	8	T10	29.84	8w	Cr 94 Fe 1028	77.64	5	Mn 42
33.27	12	—T10	80.41	20w	Fe 880 T10	30.41	4	V(Ms) T10?	78.24	0	T10?
33.77	18	T1 37 T10	81.80	13	Fe 383	31.00	0	T10(O,O)P?	78.83	-1	Fe(M)?
34.14	15	V 131 NdII 74	82.39	12	T1 74	31.43	8	Co 39	79.55	6	Fe 928 T10?
34.61	10	T10 FeII 49?	83.02	0	Gd 6?	32.04	-1	T10	80.18	0	
35.31	16	T10	83.55	16w	T1 156 Fe 553	32.60	5	—FeI031 CoI70?	81.12	17	Co 56 T1II 69
35.44	15	Fe 210,1031 T10	84.40	9	T1 74 Fe 842	32.88	35	Fe 36			FrII 37
36.16	1	Fe 1034	85.14	2	Sun	33.70	4	Sun	81.82	0	Co 196
36.46	4	T10	85.66	7	T10 Cr 192	34.15	3	SoII 30 T10?	82.34	0	Zr(M)
37.00	8	T10	86.14:	0	T10?	34.82	5	Co 191 T10	82.90	1	—T1 155p?
37.42	15	T10 Cr 206	87.12	3w	T10	35.40	2	T10?	83.38	10	Fe 1146 V(M)
37.86	0		87.65:	2		36.04	6	T10 Co 191	83.74	0	T10
38.55	25	T1 37,183 T10	88.00:	1	T10(O,O)P	36.50	3		84.17:	0	
39.95	6	Cr 59 T10	88.48	3	Fe 929	36.77	5	T1II 69	84.61	10	T1 35
39.73	4	T10	88.82	0	T1(Sun)	37.15	0	T10(O,O)P	85.13	7	Zr 26 V(Sun)
39.84	20	SoII 26 T1 37	89.28	15	T1 36	37.42:	1	T10?	85.85	2	
40.37:	BdRv	TiOa(1,1)R-head	90.48	5		37.72	2	T10(O,O)P	86.32	2	Fe 1064
40.53	15	T10(1,1)R	90.93	7	LaII 6—			FeII 48	86.64	0	Zr(M)
		Cr 237?	91.48	0		38.32	12	T1 35	86.96	3	Fe 875 Cr 191
40.86	15	V 151 T10(O,O)P	91.98	5	T10(O,O)P?	38.58	2	V(M) T10(O,O)P	87.54	3	Fe 1031 Cr 191
		T10(1,1)R			FrII 24	39.32	4	Sun	88.35	4	Ni 70
41.18	3	T10(O,O)P	92.62	10w	FrII 37 T10	39.61:	-1	Co 199?—	88.47:	2	Mn 36
41.53	6	Cr 59 T10(O,O)R	93.13	8	NdII 75 T10	39.96	12	Fe 553	89.17	10	T1 35
42.10	3	—T10(O,O)P	93.34:	0	Cr 192 T10	40.40	0	Cr 225	89.55	3	FeII45 DyII(Sun)
42.47	9	Fe 843	93.91	8	—Fe 1031	40.66	3	T1 36	89.99	5	T1 155
42.82	3	T10(O,O)P	94.56	15w	Fe 875 Zr(M)?	41.01	40	Fe37 Mn4 Sc19	90.48	8	Cr 191—
43.18	3	T10(O,O)P			T10	42.12	0				
43.36	3	Cr 201	95.29	4	Fe 1146 T10	42.60	-2		91.07	5	—T1 155p?—
43.73	10	—Fe 1089	95.78	15	T1 74	42.71:	0	Co 190	91.50	10w	Fe1062 Cr191
44.18	5	T10(O,O)P	96.19	-2		42.92	8	Sc 4	92.11	4	Sc 19 T10
44.53	0		96.70	25	Cr 18	43.40	3	Co 190	92.77	-2	
44.96:	6w	T10(O,O)P	97.32	20	Cr 94 T1 156	43.82	-1		93.23	20	Fe 553 V(M)—
45.70	2	T10(O,O)R	98.22	50w	Cr 18	44.14	8	Cb(M)? T10	93.85	0	
46.20	10	T1 282	98.82	0	Fe 875	44.69	5	T10 Cr 225	94.22	-1	T1(Sun?)p
		T10(O,O)P—	99.51	-2	Zr(M)?	45.19	-2		94.68	30	Mn 1
46.51	15	T1 37	99.95	15	T1 74	45.80	40	Cr 18	95.28	-1	T10?
47.04	35	Fe 1 T1 183				46.48	-1	T10(O,O)P?	95.55	0	
47.60	25	Cr 18	5300			46.93	-1	—T10(O,O)P?	95.93:	-1	Zr(M)?—
47.85	4	Co39 T10(O,O)R	00.74	22	Cr 18	47.39	-1		96.21	0	—T1II 80—
48.32	10	T1 37,156	01.03	10	Co 39	48.31	35	Cr 18	96.61	20	T1 3
		T10(O,O)P	01.48	0		48.87	0		97.16	40	Fe 15 T1 155
48.97	17	—Fe 1166?	01.96	10	So 4 LaII 56	49.39	20	So 17 Ca 33	97.66	0	Fe 841—
49.53	8	NdII 75	02.32	14	Fe 553	49.67	12	So 4 Fe 1163	98.30	4	Fe 1145
50.19	30	Fe 1	02.86	-1		50.29	3	T10	98.88	1	
50.64	15	Fe 66	03.27	5	T10?	50.70	3	—Cb(M)?	99.43	5	Mn 42 So(Ms)?
50.89:	14	T1 37	03.48	4	—LaII 36	51.08	3	T1 300			
51.50	12	T1 37	04.10	10w	—Cr 225	51.62	0		5400		
52.11	55	T1 4	05.16	6w	T10	52.00	3	Co 172	00.14:	0	
53.09	12	T10(O,O)P	05.95	6	T10	52.43	0	FrII(M)?	00.58	12	Cr191 FeII45
53.49	12	T10?	06.43	9	—T10—	52.89	0		01.38	11	T1 35—
54.06	1	T10?	07.36	20	Fe 36	53.39	15	Fe 1062 Ni 70	01.93	4	V 130,139
54.98	50	Fe 1	07.82	4w	T10a(3,3)R-head			Co 198?	02.10	2	T10?
54.40	15w	Mn32 NdII 43	08.46	8w		54.10	0		02.72	8	So(Ms)
55.79	15	T1 183	09.28	5	Ru 10 T10	54.56	1		03.28	OW	T10?
56.25	0		09.92	3		55.02	-1		03.85	6	Fe 1029
56.63	10	T10(O,O)P—	10.56:	BdRv		55.72	2	Sc 19	04.09:	10	T1259 FeII65
57.03	-1	Cr 205				56.08	7	Sc 17	04.93	2	—Cr 191
						56.61	-1	T10?	05.35:	0	Fe 1162p?

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
5400			51.64	0	T10	00.66	12W	T10a(1,2)R	48.44	5	NdII 73? T10
05.80	40	Fe 15	52.00	5	T1II 109 Fe870			T10a(0,1)P	48.64	5	T10
06.36	0	Sun?	3		T10	01.48N	35	Fe 15	49.17	1	T10?
06.80	2	Sun	53.72	8	T1 108 T10	3		T10	49.58	2	T10? Sc 15
07.44	25	Mn 4	3		T10	03.88	10	T1 287 T10	50.05	5	Fe 926 T10
08.11	2	Co 112 T10?	55.65	35W	Fe 15 Fe 1145	04.34	0	T10?	50.57	3	--Hf 1?
09.00	20	T1 3 Fe 1147	55.93:	15	T10 NdII 83	04.92	8	V(Sun) T10	51.05	3	T10 T1(M)?
09.68:	20	T1 155			T1(Ms)p?	05.47	4	T10	51.26	3	T10
09.80:	25	Cr 18	56.57	10W	Fe 817 T10	05.91	12	Mn 4	51.81	4	T10
10.46	0				ZrO(0,0)R-head?	06.75	38W	Fe 15	52.23	4	T10 ScII 25?
10.89	4	Fe 1165	57.05:	1	T10	07.24	3	T10	52.84	-1	T10?
11.45:	0	SnII(M)?--	57.41	15	Mn 4 T10	07.73	6	V 129	53.27	4	T10
11.93	0	T10?	6		T10	08.42	7	Sun T10	53.65	12	Fe 1161 Ni 69
12.38	1		60.53	25	T1 3	09.07	6	T10--	54.32:	-1	
12.76	1	--Fe 1162p?	60.96	10	T10 Fe 464	09.51	2	T10	54.80	10W	Fe 1183 T10
13.20	0		61.66	10	T10 Fe 1145	09.93	6	YII 19 Ni 190	55.81	8	T10
13.66	1	Mn 42	62.10:	-1	T10	10.20:	-1	T10	56.25	6	
14.08	0	FeII 48	62.48	10	T10 Ni 192	10.74	8	Ru(M)? T10	56.57:	2	Yb 1?
14.2	BdV?		62.96	4	Fe 1163	11.28	4	T10 V(Sun)--	57.08	1	Al 6
14.65	1		63.34	12	T10 Fe 1163	11.81	6	T1 108,275 T10	57.43	8	V 1
15.22	11	Fe 1165 V 130	63.96	12	T10 Cr 204	12.19:	-1	T10?	58.03	5	T10 Fe1163 Al6
15.68	-1		64.24	12	T10 Fe 1030	12.54	15	T1 106 T10	58.70	6	V 77 T10?
16.08:	0	Sc(Ms)	4		T10	13.01	5	Ca 48 T10	59.11	0	T10
16.34	0	Sc(Ms)NdII 80?	66.53	10	Y 12 T10	13.76	2	T10	59.63	5W	T10
16.61:	-1		67.03	9	T10 Fe784,817	14.43	40W	T1 106 T1 106	60.20	1	Fe 1164
17.02	1	Fe 1148	67.51	3	T10			Sc 15	60.54	11	V 1
17.16:	0		68.07	2	T10 Ni 192	15.06	10	V 2	61.14	0	T10
18.01	4	T10? V(Ms)	68.46	8	--T10--	15.55	3	T10	61.65	2	V 77 T10?
18.78	8	Zr(M)? T1II 69	68.81:	-1	T10	16.02	-1	T10 Sm 2?	62.12	0	T10
19.22	4	T1 258	69.33	8W	T10 Co 56	16.35	3	T10	62.72	8	Fe626,1163 T10
19.74	1	T10?	69.80	-1	T10	16.74	10	Mn 4 T10	63.00:	0	T10
20.36	25	Mn 4	70.23	8	T10	17.18	6	V 2 Fe 1109?	63.66	13	Fe 1062
21.04:	1W	--BaII 6?--	70.64	25	Mn 4	17.42	3	T10			Fe 112,1023p
21.27:	0		71.28	16	T10 T1 106	17.8	-2	T10	64.20	-1	T10
21.44:	2W		71.83	2	T10	18.17:	4	T10 T1(M)?	64.72	4	Sc 18 T1(M)?
21.60:	1	NdII 79 V(Ms)?	72.31	7	T10 EuII(M)?	18.39	4	T10 CeII 26?	65.56	20W	T1 229 Fe 1183
22.14:	-1	Fe 1145p?	72.73	6	T1 107 Fe 1108	4		T10	2		T10
22.50	0	T1II 80			T10	20.49	13	Sc 15 T10	67.38	15	Fe 209
23.44	0	T10?	73.09	0W	T10	21.20	1	T10	68.07	4	T10
24.08	10	Fe 1146 V(Ms)	73.47	8	T10 T1 259	21.57	5W	T10 Sr 9	68.88	-1	Fe 869
24.65	5	Ni 70	73.92	1	Fe 1062	22.06	1	T10	69.62	13	Fe 686
25.26:	2	FeII 49	74.22	20	T1 108	22.56	8W	T10 Fe 1108	70.02	-1	Fe 345p
25.41:	2	T10?	74.57	10	T10	23.22	6	Co 112 T10	70.49	6	T10 Mo 4
26.27	25	T1 3	75.26	3	V 2p T10	4		T10	3		T10
26.76	0		75.71	10	T10	25.48	5	Fell07pFel062	72.85	22	Fe 686
27.15:	-1		76.42	25W	Fe 1029 Fe 1062	26.05	5W	T10 Sc 18?	73.08	5	Fe 1061
27.44:	-1		76.95	25	Ni 59 T10	26.72	11	ScII 31? T10	73.51	2	T10
27.85	-1		77.70	7	T1 265 T10	27.1	0	T10	74.01	7	V 2 T10
28.32	0	--Zr(M)	6		T10	27.54	8	Y 12	74.50	-1	Cr 120?--
29.12	4	T1 259	80.85	6	T10 Fe 1062	27.99	2	T10	74.95	2	T10
29.70	40	Fe 15			Sr 9?	28.40	22	Mg 9	*		
30.37:	-1		81.59	20W	Mn4,31T1265T10	28.7	5	T10	76.09	10	Fe 686 T10
30.82	0		81.90	15	T1 106 Sc 16	29.17	6	Fe 872 T10	76.51	4	V(M)
31.44	2W	T10 NdII 80	82.33	3	T10 V 2	29.50	-1		3		T10
31.93	-1		82.76	3	T10	2		T10	78.73	12	Ni 47
32.54	30	Mn 1	83.01	3	T10 Fe 1061	30.80	11	Co 38	2		T10
32.96	0	Fe 1143	83.39	12	Co 39 T10	31.19	0	T10	80.31	5	T1(M)? T10
33.67	0	T10?	84.10	5	T10	31.74:	3	--T10--	80.90:	2	--T10--
34.54	35	Fe 15	84.64	2	Sc 16 T10	32.20:	3	T10	81.94	15	Ca 21 Y 12
35.16:	0	T10?	85.00	1	T10	32.74	6	Fe 783	82.52	1	T10?
35.86	8	Ni 70	85.56	15W	T10	32.96	15	T1(M)? Mo 4	83.05	6	T1(M)? T10 Sun
36.30	0	Fe 1161	86.11	2	Zr(M)? T10 Sr9?			T10?	83.66	-1	GdII 59?
36.68	20	T1 51 Fe 113	86.31:	2	T10	33.25	9	T10	84.24:	-1	
37.16	4	--Fe 1145p?	86.86	9	T10	33.81	6	V(M)	84.49	12	V 37
			87.10	6	Fell143? T10	34.30	4	T10	85.29	-1	T10?
37.75	3	Zr(M)	87.47	0	T10	34.72	7	T10 Fe871,1063?	85.70	4	T1(Sun) T10
38.30	9	T1 108	87.79	8	Fel025 T10	35.39	25	V 1 Ba 1	86.34:	-1	T10?
38.77:	0		88.28	15	T1265 T10			Fe 626, 1029?	86.78	18	Fe 686
39.05	8	T1(Ms)?	3		T10	36.02	3W	T10	87.50	2	Fe 1026?
39.80	0		90.15	12	T1 107	36.54	8	Fe 345p? T10	87.87	10	Ni 70
40.49	9	T1(M) T1 107p?	90.86	25	T1 3	37.00	0		88.46:	2	V(Ms)
		Zr(M)?	91.28	3	T10	37.40	10	Zr(M)? T1(M)?	88.76	22	Ca 21
41.03	-1		91.85	6	--Fe 1031 T10	37.74	20	Mn 4	89.30	2	T10? Ni 205
41.39	2	Fe 1144--	92.33	4	T10	38.4	BdV		89.72	4	T1(M) T10
41.92:	0		92.77	12	T10	38.60	6	Fe 839,1064	90.14	12	Ca 21
42.34:	7	NdII 76 Cr 204	93.44	6	LaII 4? Fe1061	38.80:	4	T1(M)? T10	90.74	7	Co 90
42.51:	3	Sc(Ms)?	93.92	6	T10	39.21:	3	--Fe 871	91.28	2	Sc 18 T10
42.73:	-1		94.11:	5	T10	39.59	5	T10	92.32	30	V 37 Ni 69
43.44	0W	T10	94.34:	4	T10 Fe 1024?	40.04	0	Sr 9?	92.96	9	V 1
44.15	0	Mn 31--	94.87	5W	T1 108 Ni 231	40.42	3	--T10--	93.70	3	--Ni 206
44.62	0	Co 196--	95.40	3	T10	40.93	3	T10 Sc 18	94.43	23	Ca 21
45.02	9	Fe 1163	95.94	9	V 2 T10	41.37	-1		95.03	0	
46.19	0	Sc(Sun?)	96.53:	1	T10	41.86	12	T1(M)? T10	95.89	-1	
46.65	30	T1 3 Cr 204	96.90	3	T10	42.37	-1		2		T10
46.98	35	Fe 15 Fe 37p	97.51	30	Fe 15	42.69	5	V 2	97.51	BdV	T10p(0,0)R-head
47.43:	0	Zr(M)	97.98	10	T1 51p	43.18	6	Fe 926 T10	97.77	45	T10 T1 229
47.95	0		98.50	4	T10a(0,1)P	43.57	-1	T10?	98.48	40	Ca 21 T10
48.2	BdV	T10a(0,1)R-head			T10a(1,2)R-head	44.00	7	Fe 1062 T10	*		T10
48.35	11	T10	99.00	5	T10a(0,1)P	44.72	5W	V 38 YII 27			
48.88	14	T10 T1 259			T10a(1,2)R	45.08:	-1		5600		
49.20	10	T10 T1 107	99.98:	12W	T10a(1,2)R	45.52	0	T10	00.07	18	T10 Sun
5		T10			T10a(0,1)P	45.93	6	V 38 YII 27	00.65	2	T10
51.10	20W	T10a(0,1)R-head				46.44	6W	Fe 1145 T10	01.32	25	Ca 21 V(Ms)
		Sr 9 NdII u	5500			47.07	15	V 38 Fe 1061	01.85:	1	T10?
						47.86	2W	T10			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
5600			61.47	8	Fe 1108 T10	15.12	12	Ti 228 Ni 231	87.53	0	TiO?
02.04:	0	TiO?	61.83	-1	TiO?			Fe 1061,1086	87.98	6	Cr 188
02.41:	0		62.10	12	Ti 249	15.84	1	TiO	88.49	4	TiO Cr 119 V 92
02.85	28	Ca21Fe686Fe1062	62.51	3	Fe 1087	16.40	7	Ti 249		2	TiO
03.31:	-1	TiO?	62.90	6	Ti 269 Fe 924	16.83:	-1		90.11	-2	Sun
03.54	BdRv	TiO(0,0)Q-head			YII 38	17.28	10	Sc 12 TiO	90.67	9	Sun
04.94	18	V 37	63.42	0	TiO	17.80	3	Fe 1107	91.02	10	Cr 188 Fe 552
04.94	18	V 37	64.05	8	Cr 203 T10			BdRv ZrO(0,0)Rb-head	91.44:	-2	
09.03	5	Fe 1108p?	64.55	9	Zr 47	18.28	6	-TiO	91.81	0	TiO? Cr 243
	2	TiO	65.39	3w		18.95	0w	TiO--	92.20	1	Sun
10.22	10	V 2 T10	66.20	2w	T10	19.80	6	Cr 119	93.06	2	TiO
12.71	4	Ti(M)?	66.74	-1	TiO	20.43	4	Ti 249	93.92	2	Fe 1086
14.80	4	TiO Ni 250	67.11	5	-ScII 29--		6	TiO	94.44	-1	TiO
15.26	20	Fe 209	67.55	7	Fe(Sun)	24.03	6	Sc 12	96.12	5	Ni 68--
15.67	25	Fe 686	67.91	-1		24.57	0	TiO V 135	96.80	0	TiO Cr u
17.07	5	TiO Fe 626	68.38	22	V 37	25.27	0	TiO	97.74	12	Zr 4 LaII 4
17.60:	4	TiO	69.02	3	-ScII 29--	25.65	5	V 135	98.19	0	Fe 982
17.82:	2		69.67	3	TiO	26.30	0	TiO	98.52	5	Sun Cr 17
18.38:	4		70.21	0	TiO?	27.04	28	V 35	99.04	3	TiO V 142
18.56:	5	TiO Fe 1107	70.82	25	V 36	27.66	15	V 35		5	TiO
19.05	3	TiO	71.25	-2	TiO?		6	TiO			
19.56	6	V(Ms) Fe 1161	71.82	18	Sc 12	31.24	25	V 36	5800		
20.03	BdRv	TiO?			TiO	31.79	3	Fe 1087	02.53	0	
20.16	20w	Zr 25 Fe 1026	73.36:	1	Ti(Sun)?		2	TiO		2	TiO
20.65	12w	Fe 1061 T10	74.06	2	TiO	33.98	5	V 135	04.08	5	Fe 959
22.16:	1w	V 85 T10	74.75	-1	Zr(M)? T10?	34.23:	-1		04.40	7	Fe 1087 T1 309
24.07	3	TiO Fe 1160	75.42	10	Ti 249	35.69	10	Zr 4	05.11	3w	TiO Ni 234
24.57	20	V 37 Fe 686			TiO	36.40	-1	TiO	05.77	3	LaII 4
24.86	16	V 37	78.30	-2	TiO Fe 982p?	37.04	20	V 35	06.37:	-1w	TiO
25.40	2	TiO Ni 221?	78.65	-1	Fe 113p?	38.20	2	TiO	06.67	4	TiO Fe 1180
26.02	14	V 37	79.00	3	Fe 1183	39.05	1	Ti 249p	07.24	2	TiO V 142?
26.74	0	TiO	79.46	-1	TiO	39.45	6	Ti 228	07.81	1	Fe 552p
27.09:	-1	Fe 1084?	79.88	3	Ti 269	39.97	7	Ti 228	08.29	2	LaII 4 T10
27.60	20	V 37	80.26	0	Fe 1026 T10	40.29:	-1	TiO?	09.24	6	Fe 982
28.19	2	TiO	80.68	-1	TiO?	40.78	2	TiO	09.78	1	Ti 73
28.66	5	TiO Cr 203	80.87	3	-Zr 25	41.30	4	TiO T1 280	10.37	1w	TiO(L,3)Ra-head
29.10	BdRv	TiO(0,0)Rb-head	81.26	-1	Cr u? T10	41.91	4	Fe 1086 T10	11.09	3	TiO(L,3)Rb-head
29.33	25w	TiO ZrO	81.73	1	TiO		2	TiO	11.64	5	TiO
29.75	0	TiO	82.26	1	TiO Ni 232	43.44	17	V 35	12.36:	-2	
30.10	2	TiO Y 12	82.63	15	Na 6		6	TiO	12.84	1	Ti 309
32.44	14	V 1	83.22	2	V(M)	47.79	3	TiO V(Sun)	13.52	-2	TiO
33.18	2	TiO	83.64	2	TiO	48.36	8	Ni 45 V 127	13.93	7	Ti 73 T10
33.94	9w	TiO	84.16	8	ScII 29	48.89	3	V 92	14.48	0	TiO
		ZrO(0,0)X-head	84.66	1	-TiO		2	TiO	14.79	BdRv	TiO u
35.83	2	TiO Fe 1088	85.17	0	TiO?	50.70	3	TiO V(Sun)	14.92	10w	TiO Fe 1086
36.20	3	TiO	85.73:	-2	TiO?	51.41	0	Mo 5		2	TiO
36.61	1	TiO Fe 868	85.92	3w	TiO	52.07	4	TiO Fe 1180	16.38	8	TiO Fe 1179
37.10	5w	TiO Ni 218	86.57	8	Fe 1182	52.80	6	V 92 T1 214	17.03	6	TiO V 92
37.65	-1	TiO Co 195?	86.84	10	Sc 12	53.11	6	Fe 1107 T10	17.55	0	V 142
38.21	6	Fe 1087 T10	87.54	4	TiO	53.62	4	-TiO--	23.77	6	Ti 239 T10
38.67	2	TiO	88.18	12	Na 6	54.65	20	Ni 68 Fe 866		2	TiO
39.16	-1		88.53	4	TiO NdII 79	55.82	1w	TiO	25.68	7	TiO Fe(M)?
39.54	12	Dy(M)?			Co 90	56.38	3	TiO T1 228p	25.56	5	TiO Fe(M)?
40.16	0	TiO	89.08	2	-TiO--	56.87	2	TiO	27.26	0	TiO
40.96	5	ScII 29	89.48	6	Ti 249	57.60	0	-Ca 54?	27.86	5	Fe 552p T10
41.46	6	Fe 1087 T10			TiO		3	TiO	28.45	6	TiO Fe(M)?
41.74:	0	TiO	91.48	2	Fe 1087	59.81	4			2	TiO
41.91:	1	Ni 234? T10?			TiO	60.35	3	Fe 867	30.66	4w	TiO V 142
42.23	1	TiO	94.35	7	-TiO--	60.85	2	TiO Ni 231?	31.59	1	TiO Ni 233,250
42.69	3	TiO Ni 203	94.66	8w	Cr 239 T10	61.42	8	V 35		4	TiO
43.20	4	GdI 3 T10			TiO	61.93	0	TiO(0,2)Rb-head	33.97	4	TiO Fe 209p
43.65	0	TiO	96.32	11		62.32	7	Ti 309 Fe 866	34.66	6	
44.13	20	Ti 240			TiO	63.01	8	Fe 1107	35.18:	-1	TiO
45.76	-1		98.06	2	Fe 867		5	TiO	35.56:	-2	Fe 343p
46.12	12	V 37	98.49	28	V 35 Cr 239	66.38	5	Ti 309 T10		4	TiO
46.60	1	TiO	99.13	-1	TiO		5	TiO	38.01:	-2	Sun
47.16	9	Co 112 T10				69.19	3w	TiO LaII 70?	38.57	7	Fe 959 Cr 119
48.61	10	Ti 269	5700				2	TiO V 92	39.01	-2	
49.11	3	TiO	00.17	16	Sc 12 Cu 2	72.40	4	TiO V 92	39.50	8	Sun T10
49.62	3	Fe 838 T10	00.62	-1	Cr 203? T10?		2	TiO	39.71	5	TiO T1 105p?
52.26	5w	TiO Fe 1108	00.95	0	TiO--	74.05	7	Ti 309 T10	40.41	1	TiO GdIII 112?
53.97	10	TiO Fe 1159	01.54	15	Fe 209	75.07	5	Fe 1087 T10	41.17	4	TiO Sun
54.62	7	TiO	02.27	5	Cr 203	75.58:	-1		41.71	0	TiO
55.33	10w	TiO Fe 1107	02.62	5	Ti 249 Fe 866	76.70	29	V 36	42.38	7	TiO
57.40	12	V 37 T10	03.59	21	V 35		2	TiO	43.14	1	Cr 119
57.89	4	ScII 29 T10			TiO	78.43	7	Fe 209	43.76	3	TiO
58.68	30w	Fe 686 Fe 686	05.30	1w	Fe 1087 T10?		3	TiO	44.45	3w	TiO Cr 119
59.13	8	Ti 50 Co 82	06.05	5	Fe 1183 V(M)	80.78	9	Ti 214	45.16	3w	TiO
60.74	2	TiO Fe 869	07.00	20	V 35 Fe 868		3	Cr 119,188	45.74	-2	GdIII 112?
			07.62	-1	TiO?	81.74	2	-Cr 188	46.20	3	-V 142
			08.19	9	Ti 249	82.15	11	Cu 2	46.45:	1	TiO
			08.59	7	Sc 12	82.62	2	V 35,127	47.00	9	Ni 44
			08.96	8	V(Sun)	83.15	5	Cr 188 T10	47.55:	BdRv	TiO
			09.48	23	Fe 686 Ni 46	83.83	6	TiO Cr 188	47.66	2	TiO
			10.00	-1	TiO	84.33	2	V 141	48.11	10	Fe 552,1175 T10
			10.50	-1	TiO?	84.70	2	Fe 686	48.86	0w	TiO
			11.05	9	Me 8	85.02	2	Cr 188	49.19	1	TiO
			11.79	23	Sc 12 T1 249	85.36	2	Fe(Sun)?--	49.68	4	Sun Fe 922p?
			12.07	6	Ni69 Fe686 T10	85.95	20	Ti 309 Cr 188	50.31	6	V 92
			12.80	6	Cr 119 T10			Cr 17?		3	TiO
			12.99:	4	TiO	86.20	BdRv?		52.05	3	TiO?
			13.90	7	Ti 249	86.64	-1		52.32	9	Ti(Sun)
			14.09:	1	TiO	87.02	3	Cr 119 T10			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
5800			3		TiO	6000			71.72	-1	TiO
52.77	0	TiO?	20.46	6w	TiO Fe 581	00.02	1	Fe(M)?	72.15	-1	TiO
53.18	8	Fe 35	2		TiO	00.43	0	TiO	72.58	4	ScOIII(1,1)head
53.73	15	BaII 2	22.11	15	Ti 72	00.95	2	TiO Zr(M)?	73.15	3	ScO TiO
54.43	0	TiO	3		TiO	01.51	2	TiO	73.71	0	ScO? TiO?
55.09	5	TiO Fe 1179	24.35	1	TiO?	02.19	6w	V 49 TiO	74.78	0	ScO? TiO?
56.22	3	Gd 3?	24.64	5	V(M) TiO	02.58	5	V 34 Ti 198	75.56	-1	ScO? TiO?
56.95	2	TiO	25.18	3	TiO Zr(M)?	03.03	6	Fe 959 TiO	76.30	3	ScO? TiO
57.45	15	Ca 47	9		TiO	03.62	3w	TiO	76.84	2	ScO? TiO
57.90	2	TiO	30.27	5	TiO Fe 1180	04.29	0	TiO	77.38	3	V(Sun)? TiO
58.14	3	TiO—	30.92	3	—TiO	04.96	0	TiO Co 37			ScO
59.52	9	TiO Fe 1181	31.52	2	TiO	05.55	3	Fe 207, 1079	77.77	1	ScO TiO
60.21	5	TiO	32.15	8	TiO Ti(M)?			TiO	78.48	5	Fe 1259
60.65	1w	TiO GdII 58	33.04	2	TiO Fe(M)?			TiO	78.97	-1	Fe 1176
61.02	4	TiO	33.78	3	TiO			TiO	79.29	8w	ScO(0,0)
61.78	5w	TiOa(2,4)R-head	34.41	4	TiO			TiO	80.24	0w	ScO TiO?
62.38	10w	Fe 1180	34.61	2	Fe 982	07.30	6	Ni 42 TiO?			*
62.81	8	TiOa(2,4)R-head	35.17	6	Zr 2 TiO	07.94	3	Fe 1178 TiO?	81.45	19	V 34
63.36	6	TiO	35.53	4	TiO	08.59	5	Fe 982 V 49	82.30	0	TiO?
63.97	2	TiO Cr 185	36.25	1	TiO?			TiO	82.73	10	Fe 64
66.44	30	Ti 72	37.02	3	—TiO—	12.19	4	Fe 64p	83.23	0	TiO?
66.98	0	TiO	37.78	15	Ti 72	13.46	15	Mn 27	83.60	0	TiO?
67.57	4	Ca 46 TiO?	3		TiO			TiO	84.19	2	TiO
67.96	1	TiO	40.65	20	Ti 2	15.20	2	TiO Fe 63p?	85.28	20	Ti 69 Fe 269
68.40	2w	Zr(M)? TiO?	41.34	2	TiO?	15.70	0	TiO	86.21	2	TiO Ni 249?
69.01	3	TiO	41.78	20	Ti 72	16.54	9	Mn 27 TiO	86.80	1	TiO
69.47	1	Zr(M)? TiO	42.68	6	TiO	17.02	4	TiO ScOII(0,0)	87.54	1	V 33 TiO
72.74	9	TiO Fe 552p	43.47	7w	TiO Fe 63p	17.66	2	—ScO?—	88.20	0	TiO
73.36	4w	TiO Fe 1087	44.11	3	TiO	17.86	2	V 49 TiO ScO?	88.90	0	TiO
77.23	1	TiO GdII 94?	44.65	16	Ti 2	18.36	5	Ti 198 ScO?	89.53	6	V 33—
77.78	7w	TiO Fe 1083	7		TiO	18.62	7	Ti 70	90.22	20	V 34
79.79	3	Zr 4	49.06	BdRv?	TiO u	19.34	2	TiO	90.62	-2	
80.23	10	Ti 71 Fe 1201	49.25	9	Fe 14, 1176	20.04	11	Fe 1178 Fe(M)?	90.92	-2	
80.72	4	TiO	49.51	7	TiO			TiO	91.19	9	Ti 238
81.29	7	TiO Fe 1178p	50.04	5	TiO Ti(M)?	21.78	12	Mn 27	91.87	0	TiO
81.88	5	TiO Fe 63p?	3		TiO			Fe63, 1079, 1085	92.56	BdRv	ScOII(2,2)R-head
82.35	0	TiO	52.72	1	Fe 959 TiO	2		TiO	92.78	6	Ti 153
82.67	4		53.18	11	Ti 154	23.35	3	Y 3 TiO	93.18	5	Co 37
83.29	5	TiO	53.62	1	TiO	24.09	5	Fe 1178 TiO	93.59	0	Fe 1177
83.94	7w	Fe 892 TiO	54.56	6	TiO	24.95	-2		93.87	2	TiO V(M)
85.55	3	Zr 2	55.05	BdRv	TiO u	25.27	2	TiO V(Sun)?	94.41	0	Fe 1177
86.00	0	TiO?	55.26	13	TiO Zr 3	25.75	0	TiO			TiO
86.28	3	TiO	55.73	2	TiO	26.18	0	—Sc(M)?—	96.67	3	Fe 959 TiO
86.70	1	TiO?	56.16	3	TiO	27.04	7w	Fe 1018 TiO	97.18	6	Fe 64p—
87.38	7	TiO	56.70	18	Fe 14			TiO	97.50	8	V 33 TiO
88.14	6	TiO	2		TiO	30.58	3	TiO Mo 5?	98.18	2	TiO
88.81	4	TiO—	58.26	7	TiO Fe 14p	31.16	-2	TiO?	98.67	3	Ti 304
89.87	90w	Na 1	58.96	0	TiO	31.64	12	—Ti 2			TiO
91.35	6w	TiO Fe 581?	59.58	4	—TiO—			TiO	6100		
92.14	5	—TiO			TiO	32.56	2	Zr(Sun) TiO	01.73	-2	TiO?
92.87	10	Ni 68 TiO	62.65	BdRv	TiO u	5		TiO	02.12	4	TiO Fe 1259
93.60	1w	TiO	62.80	4w	TiO	36.04	BdRv	ScOIII(0,0)head	02.72	20	Ca 3
94.08	5		63.34	6w	TiO Fe 63	36.28	12w	ScO	03.26	3	Fe 1260
94.64	2	TiO	3		TiO	37.20	2	TiO	04.02	1	TiO?
95.05	0	TiO	65.84	12	Ti 154	37.73	-1	ScO	04.65	2	TiO? V(Ms)?
95.86	90w	Na 1	6		TiO	38.08	1	ScO	05.11	-1	TiO?
96.52	3	TiO	70.77	-1	TiO?	38.61	2	TiO ScO?	05.61	0	TiO?
97.02	-1	TiO	2		TiO	39.24	0	TiO?	06.40	0	TiO
97.53	3	TiO	72.05	4	TiO	39.70	16	V 34	06.96	5	V 60 TiO
97.87	4	TiO	6		YOI(0,0)head	40.22	-1		07.52	-1?	
98.52	6	TiO	75.30	3	Fe 1017, 1260	40.61	0		08.13	13	Ni 45
99.27	40w	Ti 72	2		TiO	41.27	-1		09.18	5	TiO V(Ms)?
99.98	0	TiO	76.77	4	Fe 959 TiO	41.94	3	TiO Fe u	09.89	4	ScOIII(2,2)head
5900			2		TiO	8		TiO			TiO
00.58	8	TiO Cb(M)?	78.55	11	Ti 154	48.66	2	TiO V 49	10.20	8w	ScO TiO V(M)?
01.02	4	TiO	2		TiO	49.41	3	TiO ScO?	11.11	0	TiO?
01.70	8	TiO Fe(M)?	80.83	8	Ti 72 V 49	49.76	0	TiO?	11.64	19	V 34
02.15	0	TiO Cr 119?	2		TiO			TiO	12.35	0	TiO
02.48	2	TiO	83.49	1w	TiO Fe 1175?	52.84	1	TiO Zr(M)?	12.94	-1	ScO TiO?
02.86	2	TiO	84.00	0	Co 37?	53.74	8w	TiO CrO(0,0)head			*
03.34	10	Ti 71	84.60	23	Ti 2 V 49	54.74	0	TiO	14.08	4	ScO TiO
03.90	3	TiO	84.73	5	Fe 1260	55.22	2	TiO	14.67	3w	TiO
04.8	BdRv	TiO u	3		TiO	55.85	5w	TiO Fe 1259	15.27	-1	
10.00	6	Fe 552 TiO	87.05	4	Fe 1260 TiO	56.34	0	TiO			ScO(1,1)head
13.71	12	Ti 2	87.67	2w	TiO	56.99	8w	TiO CrO(0,0)	16.16	10	ScO Ni 218
14.14	8	TiO Fe 1180, 1181	88.46	3	Ti 154 Sc(M)?	58.12	11	V 34	16.99	4	TiO ScO Co 37
15.64	3	Co 82	88.96	0	TiO?	58.77	2	Ti 70p?	17.64	0	
15.96	0	TiO	2		TiO	6		TiO	18.21	1	TiO
16.24	4	Fe 170	90.60	7	—TiO—	62.82	8	Fe63 Zr3 Cr185	18.83	-1	TiO
18.54	14	Ti 71	91.35	2	TiO	63.36	3	TiO	19.52	17	V 34
			91.87	11	Co 90	64.63	14	Ti 69	20.25	6	Fe 14p
			95.18	3	TiO	65.48	15	Fe 207	20.98	6	Ti 153 Zr 24
			95.60	3	TiO	66.00	-1	ScO	22.21	22	Ca 3
			96.00	2	Ti 154	66.56	-1	ScO			TiO
			97.82	6w	Fe 1175 TiO	67.20	3	V(Sun) ScO	24.82	3	TiO Zr 24
			98.52	1	TiO	67.68	2	TiO ScO	26.22	16	Ti 69
			99.01	4	Ti 198	68.10	0		26.91	-1	
			99.62	3	Ti 227 TiO	68.58	4	TiO ScO	27.44	8	Zr 2 TiO
						69.27	2	TiO ScO	27.93	2	Fe 1017, 1082
						69.86	-2	TiO	28.39	3	TiO? V 33
						70.30	1	ScO TiO	29.00	6	Ni 42
						70.70	1	TiO	29.54	0	LaII 47
						71.20	0	TiO ScO	30.25	0w	TiO

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
6100			04.63	3	Ni 226?	65.16	12	Fe 62	46.89	2	—TiO—
*		TiO	05.10:	6	TiO?	66.20	14	V 20 Ti 144	47.70:	1	TiO
32.08:	1w	TiO YOY(0,0)	05.57:	4	TiO?	66.83:	1w	TiO	48.05:	1	TiO
33.22	0	TiO	05.94	3	TiO	68.11:	7	TiOa(3,6)Ra-head	48.52	2	
33.94	0	TiO	06.39	2	TiO Rb(M)?	68.84	14	V 20	49.49	6	V 84
34.56	7	Zr 2	06.86	2	TiO	69.54	2	TiO	50.43	4	TiOy(2,0)Rb-head
35.35	15	V 34	07.29	3	TiO V 20	70.28	3	Fe 342	51.19	BdRv?	TiO
35.93	0			4	TiO	71.23	5	Fe 685 TiO-			ZrOy(0,0)Rb-head
36.59	17	—Fe 169	09.37	2		71.85:	0	TiO	51.33	5	ZrO TiO
37.01	9	Fe 62	09.94	12	TiO	72.56:	3w	TiO	52.12	3w	TiO
37.71	15	Fe 207	10.74	30w	Sc 2 TiO	73.39	12	Ti 1	52.96	1	TiO
38.43	3	Y 3 Ti 197	11.53:	2	TiO?	74.69	13	V 19	53.36	0	TiO
2			11.85	11w	TiO—	75.55:	3	TiO	53.85	3	Fe 13p
40.42	3	Zr 24 TiO	12.92	11w	TiO Zr 24?	76.33	3	Sc 2	54.48	1	TiO
41.01	1		13.43	4	Fe 62	77.15:	3	TiO	55.08	7	Fe 342
41.73	18	BaII 2 Fe816	13.88	9	V 20 TiO	77.68:	4	TiO Ti 144	55.83	2	TiO
42.48:	0		14.12	0	TiO	3			56.41	1	TiO
43.23	10	Zr 2	14.81	BdRv	TiOa(2,5)Rc	80.65	20	Fe 13	56.97	0	TiOy(3,1)Ra-head
44.08	0w	TiO	15.09	20w	TiO Fe 1018	81.34	0	TiO	57.45	10w	TiO V 84
45.53	-1	TiO			Ti 293	82.36	6	TiO	58.14	0	TiO? LaII 47?
46.22	8	Ti 153	15.85	5	TiO	82.76	7	TiO Co 37	58.72	21	Fe 13
47.80	3	TiO Fe 1016	16.36	22	V 19 TiO	83.50	0	TiO	59.40	1	TiO?
48.60	BdRv	TiOy(5,2)Rb-head		4	TiO	84.08	4	TiO	59.92	13	Ti 1
48.82	20w	TiO	19.27	12w	Fe 62 TiO	85.16	18	V 19	60.34:	-1	TiO
49.25	5	TiO	19.95	3	TiO	9			60.71	0	TiO
49.70	5	Ti 197	20.44	2	TiO Ti 293	92.85	22	V 19	61.43	3	TiO Ti 196
50.18	16	V 20	21.00	6w	TiO Fe 958	94.02:	0	TiO	61.74	2	TiO
51.61	13	Fe 62 V 33	21.55:	7	Fe 13 Fe 981	95.16:	7w	Ti 1 TiO	62.31	1	Zn 6? Sc 1?
5			22.13:	BdRv?	TiO u	96.58	30	V 19 Ti 1	62.92	15	Cr 6 Fe 1019
54.21	4	Na 5	22.51	20w	TiO Y 2	97.81	10	Fe 62	63.75	5w	TiO
3			23.36:	9w	TiO	3			64.24:	0	
56.00	3	TiO Ca 20	24.00	2	Ni 228 TiO				64.85	16	Ti 1 Fe 1229?
2			24.49	11	V 20 TiO	6300			65.55	3	—TiO?—
57.74	10	Fe 1015 Zr(Ms)		4	TiO	01.48	4	Fe 816	66.37	18	Ti 103
58.44	BdRv	TiO	26.68	4	TiO Fe 981	02.39	3	—Fe 816	66.97	0	TiO
58.65	10	TiO	27.11	2	TiO	03.19:	-1	TiO	67.33	1	
59.23	20w	TiOa(1,4)Rb-head	27.67	8	TiO Ti(M)?	03.77	8	Ti 104	68.03	6w	TiO
59.68	-1		28.04:	0	TiO	05.66	30w	Sc 2 Sc 3	68.64	1	
60.18	6	TiO Zr(M)	28.53	3	TiO	06.59:	6		69.33	5	—TiO
60.75	10	Na 5	29.20:	7w	TiO Fe 342				69.86:	0	
61.32	8	Ca 20		4	TiO	6			70.29	0	—Ni 127?
62.18	28	Ca 3	30.3:	4	TiO	11.48	2	TiO Fe 342	70.91	3	—TiO—
62.94	3	TiO	30.80	16	V 19 Fe 207	12.22	9	Ti 104	71.46:	2	TiO
63.66	15w	Fe 64 Ca20 TiO		Co 37	12.94	3	TiO Zr 65	71.78:	4	TiO	
2			31.71	4	TiO Sc 3	13.50:	0	TiO	72.21	0	
65.44	8w	TiO Fe 1018	32.16	2	TiO	14.16	1	TiO	72.77	4w	TiO Cr(M)?
66.05	6	TiO	32.75:	0	Fe 816?	14.66	6	Ni 67	4		TiO
66.59	10w	Ca 20 TiO	33.18	11w	V 20 TiO	15.30	3	Fe 1015	75.73	1	
3			34.07	6w	TiO	15.80	0	Fe 1014	76.20	4	TiO
69.10	16w	Ca 20 TiO	34.78	5w	TiO	16.45	6w	TiO	76.85	3	—TiO
69.69	16w	Ca 20 TiO	35.32	2	TiO Ti(M)?	17.32	2	TiO?	77.33	1	TiO—
70.46	23w	V 20 TiO Fe1260	35.94	9w	—TiO	17.98	10	Fe 168 Ti 103	77.96	0	
2			36.46	1		18.65	1w	TiO	78.30	6	ZrOy(1,1)Rb-head
72.75	11	TiO LaII 4	37.14:	8w	TiO	19.26	2	TiO			TiO Ni 247
73.44	20	TiO Fe 62	37.70:	0	TiO?	19.83	1		78.82	10	Sc 1
74.13	BdRv	TiO u	38.10:	0	TiO?	20.34	2	LaII 19	79.46	3	TiO
74.39	25w	TiO	38.56:	-1	TiO?	20.78	4	TiO	80.27	2	TiO
75.27	10w	TiO	38.98	3	TiO	21.32	3	Zr(M)	80.74	4	Fe 1015
75.60:	-1	TiO?	39.30	4	—Sc 2	21.75	3	TiOy(2,0)Rb-head	81.38	5	TiO Ti 196
75.94:	-1	TiO?	39.82	5	Sc 3	22.24	1	TiO	82.03	2	
76.29:	-1		40.23	4	V 20 Fe 1015	22.70	5	Fe 207	82.73	6	TiO
76.80	2	Ni 228? TiO	40.70	7	Fe 64 TiO	23.23:	0	TiO	83.26	-1	
77.31	2	Ni 58? TiO	3			24.45	1	TiO	83.89	0	
5			42.96	40w	V 19 V 19	25.16	20	Ti 1	84.22:	BdRv	TiOy(3,1)Rb-head
80.36	10	Fe 269 TiO	43.78	1	TiO?	26.02:	0		84.45:	8w	TiO
2			44.22:	2	TiO	26.89	6	V 84	85.16	6	NdII 85
81.81	2	V(Ms) TiO	44.80	2	TiO	27.59	12	Ni 44	85.74:	0	Fe 1253p?
2			45.19	5	V 20	28.42	5w	TiO	86.42	5	Zr? Sr(M)? TiO
83.21:	BdRv	TiO u	45.66	3	ScII 28?	29.04	1		87.10	1	
5			46.33	7	Fe 816 TiO	29.54	1		87.61	4	TiO
86.25	BdRv	TiO u	47.47:	2w	TiO FeII 74?	30.14	15w	Cr 6	87.92	2	
86.50	25w	TiO Ti 197	3			30.84	1	Fe 1254	88.48	2	
88.15	8	TiO Fe 959	49.64	7	Fe 685p?	32.36:	0w	TiO Sc(M)?	89.08	5	TiO
88.80	9w	TiO Co 37		2	TiO	33.27	0	TiO	89.87	1	SmII(M)?
89.38	5	V 20	51.84	22	V 19 TiO	33.89	1		90.32	5	TiO LaII 33
2			52.56	8	Fe 169	34.18	1		90.66	4	Zr?
91.18	12w	Ni 45	53.12	4	TiO	34.81	2	TiO?	91.20:	-1	Mn 39??
91.69	16w	Fe 169 Y 2	53.83	3	TiO	35.38	17	Fe 62	91.62:	0	
92.27	6	TiO	54.30	18w	Fe 111 TiO	36.11	9	Ti 103	92.08	4	TiO
92.86	8	TiO Zr 24	3			36.85	8w	Fe 816	92.62	5	Fe 109—
93.43	5	TiO	56.37	11	Fe 169 Ni 43	37.77:	0	TiO?	93.58	20	Fe 168
93.76	7	TiO Sc 3	57.42	1	TiO?	38.14:	-1		94.35	2w	
7			57.80	4	Ti 1 NdII 72	39.06	5	V 84	94.89	5w	TiO
97.74	11	TiO Fe(M)?	58.06:	11	Ti 104	40.04	3w	TiO	95.50	1	
98.42	10	TiO Sc(Ms)	58.68	20	V 19 Ti 104	40.96	2		95.90	4	TiO
99.16	18	V 19 TiO	58.86	18	Sc 3	41.90	4	TiO	96.38:	2	Fe 921p?
99.66	4	TiO	59.59	0	Ni 216?	42.43	0		96.56:	3	Co(M)?
99.92:	3	TiO	60.04	1		42.90	2	TiO	97.34	5	V(M)??
			60.48	1		43.41	0	TiO	98.00	4	TiO?
			61.11	30	Ti 104 V 20	44.10	9	Fe 169	98.44	1	
6200			61.82	1	TiO	44.67:	BdRv	TiO?	98.96	2	LaII 104?
00.36	7	Fe 207 TiO	62.28	2	LaII 33 TiO			ZrOy(0,0)Rb-head	99.46	0	TiO?
6		TiO	4			44.97	14w	TiO Sc 1	99.96	8	Fe 816
			64.70	6	TiO Ti 144	46.05	2	TiO			
						46.54	2	—TiO—	6400		

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
6400			59.12: 2			16.11 10		FeII 40--	81.22 8		Fe 34
00.31 20		Fe 13	59.90: 2			16.97 10		-TiO(5,3)Rb-head?	81.83 -1		TiO
00.94: -1		Bu(M)?	60.75: 2			17.67 4		TiO Ti(M)?	82.41 -1		
01.35 5		TiO	61.74: 0			18.55 12		Fe 342	83.18 3w		TiO
02.02 2		Y 2	62.63 25		Ca 18 Fe 13p	5		TiO	83.54 -1		Cu(M)?
02.31 3		TiO	63.89 5			21.78 3w		TiO	83.94 2		TiO?
03.42 4		TiO	64.73 10		Ca 19	7		TiO	84.53 3		TiO
03.76 2			65.62 3			25.83 3w			85.40 4w		TiO Y 1
04.44 5			66.42: 1			26.26 -2			86.25 25		Ni 64 TiO
05.48 7w		TiO	67.01: 5		V 32	26.62: 0			87.17 5		TiO
05.76: 1		TiO	67.48: 0		Ti(M)?	26.96 10		LaII 33 TiO	87.67 -2		
06.71: -1			68.36: 1			*		TiO	88.65: 0		
07.03 1		Zr 2	68.13 20		FeI68p FeI258	28.02 3		TiO?	89.32 3		TiO
07.45 0			70.14 3w		-Zr 65	28.57 5		Fe u	89.99 1		TiO?
08.00 13		Fe 816	70.70 4			29.14 0		Cr 265?	90.75 4		TiO
08.54 0		Sr 8--	71.62 18		Ca 18	29.56 8		TiO	91.44 4		TiO Fe 1289?
09.04 1		TiO?	72.7 3w		TiO	30.38 8		TiO	91.94 0		TiO
09.65 0			73.54 Bdrv		ZrO(0,0)Rb-head	30.92: -1		TiO?	92.42 3		TiO
10.14 1		TiO?	73.78 12		ZrO TiO?	31.42 20		V 48	92.92 14		Fe 268 Ti 102p
10.37 1		TiO	74.36 4		ZrO	31.86 -1		TiO?	93.92 20		Fe 168
10.87 0			75.00 3		TiO	32.30 0		TiO?	94.53: -1		
11.25: -1			75.61 12		Fe 206	32.86 9		Ni 64	95.12 0		TiO
11.65 10		Fe 816	76.22 9		TiO ZrO	33.38 -1			95.90: 2		TiO Co 174
12.30 2		Fe 169p	76.85 1			33.95 7		Fe 1197	96.65 1		TiO
		ZrO(2,2)Rb-head?	77.58 3		TiO	34.50 2			97.50 3		TiO Fe 1253
12.81: 0		TiO	78.07 2			34.96 4		TiO	99.13 25		Ti 49
13.25 25w		Sc 1 Ti 1p	78.90 4w		TiO	35.66 5		TiO	99.96 3		TiO Zr(M)?
13.76: -1			79.60 1			36.54 7w		TiO			
14.12 2			80.05: 2w		TiO	37.12 -1			6600		
14.80 6		TiO(3,1)Rb-head	80.57 0		ZrO(0,0)Rb-head	37.92 18		Cr 16	00.59 2		TiO
15.5 5w		TiO	81.00 1		ZrO	38.42 1		TiO	01.04 1		TiO
16.40: -1			81.86 18		Fe 109	38.88 0		TiO?	01.41 1		Atm? Sun
16.68 3		TiO	82.42 0		TiO?	5		TiO	01.86 0		
17.01 5		V(Ms)	82.82 16		Ni 66	42.75 3		TiO SmII(M)?	02.34 1		TiO
17.80 5		Co 111	83.43 1		TiO(4,2)Qb-head			ZrO(2,2)Rb-head	02.76 0		
18.49 6w		TiO	83.88 8		-Fe 34p	43.50 11		V 48	03.22 3		Fe 862p
19.11 4		Ti 196	84.73 4		TiO	44.16 7w		TiO(5,3)Rb-head	03.75 2		Fe 860p
19.82 15w		Fe 1258 TiO	85.12: 1		TiO			Fe 1139	04.62 13		ScII 19 FeI254
20.67 6		-TiO	85.76 6		TiO	44.81 3w			05.57 0		TiO?
21.37 18		Fe 111	86.40 4			46.27 25		Ti 102	05.97 15		V 48
22.08 0			87.09 2w			46.62 2			06.48 4		TiO
22.54 4		TiO	87.84 8		TiO	47.61 7		Fe 13p	07.09 6w		TiO
23.15 3		TiO? V(Ms)	88.51 7		TiO	48.20 3		TiO	07.88 15		V 59
23.95 2		TiO	89.01 0		TiO? Yb 3?	48.9 1			08.44 2		TiO
24.75: 1		TiO	89.68 3		Zr 65 TiO	49.46 Bdrv?		TiO	09.12 17		Fe 206
25.43 4		-TiO	90.29 10w		Co 81	49.62 10		TiO	09.72 17		Fe 13p TiO
25.97 3			91.02 2			50.16 0		Sr 12??	10.50 5		TiO
26.52 2			91.71 15w		Mn 39 TiII 91	50.58 -1		Zr(M)	11.28 8		TiO
26.90 2		TiO	92.50 10		TiO?	51.06 6		TiO	12.03 10		TiO
27.32 3			93.28 2		TiO	51.66 18		Fe 13p	12.82 9		TiO
27.71 1			93.78 18		Ca 18	52.14 2		TiO?	13.77 20w		Fe 13p YII 26
28.06: 0			94.23 1		TiO?	52.61 2		TiO	9		TiO
28.71 6		TiO Fe 1138?	94.96 30		Fe 168	53.11 1		TiO?	21.12 3		Sun Ni 97
29.52 3		TiO	95.33: -1			53.70 3		TiO?	21.72 4		TiO
29.90: 0		Co 81	95.71 2		Fe 1253	54.23 20		Ti 102	22.80 6		TiO
30.40 1		TiO? V 107	96.16 1		TiO	54.86 1		TiO	23.45 2		-V(M)?
30.87 20		Fe 62	96.90 30		BaII 2	55.38 0		TiO	23.86 3		Co(M)?--
31.79: 2		TiO V 107	97.67 15		Ti 102	56.06 30		Ti 102	24.39 0		TiO
32.00 2		TiO Ni 126?	98.34 3		TiO	56.96: 3		TiO?	24.95 22		V 48 Fe 13
32.78 1		TiO?	98.93 25		Fe 13	57.14: 3		TiO Y 1	26.03: 9w		TiO
33.20: 0		TiO? V 107	99.62 15		Ca 18	57.99 10		V 59 Sc 24	27.03 6		TiO
33.73 1		TiO						Sc 24	27.64 0		Fe 1174
34.45 2		TiO	6500			58.71 5		TiO	28.23 9		TiO
35.04 10		Y 2	00.40 6		TiO	59.50 6		-TiII 91	29.38: 0		TiO
36.13 1		TiO V 107	01.21 18		Cr 16	60.25 6		TiO	29.99 15		Cr 16
36.75: 3		TiO	01.66 4		Fe u	61.07: -1		D 1?	31.06 4		Sun?
37.14 1			02.34 3		TiO	61.46: -1			31.77 7		TiO--
37.75 3w		TiO? BuII 6	02.82: 1			62.81 100w		H 1	32.45 7		Co 111
39.05 25		Ca 18	03.19 1		TiO	63.95 2		TiO	33.31 8		V(Ms) FeI258
39.84 3		TiO	03.54 5		-TiO--	64.86 3		TiO	33.74 0		Fe 1197
40.90 3		Mn 39 TiO?	04.15 7		V 48	65.75 12w		Ti u V 48	34.25 7		TiO FeI258p?
41.47 1		TiO?	*			66.80 7		TiO	35.03 Bdrv?		TiO
42.09 4			05.43 5		TiO	67.47 4		TiO	6		TiO
42.67 0			05.84 1			68.02 4			39.70 9		TiO Fe 1195
43.24 -1			06.38 6		Zr(Sun)	68.62 2w			40.39: 2w		TiO
43.86 2			06.89: 0		TiO?	69.15: 9		FeI253 SmII 62?	41.63 5		TiO? Atm?
44.30: 2			07.02 1		TiO?	69.50: 7		TiO	42.41 1		TiO?
45.15 4		TiO	07.50 1		TiO?	70.11 2		TiO	42.94: -2		Atm?
45.77 5		Zr 57	08.13 18		Ti 102	70.65 0		SmII(M)?	43.64 25		Ni 43
46.58: 5w		-LaII 104	08.87 4		TiO ZrO Ca18	71.22 12w		TiO Fe 1121?	44.23 -1		TiO
47.98 10		Sc 1	09.24 0		ZrO?	71.73: -1			44.67 -1		TiO
		TiO(4,2)Rb-head	09.80 4		TiO?	72.29 1			45.17 2		BuII 8 TiO?
49.04: 2			10.51 6		TiO	72.77 45		Ca 1 Cr 16	45.69 1		TiO
49.86: 10		Ca 19	11.17 6		TiO?--	73.68 4		TiO	46.13 1		TiO
50.24 20		Co 37	11.83 5		TiO	74.22 20		Fe 13	46.91 4		TiO Fe 206
51.57: 5w		-Fe 921--	12.48 6		-TiO(5,3)Rb-head	75.06 21		Fe 206 Ti 286	47.43: -1		TiO
52.28 12		V 48	13.2 1			76.47 8		TiO Zr(M)?	48.06 14		Fe 13p
53.64: 0			13.8 -1			77.14: 0			48.77 2		TiO
54.94: 0		Co 174	14.06: -2			77.76 4		TiO	49.55 7		TiO
55.60 10		Ca 19	14.41 0		TiO	78.01 4			50.31 3		
56.35 2		FeII 74?	14.81 8w		ZrO(1,1)Rb-head	78.93 8		V 32	51.06 Bdrv		TiO(0,0)Rb-head
57.11: 2			15.24 0		ZrO?	79.67 1		TiO?	11		TiO
57.67: 2		Zr(M)				80.30 2		TiO	57.78 5w		TiO(1,0)Qa-head
58.49: 2						80.93 8		Cr 16 TiO?			Cr 282

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
6600			38.42	4	—T10—	35.18	5	T10? So u?	19.89:	-2	T10
4			38.92	5	T10 Sun?	36.66	6	—T10—	20.48	7	T10?
59.98	0	Sun?—	39.48	10	Fe 34	36.23:	7w	—T10—	21.01	3	T10?
60.48	8	—T10—	39.99	3	Sun?	36.72:	-2	Sun?	21.84	10w	T10
61.16	10	T10 Cr 282	40.60	5	T10	36.99:	6	Fe 1225?	23.14:	0	T10?
3			41.07	2		3			24.21	15	T10 Cr 222
63.17:	BdRv?		41.68	7	—T10—	38.92	25w	T10 Fe u?	24.82	0	Atm?
63.49:	18	Fe 111	42.16	7	T10	39.82	30	Fe 205	25.30	10	Cr 222
63.81:	10	T10	43.10	25	T1 48	40.25:	2		26.11	30	T1(N) Cr 222
64.54	17	T10 Y(M)?	43.94	1	T10	41.36:	20w	Fe 1195 T10?	26.88	4w	T10?
65.36	17	T10 Fe 34p	44.50	8	T10	41.96:	25w	T10? V 31	27.72	4	
65.92	0		44.72:	0	—T10—	42.68	10	T10 Fe 1197	30.08:	-3	T10
66.33	17w	T10 T1 101	45.15	5	T10 Fe 1227?	43.22:	0	T10	30.64	0	T10
67.07	6	T10	45.54	6	T1 226p	43.62	8	Fe 1173	31.34	15	T10 Atm
67.41	6	T10 Fe 168p	46.36	8	T10 T1 152	44.07	1	SnII l?	32.44	3	Atm T10?
12			46.85	5	T10	44.60	25	Fe 34p	33.79	25	Fe167,1006 AtmO
74.31:	0		47.64:	15	T10y(2,1)Rb-head	45.23	3	Y 16	34.24	1	
74.98	15	T10	3			45.37	6		35.34:	0	T10 Atm
75.42	-1	T10?	50.16	18	Fe 111 T10	46.39	4	T10?	35.98	0	T10?
75.99	13	T10	50.76	6	—T10—	46.86	4	T10?	36.45	3	T10
76.32	5		51.21:	-2	T10	47.58	4	T10?	37.72	2	Atm vv Co 139
76.86	-1	T10?	51.68	5	T10	49.80	BdRv	T10y(3,2)Rb-head	38.29	5	
77.13	13	T10 T1 274?	52.17	6	T10	50.99	5		38.76:	0	T10
78.00	14	Fe 268	52.86:	15	V 31 Fe 1195	51.60:	11w	Fe 34 T10	40.05:	3	Atm?—
3			53.52	10	T10	52.26	12		41.55	BdRv	T10 u
80.64:	BdRv	T10y(1,0)Rb-head	54.60	12w	T10 SmII(M)?	3			41.72	25w	T10 Atm vv
7			56.11	15w	T10—	55.10	5	Fe 1195	42.29:	10	T10
85.29	1	T10?	56.85	4	T10	55.66	10	T10 Fe 1194?	42.72:	5	
85.89	12w	T10y(1,0)Rb-head	57.53	13	T10	56.29	8	T10	43.23	8	Atm vv T10
86.42	1		58.18	6w	T10—	57.21	10	T10	43.75	7	T1(M)?—
86.88:	0		4			58.19	12	T10 Fe 1173	44.20:	3	
87.55	10	T10 Y 1	62.23	15w	T10 Zr 1	58.70	3	T10	45.19	30	Fe 111
3			7			59.16:	-2	T10	45.94:	-2	T10? Sun?
89.94	0	T10?	66.54	12	V 31 T10	59.74	5	Sun?	46.47	5	—T10—
7			67.09	4	T10	60.31	15	Fe 205 T10	47.54	0	Fe 1221,1224
94.88:	1	T10?	67.78	20	Ni 57	60.92	5	Fe 34lp?	48.00	2	
95.31	7	—T10—	4			61.49	8	T1 237	49.29	5	T10
95.90	BdRv?	T10 u	71.05	12	T10 Co 54	61.98:	5	Fe 109	49.61	6	
96.10	13	T10 Al 5	71.77	10	T10	62.48:	3	Fe 1191 T10	50.79	8	Fe 205 Atm
97.07:	3	T10?	72.26	13	T10 Ni 127	62.88:	-2	SmII 55?	51.27	4	Fe 1186,1193
97.96	5w	T10	4			63.52	20w	T10	52.14:	-1	
98.60	11	T10 Al 5	75.36	10	—T10—	64.26	3		52.92	5w	T10? Fe 815p?
99.18	4	T10	75.98	8	T10	64.78:	0				
99.64	13	T10	76.54	9	T10	65.06	15w	T10	53.81	15	T10 Zr(M)?
			77.44	20w	T10 Fe1010,1013	65.62:	2	Sun?	54.44	4	AtmO?
			78.64:	10w	T10? SmII(M)?	66.08	3	Sun	55.10	5	Ni 157 Atm vv
			79.34	8	T10	12			56.32:	-2	T10?
			79.82	1	Tm(M)?	74.11w	30	AtmO T10	56.82:	-2	Atm?
			80.30	12	T10	74.48	0	T10	57.25:	-1	
			81.00	BdRv	T10	75.47	1	Fe 167	57.83:	-1	Atm?
			81.22	6	T10 SmII(M)?	76.70:	-2	Ni 97	58.53	0	T10?
			81.85:	20w	T10y(2,1)Rb-head	77.70	2	T10?	59.64:	-2	T10?
			82.51:	5	T10	80.04	6	T10	61.55:	0	T10
			83.11:	30w	T10	80.74:	-2	T10	62.14	15	T10—
			83.69:	20	T10 Fe 205	81.75	10	T10 Cr 222	62.89	10	
			84.30:	25	T10	82.52:	15	Cr 222	63.65	6	
			85.04	25	T10 V 31	84.05	5	T10	64.22:	-2	
			91.16:	20w	T10 Sr 3	85.74	3	Fe 1173	64.72	10w	K 7?—
			12			86.67	4	T10	65.32	9	
						87.52:	0	AtmO—	65.82:	-2	
						87.98:	1	T10	66.37	10	Zr(M)?
						90.04	0		66.83	15	—T10—
						90.67	4		67.43	2	
						90.88:	2	T10	68.40	15	—T10—
						91.08:	2	T10	69.10	5	
						92.78	25	AtmO Sr 1	70.06:	10	T10 Atm vv
						93.58:	-1		70.41:	10	T10
						94.35	8w	T10	70.96:	-2	T10?
						94.93	5	T10	71.26:	-2	T10? AtmO
						97.30	2	T10	71.96	15	Fe 404
						97.97	0	T10?	72.82	0	T10? Atm?
						98.38	8	T10 Fe 1078	73.68:	8w	T10
						98.82:	0	T10?	74.66	3	
									75.26	10w	T10
									76.18	10	T10
									76.97	20	T10 Atm vv
									77.34	-1	T10
									77.93:	-2	T10
									78.40	22	Cr 222
									78.93	30	Fe 111—
									79.79	22	Cr 222
									80.37:	-1	T1(M)?
									80.92	20	Cr 222 Atm vv
									81.91:	-1	T10?
									82.68	3	T10?
									83.19:	-1	T10?
									83.65	4	T10?
									84.64	12	T10
									85.59	-1	
									86.62	10	
									88.56	30	Fe 167 Atm vv
									89.75	6	T10? Atm vv
									90.32	4	T10
									90.86	10	Zr(M)
									91.28	4	AtmO—

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
6900			63.91	25	T10	71.13	15	T10	48.73	3	T10 Atm
91.86	6	Gd 1? Atm?	64.55	30	T10	71.64	30	T10	49.28	20	T10 Ti(M)?
92.87	15w	T10 Atm vv	65.22	33w	T10 Ti 100	72.27	20w	T10 Atm	50.15	5	T10
93.96	-1	Atm?	65.78	5		72.27	6	T10 Atm	50.76	9	T10
94.38	15	Zr(M)?	66.33	30	T10 LaII 1	76.99	30	T10 Fe 1276	51.74	40	Atm vv Ti 99
94.93	14	--T10--	67.14	23	T10			Atm vv	52.18	6	T10?
95.62	-2	--AtmO	67.94	23	T10	77.74	25	T10 Atm	52.64	9	Atm vv T10
96.10	3		68.55	23w	Fe 1004--	78.26	8	T10	53.16	10	Atm T10
96.61	20	Ti 256	69.07	14	T10 Ti 307	79.36	23	T10	53.73	7	Ti 143
96.99	-2		69.51	15	Fe 205p	80.00	21w	Fe 33			T10
97.49	8	--Atm--	70.15	30w	Sr 5 T10	80.56	1	T10?	56.78	10	T10 Ni 97
98.93	8	T10?			T10	81.08	35	Atm T10 Fe 1078	57.71	7	T10
99.25	8	T10?	76.44	18	--T10	82.02	5	Ni 126	58.92	8	--Atm--
99.82	11	--Fe 1051	76.96	18				T10 Atm	59.29	2	T10
7000			77.65	23	T10? Atm	84.97	0		59.64	3	T10? Atm
00.57	8	Fe 1005	78.04	23	T10			T10 Atm	60.50	7	T10
01.47	22	--Ni 64 AtmO	79.00	35	--T10 Atm	87.36	15	Fe 1051 Atm	60.96	5	Fe 267p? Atm
02.03	-1	Atm?	80.22	28	T10	87.85	2		61.45	12	T10 Fe 1188?
02.65	0		80.76	-1		88.17	15	T10	61.89	14	Ni 62
03.52	5w	T10? Si 60?			T10	88.60	12	T10 Ti 99?	63.35	13	T10
04.14	3	--Atm vv	84.57	-3		89.13	15	T10 Fe 463	64.03	20	T10 Atm vv
04.62	20	Ti 256	85.01	25	T10 Co 54	89.47	9	T10	65.35	3	T10
05.11	-1	Atm?	85.48	13		89.95	15	T10 Ti 285	66.28	15	T10 Ti 143
05.96	15	Si 60?	86.38	13	T10			T10 Atm vv	67.09	15w	T10 Fe 61p
		GdII 130?	86.83	12	T10 Fe 815	93.80	11	T10 YII 33	67.95	0	
06.68	10w	Ti(M)--	87.34	BdRv	TiOy(0,0)Rb-head			T10 Atm	68.47	2	T10?
07.94	13	Ti 100 Fe 1078	90.52	25w	T10 Fe 1051	97.04	15	T10 Ni 62	69.21	28	T10 Atm vv
08.36	15	Ti 256			T10			T10 Atm vv	69.91	0	Sum?
09.74	0		93.85	BdRv	TiOy(0,0)Rb-head	99.35	-1		70.29	1	T10
10.42	15	T10 Fe 1221			T10	99.91	32	T10 Atm vv	70.81	-1	T10
10.96	15	Ti 256	97.84	25	T10 Zr 42				71.27	1	T10
11.87	0?				T10	7200			71.62	7	Ti 97--
12.32	-3		7100			01.55	12w	T10 Zr(M)?	72.88	-2	T10
12.90	-2		00.21	10	T10 Fe 61p?	02.26	15	Ca 29	73.21	2	T10
13.36	-2				T10			T10	73.74	8	Ti 212
14.10	-1		02.84	10	T10 Zr 42	05.22	2	T10?	74.23	0	T10
15.04	5w	Fe 167	03.24	25	T10	05.90	30	Atm vv T10	75.30	8	T10
16.05	30	Fe 109 Atm vv	03.84	22	T10 Zr 42	06.48	2		77.49	14	T10 Atm
16.60	35	Co 54	04.23	19	T10?	06.81	3	T10	78.24	8	T10? Atm
17.61	2	T10?	04.74	10	T10?	07.32	7	Fe 1051 Fe 1001	78.45	8	T10 Atm vv
18.18	2	T10?	05.29	8	T10?	07.85	6	T10	79.07	6	--Atm
19.41	2		05.67	8	T10? Atm	08.34	6	T10			T10
20.24	10w	--SmII 59	06.08	15	T10	09.44	23	Ti 99 T10	82.30	1	LaII 1
20.95	1	Atm?	06.52	18	T10			T10	82.61	10w	T10 Fe 1274
21.46	1		07.42	18	T10 Fe 1005			T10	82.86	-1	Sum
21.96	-2	Atm?			T10	13.40	15	T10 Ti 143	83.97	7w	T10
22.51	-2		10.86	18	T10 Ni 64	13.97	8	--T10--	84.35	8	T10 Fe 1004
22.94	20	Fe 1051 Atm vv	11.49	22	T10	14.57	15	T10	85.85	2	Fe 1188
23.52	-2		12.11	15	T10 Fe 404	14.92	10	Ti 314	86.24	5	T10
24.09	8	Fe 1003	12.60	12	T10	15.49	2	T10	88.50	8	T10?
24.65	15	Fe 1187	13.20	22	T10	15.91	10	T10 Atm vv	88.72	8	Fe 1077
25.27	0		13.64	20	T10 Co(M)?	16.21	20	Ti 98 T10	89.25	0	Si 24?
26.06	0	V(M)? Atm	14.09	2		16.75	11	T10	90.74	5w	T10
27.43	12	Zr(M)	14.46	25	T10	16.97	0		92.15	2	T10 Atm
29.03	15	Sum	15.24	25	T10	17.51	13	T10 EuII 8?	92.85	12	Fe 1189 Atm
30.06	5	Ni 126	15.93	30	T10	18.52	8	Sum Cr 264?	93.15	12	Fe 1077 T10
30.63	5w		16.49	-1				Atm	93.89	4	T10
31.16	5w				T10	19.00	3	T10	94.49	18	Atm T10
32.31	5	Sum	22.23	30	T10 Ni 126	19.59	25w	Fe 1001 T10	95.04	2	Fe 1187 Atm vv
32.84	-2		22.82	20	T10	20.43	6	T10	95.66	12	T10 Atm
33.30	12		23.46	17	T10	20.74	4	--Ni 294?	97.12	10	T10
33.77	-1		23.86	5		21.34	15	T10	97.91	8	T10
33.92	-2		24.29	10	T10	21.82	8	T10?	98.52	0	Sum?
34.36	4	Ni 97	24.73	BdRv?	T10	22.46	12	T10	98.92	0	T10?
34.91	0	Si 50?	24.89	25	T10	23.66	20w	Fe 463 T10	99.33	10	--T10 Atm
35.87	20w	Ti 307	25.29	BdRv	TiOy(0,0)Rb-head			T10	99.67	20	Ti 97
36.58	-2	Atm			T10	26.23	10	Si 26?	7300		
37.04	4	Fe 61p Atm	48.17	22	T10 Ca 30	27.32	4		00.45	8	T10
37.40	7	Ni 288?			T10	28.08	3		01.22	5	T10 EuII 8?
38.24	10	Fe 1051	51.49	0	Fe 109 Co 89	28.64	20w	Fe 267 T10	03.02	0	T10
38.82	16	Ti 256 Fe 1078	52.03	20w	T10	29.23	5	T10?	04.54	0	T10
39.29	10	Ti 307	53.13	35	T10	29.70	3	T10	05.07	0	T10
		SmII 57.61 Atm	53.34	2		30.00	0	--Atm	05.48	1	T10
42.16	0w	--SmII 58			T10	30.53	10	T10	06.03	6	
42.66	-1		60.31	15	T10 Ti 98	30.97	6		06.55	5	Fe 1077
43.07	0w				T10	31.31	5	T10	07.12	5	T10 Ti(M)?
44.48	3w		62.84	5	Atm--	32.33	20	Atm vv	08.06	23	Sum Atm T10
45.37	0		62.95	17	T10	32.97	4	T10	08.47	0	
46.92	1		63.60	10w	T10	33.41	5	T10 Ti(M)?	09.68	1	T10? Atm
48.65	-2		64.37	20	T10 Fe 1051	34.24	22	Atm vv T10	10.55	3	T10
49.12	2		65.12	12	T10	35.16	1	--T10--	11.08	7	T10 Fe 1077
50.04	2	Atm--	65.74	15	T10	36.03	8	T10	11.53	-1	T10
50.71	14	Ti 256	66.10	-3	Atm?	37.10	4		12.60	2	T10 Atm
51.41	0	--SmII(M)?	66.61	14	T10			T10			T10
52.92	25	Co 54	67.25	37w	T10 Ti(M)	39.15	10	--Atm	15.57	9	Ti(M) T10
53.60	-3				Ni 109	40.10	35	Atm vv Fe 1105	16.77	10	Atm Fe 267p
54.05	BdRv	TiOy(0,0)Rb-head	68.05	9	T10	40.85	2	T10?	17.44	3	T10 Atm
		T10	68.56	16	T10			T10	18.03	BdRv	
58.16	10		69.12	15	T10 Zr 42	44.49	1	T10?	18.39	15	Ti 212
		T10	69.57	15	T10	44.92	37w	Ti 99 Atm	18.82	4	T10
62.98	30	T10 Ni 64	69.99	20	T10 Atm	45.46	-1	Atm?	19.80	1	T10
63.59	6	T10? Sum?	70.64	10	T10 Atm			T10	20.77	8	Fe 1188, 1276--
						47.59	6	T10 Mn u?	21.45	1	T10
						48.27	6	T10 Atm			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
7300			94.34: -1	T10		78.18: -2			41.40: 0	T10	
22.21 6	T10		95.36: 1	T10? Atm		78.89 3	Fe 683p?		42.06 7	T10? AtmO?	
23.20: 2w	T10? TIII 101p?		95.62: 0	T10		81.78 8	Fe 266p		42.54 -2	T10	
23.90 20	--Sun--		96.49: 0w	T10? Atm		83.50: 5			42.77 -1	T10	
24.53: -1	T10?		97.12: -1	T10?		83.55: 8	LaII 1		46.04 8	T10	
24.83: -1	T10?		97.88: -1	T10?		85.99: -2			46.79 2		
25.40: -1	T10?		98.86: 0	T10		88.02: 0	Sun?		47.83 3	T10	
26.17 18	Ca 44		99.57 -1			88.70: -1	Ni 157?		50.54: 2	T10	
27.11: 0	T10					89.59 15	Ti 225		50.92 8	Fe 266p	
27.63 1	Ni 140		7400			90.38: -1			51.72 9w	T10	
28.10: 0	T10?		00.18 25	Cr 93		91.37 12			52.63 3	--T10--	
28.75 1	T10		00.92 2	Fe 204p		91.64 10	Fe 1077		55.56 2	T10	
29.46 1	T10		01.71 4	Fe 1004		94.71 8	Fe 33p		56.01: 0	T10	
30.16 2	--Atm		02.28: -2	T10?		95.10 15	Fe 1077		57.04 2	T10	
30.99 16	Ti 143		02.72: -1	T10?		96.11 10	Ti 225		57.62 11	Mg 22	
31.69: -1			03.35 0	T10?		98.57 5	Fe 1001		58.32: -1	T10	
32.28 17	Ti 143		03.96 -1	T10?					59.17 0	Sun?	
33.55 0	T10 Fe 1078		04.48: -2	T10?		7500			60.82 2		
34.19 -1			05.00: -3	T10?		00.28: *	Sun?		61.18 8	Fe 1077 T10	
35.33 3			05.78 4	Si 23?		04.97: 0	Sun?				
35.94 5	Zr 23		09.30 18	Ni 139 Si 23?		06.06 1w	T10?		64.24N 50	AtmO Fe 402	
36.98 2	T10		10.04: -1	T10?		07.27 10	Fe 1137		64.83 25	K 1	
37.74 8	Ti 212p		10.56 -2	T10?		09.72 3			66.01 10w	TiOy(0,2)Ra-head	
38.94 10	V 117 T10?		11.15 20	Fe 1077		11.04 20	Fe 1077		5	T10	
39.60 0	T10		13.01 2			13.74 0	Sun?		71.59 Bdrv	TiOy(0,1)Ro-head	
40.82: 1w	T10 Fe 684p?		13.92: -1			14.22 -1	Sun?		6	T10	
41.59: -1	T10		14.50 22	Ni 62		22.79 16	Ni 126		75.56: -1		
42.37: 0	T10		15.32: -1	T10? Si 23?		25.13 14	Ni 139		77.75 0		
43.39: -2	T10		15.98 2	Si 22		26.09: -2	Sun?		78.43 6	T10	
44.15 6	Fe 266p		17.00 8			26.60 0			79.04 5	TiOy(0,1)Qc-head	
44.74 25	Ti 97		17.40 13	Co 89		31.16 17	Fe 1137		79.70 4	T10	
45.91: -1	T10		18.00 0			33.39: 0	FeII 72?		80.32 9	T10 Si 36?	
46.59: 0	T10		18.69 8	Fe 1001		*	Sun?		81.08 7	T10	
47.17 4	Fe 266p		21.60: -1	Fe 1188		37.45: -1	Fe 1000p?		81.80 8	T10	
47.86: 0	T10		22.32 14	Ni 139		38.23: -1			82.63: -1		
48.66 3	T10		23.19 8	Ti 97		39.41: 0	Sun?		83.12 10	--AtmO	
49.47 2	T10 Atm		23.54: 0	Si 23		40.45 10	Fe 266p		83.56: 0	T10	
49.89: 1			24.61N 15			41.09: -2			84.11 3w	T10	
50.44 3	T10		25.32 -1	T10? Atm		41.93: -2	Sun?		84.72: -2		
51.06 1	--Fe 1275		25.95 -2	T10?		46.19 9	Fe u		85.23 5	T10	
51.55 1	Fe 1275		26.59 -1	BuII 8		47.08 -1			85.80 2	T10	
52.14 8	Ti 272		27.49: -2	--Sun?		47.92 0	Fe 1306p?		86.55: -2		
53.16: 4	T10		28.27: -1			51.15: -1	Fe 1303p?		6	T10	
53.38: 6	T10		28.85: -3			52.88 2	Sun?--		91.62 13	Mg 29p T10	
54.20: -2			30.61 9	Fe 204 T10		53.96 2	Co 183		92.14: -1	T10	
54.58 7	Co 53		30.92: 0	Fe 1189		54.73 8	Zr 23		92.68 5	T10	
55.36: -2	--TIII 101?		31.62: -2	V(M)?		55.63 16	Ni 187		93.40 8	T10 Zr(M)?	
55.89 22	Cr 93		32.03 1	Ti 142p?		58.41: -1	Zr(M)?		5	T10	
56.51 7	V 117 Atm T10?		32.69 9			59.73 3	Fe 1308--		97.61: -2	T10?	
57.00 3	Sun?		33.33 6			60.49: -1	Sun?		98.22 9	T10	
57.75 25	Ti 97		33.90: -1	V(M)?		62.13 0	Zr(M)?		98.94 50	K 1	
60.16: 0	T10		35.62: 0	T10 Sun?		62.02 2	Fe 1251		99.64 8	T10	
60.48: 1	T10		36.11: -2	T10?		65.37 2w			7700		
61.49 10	V 117 Ti 212p		36.57: -2	Bu(M)?		67.50: -1w	Sun?--		00.39 4	T10	
62.23 7	--Al 11		37.15 3	Co 53		68.92 15	Fe 1077		4	T10	
63.14 7	V 117 Atm		39.86 8	Zr 23 T10?		69.55: -1	Sun?		04.42: -2		
64.10 30	Ti 97		40.56 10	Ti 225		74.06 10	Ni 156		9	T10	
64.70 -2	T10		40.97 1	Fe 1273		74.70: -2			10.42: 15	Fe 1077	
65.21: -1	T10? Atm		41.52: 1			75.34: -1	--Sun?		11.06: 3	T10	
66.08 -2	T10		41.88: 0	Gd(M)?		76.51 -1			11.47: 14	T10	
66.57 22	Ti 96		42.46: -1	Rh(M)?		77.43 1			12.71 17	T10 Co 126	
67.19 0	Zr(M)?		43.04 8	Fe 1002		78.65: -2w	--Co 53 V(M)?		13.20: 0?	T10?	
68.67: 3	Atm T10		43.92: -2			80.25 8	Sun		14.33 23	Ni 62 T10	
69.33 0			44.47: -2	T10?		82.15: -2	Sun		15.67 8	Ni 109 T10	
70.20 4	BuII 8 T10		44.67: 0			83.79 20	Fe 402		11	T10	
70.91: -1	T10 Atm		45.76 20	Fe 1077 Fe 107p		86.03 16	Fe 1137		23.26 20	Fe 108	
72.10: -1w	T10		47.38 3	Fe 1273		86.75 0	Co 139		7	T10	
72.97: 4w	T10 Fe 108p?		48.04: -1			87.63: -2			27.63 20	Ni 156	
4			48.89: -2			88.34 0	Fe 1306p		28.33 18	T10	
76.50: -1	T10 Fe u?		49.47: 0			89.19: Bdrv	TiOy(0,1)Ra-head		29.41 9w	DyII(M)?	
77.17: 2	T10		50.29 0	YII 25		89.37 15	T10		14	T10	
77.80: 1	T10 Zr(M)? Sun?		51.43: -2			90.67 10w	Co 89--		40.62: -2		
79.28: 1w	T10 Zr(M)?		52.14 -1			91.88: -1			41.02 6	T10	
80.76 0	T10		52.99: -1			92.44 -1			41.24 4	T10 Se(M)?	
81.88 4	Ni 292		54.04 2	Fe 1001		3	AtmO		42.00 1	T10	
82.60 8	Fe 266p T10		55.95: -1			94.44: 8	AtmO--		42.71 14	Fe 1306 T10	
82.98 5	Fe 1188 T10		56.59N 20			22	AtmO		43.23 15	T10 Co 183?	
83.63: -1			57.36: -1	Co(M)					5	T10	
84.09: 0			61.52 11	Fe 204		19.20 9	Ni 156		48.29 20	Fe 402	
84.48 1			62.33 24	Cr 93		19.81: N 0	T10? AtmO?		48.88 15	Ni 156 T10	
85.35 12	Ni 84		64.23: -1	Sun		21.16 2	T10? AtmO		49.59 10	T10	
86.33 8	Fe 1275 Ni 286?		64.85: -2			25.21 -1	T10		50.21 5	T10	
86.99 0	T10 Fe(Sun)?		66.45 -1	Ti 142p		27.64 40w	AtmO-- T10		51.10 20w	TiOy(1,2)Ro-head	
87.68 6	Mg 30		67.49: -2			28.46: -1	T10 AtmO		5	Fe 1304	
88.71 3	Co 139		67.75 0			29.56 4	TiOy(0,1)Ro-head		55.17 20w	T10 SmII(M)?	
89.39 22	Fe 1077		69.95 20	Sun		32.51 5	T10 AtmO?		55.80: 0	TiOy(1,2)Ro-head	
90.00: -2	T10		71.23 17			33.60 -1	T10		56.58 16	T10	
90.41 6			72.82 -1	Fe 267p?		35.63 1	T10 AtmO?		57.35 5w	T10 Rb(M)?	
91.00: 0	T10		72.86 0w	--Sun?		36.63 1	T10 AtmO?		27	T10	
91.55 9	--T10--		73.54: 0	Fe 1188		37.00 0	T10		80.53 20	Fe 1154	
93.61 20	Ni 109		74.46: -2	Fe 957p?		37.56 7w	T10--		12	T10	
			76.33 3			40.98 2w	T10 AtmO?		88.93 50	Ni 62	
			77.54 6	Fe 957p							

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
7700			09.50	8	T10	14.38:	0	T10	12.16	6	Fe 265p
89.71	0	--T10	3		T10	14.96:	-1	T10	14.30	-1	Zr(M)?
90.35	10	T10	12.90	30	Fe 12	15.77	3	T10	14.99	3	
91.47	15		4		T10	17.01:	3w	--T10?	15.88	0	Sun?
92.76	3	T10	16.96	5	--Atm wv?	18.06	3	Cr 299--	16.82	50w	V 30
93.20	2	T10	17.55	3w	T10 Ni 109	20.20	5	T10	18.21	2	
93.86:	-2		18.30	3w	T10	21.37:	3w	T10 Atm	19.55	5	--Atm
5		T10	5		T10	23.98	0	T10	22.08	9	Atm--
97.59	20	T10 Ni 201	26.24	7	T10 Ti 308	24.85	22	Ti 151	23.30:	1w	
5		T10	27.06:	-1	T10?	25.61	7		24.11:	0	
7800			27.97	10	T10	28.35:	-1	T10	25.60:	-1	
00.29	10	T10 Rb 1	28.54:	0w	Atm?	27.35	30	V 30	27.02:	2	
00.79	2		29.68	1	Sun?	27.94	9	Fe 623p Atm wv	27.89:	2	Sun?
01.42	4		30.35:	1		28.34	14	Fe 1154	29.34	9	Fe 265 Atm
12		T10	30.78	8w	T10 Mg 42p?	29.06:	4	T10?	30.55	9	--Atm--
11.11	16w	T10 Mg 45p?	32.40	3	T10	30.58:	0w	--Atm wv	30.95:		
9		T10	33.09	8	T10 Cu 6	32.71	2w	--Atm	31.45	5	Ti(M)?
19.37	7	T10 Zr(M)?	33.61	5	--T10--	33.59:	1	T10? Atm	32.05:	0	
20.15	5	T10	*		T10?	34.05:	-2		33.07	30	Atm Zr 40
20.93:	6	T10	36.40:	0	T10	34.99	5	T10 Sun	*		Atm?
4		T10	37.14	20	Fe 1136	36.01	1w	T10	38.07	2	Atm Sun?
26.77	10	T10 Ni 109	4		T10	37.05:	-1		41.82:	-1	
3		Zr(M)?	41.09	13	Fe 623 T10	38.13:	0		44.57	25	V 30 Atm
30.02:	8wv?	T10	41.93	10	T10	40.17:	3w		45.55	10	Atm Fe u
30.32:	10w	T10	43.27:	0	T10	40.96:	0	T10	4		Atm
*		T10	43.92	10	T10 Ti 308	41.71	2		49.68:	-2	
32.25	20	Fe 1154	44.60	12	Zr 40	43.16:	0	T10	50.08:	0	
33.21	0	T10	45.91	24	Fe 1154	43.82	0	T10 Atm wv	50.66:	-1	
34.55	9	--T10	46.77:	0	T10	44.39:	-1		51.88	17	Atm wv--
35.38	8	Al 10 T10	47.18:	0	T10	46.06	20	Fe 1136	52.62	0	Zr(M)?
36.20	10	Al 10 T10	47.65	3	T10 Rb 1	46.84:	-1	T10	54.70:	0	V(M)?--
37.08	8	T10	49.17	32	Ti 125	47.63	30	Fe 12	55.34	2w	
37.66	2	T10	50.51	2w	T10 Atm	*		T10	57.92	-2	
38.24:	0	T10 Co(M)?	51.10:	-2	T10	49.99	5	T10	59.10	0	
39.36	5	T10	51.67	7	--T10--	51.14	0	T10	61.11:	35	V 30
39.93	1w	T10	52.60:	-1	T10	51.92:	0	T10	62.21	-1	
40.53	15	--T10--	53.27	1w	T10	53.12	6	T10 Sun	62.90	0	
5		T10	54.01:	0		54.16	9	T10	63.80	35	--Atm wv
44.61:	1	T10 Fe(M)?	54.79	8w	T10 Fe 402p	55.30:	0	Zr(M)?	64.65	3	--Atm wv
44.86	7	T10	55.77:	0	T10 Fe 1305	56.88:	0	T10	65.93	2w	
45.49	3	T10	56.83	8	T10? Zr 41	57.52:	0	T10?	66.82:	-1	
46.29	4	T10 GdII u?	58.31:	0		58.11	11	Zr 41	67.57:	-1	
47.09	6	T10	59.10	0	T10	62.62:	0	T10? Atm?	70.70	0	
47.55:	-	T10	59.69:	-2	T10 Atm wv	63.15	15	Zr 40	71.29	0	V(M)?
48.07	0	T10 Atm?	60.03	10	Atm wv Zr(M)?	64.21	1	--T10	6		Atm
48.86	3	T10	60.73:	0		65.16:	-1	T10	79.12	1	Fe 1136--
49.36	8	Zr 40 T10?	61.59	9	Ti 308	66.07	10	Ti 151p	80.29	1	
50.86	4	T10	62.52	8	T10 Atm wv	67.26	2		83.26	35	Na 4
51.74	2	T10	63.28	8	T10? GdII u?	68.27	19	Ti 151	84.59	2w	
52.71	25	Ti 34p	65.13:	2		69.83:	0	T10	86.73	30	V 30 Fe 1272
5		T10	65.73	10		70.11	20	Zr 40	93.01:	-2	Co(M)?
55.42	2	T10 Fe1305p?	66.67:	-1	T10?	70.74	2	T10	93.87:	-1	
55.81:	-1	T10	67.66:	1w	T10 Atm	71.58:	-2	T10	94.82	40	Na 4
56.33:	-2?	T10?	68.52	5	T10	72.23	16	Fe 108p	95.62:	-1	
56.90	10	T10	69.32:	-1	T10	73.81:	-1		98.87	20	V 30 Fe 1154
57.66:	-2	T10	70.18	7	--T10	74.43:	0	Sun?	8200		
58.21	0	T10	71.80:	-2		75.19	30	Fe 12	00.94	0	Atm? Sun?
58.72:	-1	T10	72.36:	3	T10	76.51:	-1	T10?	01.70	11	Zr 40
59.34	10w	T10	73.12:	-3		77.09:	0	T10?	02.99	1	V(M)?
69.76:	20w	T10 Zr 41	5		T10	77.95:	1	T10?	03.25	0	
11		T10	77.22	1		79.45:	-1	T10?	04.11	20	Fe 12p Atm
79.69	2		78.77	25	Ti 151, 308	80.57	19	Ti 195p	04.85	24	Fe 12p
80.68	4	T10	79.84	0w	Atm wv?	81.61	6	T10 Sun	05.78:	0	TiOy(0,2)a-head
81.69:	4	--Sun--	80.60:	0	T10	82.50	0	T10	07.04:	0	T10?
82.18:	4	T10	81.17:	2	T10	83.22	1	T10	07.74	4	Fe 1136
82.90:	0	T10	82.06:	0		84.54	3	T10	08.58	1	Co 193
83.51	0	T10	5		T10	85.19	22	Fe 1136 T10	09.66	7	Atm?
84.30	4	T10 Atm	87.40	23	Co 89	86.23:	1		10.20:	-1	
85.01	25	Ti 34p	88.00	3	T10	87.36:	0	T10?	11.45	15	Atm wv Si 19?
85.65	0	T10	89.37	1	T10?	88.41:	0	T10?	12.57	18	Zr 40
86.15:	-1	T10?	90.00:	-1	--Atm wv	89.65	7w	T10	13.05	16	Mg 28p
86.67	2	Atm wv T10	90.75:	-1		90.48	0	Sun	13.81	0	Atm?
87.80	8w	T10	91.68:	-2		92.64	6	Cu 6 T10	15.22	2	Sun--
89.23	7w	T10--	92.31	8	Sun	93.51	30	V 30	17.27	5	
90.09	2	T10	93.61	7	--T10--	94.00:	3		18.59:	1w	--Atm wv
90.65	0	Atm--	94.38	8	--Fe u	94.88:	-2	T10?	20.36	22	Fe 1136
5		T10	95.05:	1w	T10	95.02	6	T10? Atm	21.35:	-1	
95.47	12	Ti 34p Atm wv	95.67:	1	T10?	96.88	9	Fe 999	22.19	0	
5		T10	96.45	10	Ti 308	97.64	1	T10?	22.76:	-1	Atm
7900			98.97	20	Fe 1136	98.36:	0	T10	29.75	0	--Atm
01.09:	6	T10 Atm wv	8000			98.81	4	T10 Mg 41p?	30.61	30	Atm wv Si 19?
01.86	1	T10	00.23:	1	T10?	99.49	-1		31.62	3	
02.44	0	T10	00.85:	0w	T10	8100			32.29	10	Fe 1136
03.05	3	T10 Sun	01.64:	-1w	T10	00.10	3	T10	34.49:	-1	
04.04	6	T10 Ti(M)?	02.52:	5		*		T10?	35.95:	-1	
6		T10	03.31:	1w	T10 Atm wv	02.50	2	V(M)? Atm	37.38	0	
08.13	5	Atm wv	03.80:	0w		03.17:	1		39.17	30	Fe 108 Atm wv
08.60	10w	Atm wv Co 189?	04.75:	0	T10	03.97:	8w	T10 Atm?	41.03:	-1	SmII(M)?
			05.25	0	T10	05.79:	M 1		41.64	25	V 30
			07.54:	4	T10	07.23	6	Atm--	43.82	0	
			08.49	5		08.36	10	Fe 265p Atm	46.76:	-1	
			09.33:	0		09.07:	-1	V(M)? Atm	47.59	0	Fe(M)
			10.17	8	T10? Atm?	09.99	2	Atm?			
			12.02:	2w	T10	11.90:	2	Atm wv			

TABLE 5 -- Continued

λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.	λ	Int.	Ident.
8200			70.16	Sw	Zr 40	97.08:	3	Fe 1172	62.12	90W	CaII 2 Fe 60
48.14	7	Fe 1156	70.62:-	-1		98.07	75W	CaII 2 Zr 40	67.36	4	Fe 166p
50.80:	1w		71.36:	3w		*			71.95	4	Fe 1272p Mn 49
53.53	35w	V 30	72.81	6w	-Co 193-				74.73	25	Fe 339
55.90	45	Atm vv V 30	73.99	5w		8500			75.40	32	Ti 68
57.30:	0w		75.54:	5w		02.58:	1		78.90	2	
61.85	2		76.45:	4w	Sun? Atm	05.08:	0	Atm?	79.81	1w	
63.21:	0	Atm?	77.84	35	Ti 33	08.77:	0w		80.78:	0	Fe 999p
63.98	1	Fe 1272p	80.93:-	-1	Atm--	14.07:	30	Fe 60	83.00	32	Ti 68
65.02	1		82.59	5W	Ti 33 Fe 12p	15.15:	20	Fe 401	84.25	2	
66.42	3	Sun	84.77	1	Atm? Sun?	18.03:	25	Ti 182	86.30:	-1	
67.06:	0	Sun?	85.63:	0		18.42	30	Ti 150	88.66	40	Fe 60
67.59	2	TiO u?	86.34	11	Ti 182p--	19.04	-1		89.79	1	Fe 507p
69.56:	4		87.79	30	Fe 60	21.26:	-2		92.35	30	Ti 68
			89.45	13	Zr 40 Ti 182	26.02:	-1	Ti(M)?	94.05	0	
			90.51:	1	Sun?	26.69	-1	Fe 1270	95.18	-1	
3		Atm vv				28.07	1w		97.35	-1	
74.51:	0	Atm?	92.30	0		30.25	2W		98.73	7	Fe 400p
76.71	1		96.91	35	Ti 33	31.41	5	Ti 141	99.50	4	Fe 1267
80.04:	1		97.85:	0		38.09	-1				
82.54:	0		98.57:	1		38.67	-1				
83.64:	0		99.20:	0	--Atm	39.31:	5	Ti 209	8700		
84.55:	0	Atm?				42.13	85W	CaII 2	01.06	0	Mn 49
84.92:	-1		8400			46.40:	0		02.49	2	Ni 83
85.49:	-1		01.42	13	Fe 108	46.91:	-2		03.73:	0	Mn 49
88.00:	0		01.86	0		48.10	17	Ti 150	05.89:	-1	
92.04:			02.65:	4	Ti 224--	48.95	7	Cr 56--	07.45	2	Cr 56
93.50	23	Atm vv Fe 623	03.48:	-2		50.62	1	Ti 141	10.38	13	Fe 1267
96.12:	2		04.22:	-1	Sun?	55.54	5	Cr 56	11.18:	-1	
97.75:	-1					58.85	8		12.68	1	Sun
97.41:	1	--Atm--	07.15:	0		59.91	-1	--Fe 1321	13.22	17	Fe 400, 1267
98.02	0		11.34	1	Sun?	60.77:	0		14.08:	1	
98.53	2		12.38	35	Ti 33	62.20	2		16.15:	0	
8300			14.03	6	Zr 40	64.55	0		16.55:	0	
01.29:	1w	TiO(0,2)e-head	16.97	10	Ti 224	65.54	6	Ti 141	17.18:	-2	
03.15	1	Fe 265p	17.62	8	Ti 182 Atm	67.09:	-1		17.78	7	Sun
05.94	17	Zr 40	18.73:	0	Ti u	67.76:	-1	Fe 1269p?	18.67:	0	
08.29:	0	Ti(M)	19.88:	1	Sun?	69.73:	8	Ti 209	19.61	12	Ti 140
06.82	1	Atm vv Si 19?	20.49:	1	Sun?	71.90:	0		25.75	0	Ti 139
07.47	35	Ti 33 Fe 12p	21.32:	3	V(M)?	73.32	1		29.17	10	Fe 713p
08.27	-1		22.32:	1		74.54	5	Co(M)?	31.27:	0	
10.14	0	--Atm--	23.04:	12	Fe 999 Ti 150	75.54	3	Co(M)?	34.73	50	Ti 68
11.03	11	Fe 12p	24.41	17	Ti 182	77.52:	-1		36.07	12	Mg 39p
11.73	1	Ti(M)	25.90	4	Fe 12p	78.46	11	Ti 141	38.69:	0	
12.41:	0		26.54	35	Ti 33	82.27	25	Fe 401	40.94	1	Mn 49
14.73:	1		34.94	35	Ti 33	83.32	0	Sun	42.41	1	Si 44
15.30:	0		35.75	33	Ti 33 Atm	*			43.57:	-1?	Fe(Sun?)
21.65	2		36.78:	0		91.52:	-1		45.35:	0	
23.32:	5w		38.91	17	Ti 224	93.02	7	Fe 1267	47.42	8	Fe 401
25.47:	0		39.61	14	Fe 1172	95.61	1		47.91:	-1	
26.05	1		42.97	3	Ti 210	96.18:	0w		48.60:	0	
27.05	37	Fe 60	45.04:	1		98.22	8	Ti 256	49.28:	1	
28.52:	-1		45.74:	0	Sun?	98.83	6	Fe 1153	52.01	3	Si 43
30.41:	0		46.48	3		99.97:	0		57.17	25	Fe 339
31.33:	-1	V(M)?--	47.67	19	Fe 12p				61.40	9	Ti 139
31.94	20	Fe 1153	49.91:	3		8600			63.95	13	Fe 1172
32.93	4	Atm Sun?	50.24	8	Cr 56	00.99	3	Ti 141, 209	64.83:	-1	
34.40	30	Ti 33	50.92	10	Ti 224	02.56	1		66.68	25	Ti 68
37.34	0		51.93:	2		03.96:	-1		71.27	9	
39.41	14	Fe 1153	53.09:	1w		05.96:	2	Fe 1272p?	72.14	2	
42.85	10	Fe 401p	55.25:	4	Cr 56	07.10	2		72.88	20	Al 9
42.84:	0	Fe 1270	57.10	10	Ti 141	07.80	0		73.92:	23	Al 9
43.45:	0	Atm--	60.31	1		10.65	1	Fe 1153p?	78.68	18	Ti 140
43.96:	0		62.43:	0	Fe 814p?	11.82	25	Fe 339	79.47:	-1	
44.60:	0		63.56:	2		12.62	1	Ti 141	80.07	1	
45.20:	1	Fe 265p	64.10:	-1		15.98	0	Fe 1272p	84.41:	1	Fe 1270
46.30	15w	Mg 40p?--	64.66	5	Zr 40	16.21:	-1		90.44	12	Fe(Sun)
47.32	1		65.73	0		16.26	5	Fe 1266p	93.32	20	Fe 1172
48.31	12	Cr 56	67.17	13	Ti 182	17.11:	0w		8800		
49.02	30	Fe 12p	68.45	33	Fe 60 Ti 150	18.39	5	Ti 209	00.55:	3	Sun
50.85:	0		71.76	5	Fe 1270	21.62	23	Fe 401	04.63	15	Fe 106
52.34:	-1		75.79:	0		23.40:	-1		06.83	40	Mg 7
53.17	30	Ti 33 Atm	77.59	1		31.18:	0		09.46	3	Ni 200
54.43	1w	Sun?--	78.45	4		33.42:	1w		19.51	10	Ti 68 Fe 1266p
55.44	0		82.04	1	Sun Fe 999p?	33.98	3w	Sun			Fe 1266p
58.58	12	Fe 401p	83.17	10	Ti 150	35.08:	-1		20.21:	-1	
60.82	14	Fe 1153	85.72:	0		36.20	2	Cr 56 Ti(M)	21.25	15	Ti 139
62.06	10	Sun	88.02:	1w		37.01	0	Ni 186	22.90:	-1	
62.66:	1		91.54	2		43.07	4	Cr 56	24.27	35	Fe 60
63.59	9	Ti 182p	92.98	3		48.49	3	Si u	38.46:	30	Fe 339
64.28	37	Ti 33	94.43	13	Ti 141	50.01:	-1		39.19:	20	
65.66	13	Fe 623	95.42	1	Ti 210	52.55	0	Fe 1050p			
			96.05	5	Ti 209	54.49	3	Fe 623p?			

NOTES TO TABLE 5

- 3403.32 The principal contributor to this line is Fe (304), which was inadvertently omitted from the table.
- 3435.47 Another secondary contributor to this line is Cr (53).
- 3440.62 For 2 A on either side of this strong Fe line and the following line, lesser lines are lost in the wings.
- 3695.62 This line is as strong as λ 3695.52 on Plate 2376, but barely visible on Plate 2396.
- 3933.52 The intensity given here applies only to the absorption core which is about 0.5 A wide.
- 3977.76 Intensity scale of Plate 2083 (enlargement) was adopted for the following 40 lines. After that, intensities are simple means of intensities estimated on enlargements of Plates 2083 and 2376. This holds until λ 4900.
- 4084.49 General background absorption extends from this line to approximately λ 4088. This absorption coincides in position with the (1,0) band of CaO.
- 4143.11 The (1,1) band of SiN starts here, producing the characteristic background absorption which is seen in Figure 4,a.
- 4272.89 On Plate 2083 this line is fainter than λ 4272.78!
- 4464.70 General background absorption begins here and continues up to λ 4465.81.
- 4639.01 General background absorption begins here.
- 4932.06 Weak general absorption begins here. It may be due to an unknown molecule.
- 4954.06 Intensity scale based on Plates 2376 and 2083 ends here.
- 4959.66 From this point on, wave lengths and intensities of pure TiO lines occurring in groups are omitted. The number of TiO lines and features which have been omitted is indicated by a numeral in the first column.
- 5040.63 Strong absorption from here to λ 5041.74.
- 5075.81 Strong absorption from here to λ 5077.21.
- 5129.32 Absorption is practically continuous from λ 5129.0 to λ 5130.6.
- 5276.11 After this line, the continuous background suddenly gets stronger on Plate 2621 than it is for 100 A on shortward side. Such a rapid change is not evident on Plate 671. The phenomenon may not be real, or it could be due to uneven plate development.
- 5501.48 After this line, vibrational quantum numbers for TiO are again omitted because the (1,2) band can no longer be identified (inadequate laboratory data!).
- 5888.14 General absorption begins here and continues until λ 5891.5. There is no similar absorption associated with the D1 line.
- 5899.27 Appears double, the longward component being stronger.
- 6782.51 From here on, intensities are less reliable, being from Plate 2899. These intensities are, on the whole, somewhat larger than the preceding ones. Furthermore, the spectrum appears to fade out for about 100 A.
- 6874.11 After this line the atmospheric oxygen lines (B-band) are not included in the table unless they are definitely blended with stellar lines.
- 7424.61 A line of Si, multiplet (23), coincides with this stellar line but could hardly contribute more than one unit to the intensity.
- 7456.69 This line may be the same as the sun-spot line of approximate intensity 1 observed by Mr. Babcock, who believes the latter is a predicted Ti line.
- 7619.81 Beyond this line, atmospheric oxygen lines continue to interrupt the stellar spectrum, but individual lines are not cited unless they are blended with stellar lines.
- 7664.2 This is the atmospheric oxygen line which masks the K line in the solar spectrum.
- 8105.79 From here on there appear to be no more TiO lines on the laboratory comparison plate.